Built to Sell: Strategic Entry in Pharmaceutical Markets *

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October 16, 2026

Most new cancer drugs are invented by small biotechnology firms and brought to market by large ones, with control changing hands during clinical development. For a potential entrant, the prospect of acquisition acts as an entry subsidy. Antitrust authorities read the same transactions as the removal of future competitors, and the remedy under debate bars incumbents from acquiring development-stage compounds in markets where they hold overlapping products. Using a quarterly panel that links oncology drug development to ownership of control rights, we show that acquirers are a small, identifiable set of large firms whose demand rises predictably as their own products approach loss of exclusivity. We then model the sale of a development-stage compound as a first-price auction among the incumbents that engage, so that the set of permitted acquirers enters the price, and estimate how the acquisition payoff affects whether small firms enter development and which cancer they target. The ban cuts the expected acquisition payoff in half, redirects about 3 percent of small-firm entry away from the most contested cancers, and lowers the number of entering compounds by about 2 percent. Valuing deterred entrants and prevented killer acquisitions at the same approval odds, the ban raises expected approvals only if more than a fifth of the acquisitions it prevents would have been killer acquisitions, about three times the highest estimate in the literature.

*We thank Zach Brown, Francine Lafontaine, Sarah Miller, and Charles Murry for their invaluable advising and guidance. We also thank Brian Wang for excellent research assistance. Zoey Chopra and Eric Parolin's effort was supported in part by a National Institute on Aging training grant to the Population Studies Center at the University of Michigan (T32AG000221). The content is solely the responsibility of the authors and does not necessarily represent the official views of the National Institutes of Health or the Department of Veterans Affairs. Nothing in this paper constitutes medical advice or clinical guidance. The therapeutic examples are illustrative, are drawn from the public record of clinical development, and describe where compounds have been studied rather than where they are effective.

1 Introduction

The pharmaceutical industry, particularly the market for oncology, operates differently from other industries. In a traditional pipeline for product development, companies vertically engage in research and development (R&D) and product sales. However, in the market for cancer drugs, one group of firms manages R&D, while another group of firms brings products to market. Innovation is dominated by small biotechnology firms that are typically financed by venture capital to develop one drug. Of the 50 first-in-class cancer drugs approved in the United States between 2010 and 2020, small biotechnology firms were the sole originator of 46% (Kennedy et al., 2023). Commercialization, however, belongs to large, established, often publicly traded firms, which owned or licensed 38 of those 50 drugs by the time they reached the market. Of the 23 drugs that small firms originated, only four were developed and launched by the firm that discovered them. Whether this division of responsibility is *efficient* – where firms operate according to their comparative advantage – or *predatory* – where large firms oligopolize commercialization – is an open question. We study the incentives driving this unique market structure and associated welfare consequences. With estimated spending on oncology therapeutics expected to be $180 billion by 2028 (Jazowski et al., 2024), understanding these questions is paramount.

A primary driver of the oncology market structure appears to be cost. The development of oncology drugs is resource-intensive across time, capital, and probability of success. The duration from first synthesized dose of a drug to approval takes about 10 years, estimates of the capitalized cost of development run from several hundred million dollars to over $1 billion, and only about 5% of compounds that enter human clinical trials reach approval (Wong et al., 2019; DiMasi and Grabowski, 2007). While the probability of approval increases as drugs progress through clinical trials, so too does the cost of financing development. These challenges are difficult to surmount for smaller firms without internal cash flow. In 2024, 39% of public biotechnology firms held less than 1 year of cash (EY, 2025), which is insufficient for funding required stages of development. Even after approval, assets required for commercialization – manufacturing capacity, regulatory function, payer contracting, and specialty sales – are large, expensive, and indivisible, generally pricing out smaller firms from marketing drugs on their own (Teece, 1986).

To continue innovating, these small firms rely on large pharmaceutical firms for funding throughout the R&D process. In 2024, for example, funds raised through licensing deals with large firms were roughly three times the total venture capital raised by biotechnology firms (McLaren, 2025). However, this financing is not

provided cheaply and often comes at the cost of acquisition, either through exclusive licensing or outright purchase of company assets.

Given the timeline and funding constraints inherent in this market, acquisition is the expected outcome of successful early development, rather than a deviation from the norm. Cougar Biotechnology illustrates the path. Founded in 2003 around a single licensed compound, abiraterone, Cougar ran its Phase II trial in prostate cancer. After promising results in 2008, its board hired an investment bank rather than a sales force. Public filings show that 17 potential buyers signed confidentiality agreements, 2 submitted formal bids, and ultimately, Johnson & Johnson (J&J) acquired Cougar for about $1 billion. After acquisition, the drug completed Phase III trials, and J&J launched the drug as Zytiga. Anecdotally, venture investors describe choosing which biotechnology firms to fund by asking who among the incumbent players in the pharmaceutical industry would be likely buyers. For a potential entrant, the prospect of acquisition acts as an entry subsidy, raising the expected value of beginning development even for a firm that will never commercialize a drug.

Antitrust authorities increasingly consider the risks inherent to a market structure that requires acquisitions of entrants by incumbents to bring new products to market. In particular, there are concerns over *killer acquisitions*, in which setting an incumbent firm acquires a development-stage competitor's product with overlapping demand to shelve that product and preempt future competition. In the pharmaceutical industry, Cunningham et al. (2021) estimate that 5.3-7.4% of acquisitions are killer acquisitions. Because many of these deals fall below the Hart-Scott-Rodino (HSR) notification threshold (Wollmann, 2019), they are not reviewed by the Federal Trade Commission (FTC) for antitrust concerns.

In recent years, there has been policy response to target the theoretical origin of killer acquisitions: the acquirer's existing position. If killer acquisitions are meant to deter future competition, acquiring incumbents likely have pre-existing products on the market or in development with overlapping consumer bases. The 2023 United States merger guidelines treat an acquirer's position in the target's market as a basis for scrutiny (U.S. Department of Justice and Federal Trade Commission, 2023), and the European Commission's 2024 ex-post evaluation concludes with recommendations aimed at preventing incumbent acquisitions of overlapping pipeline products (European Commission, 2024). These remedies are forward-looking, preventing horizontal acquisition before the to-be-eliminated competition exists. However, these proposals have not confronted externalities of such policy on entry. If acquisition is an entry subsidy, restricting the likelihood of acquisition decreases the subsidy. Smaller firms whose innovation the policy is

designed to protect may respond by substituting to different markets (i.e., different cancers) or by not entering at all. When designing such policy, understanding the equilibrium effects is necessary to ensure both welfare optimization and continued innovation.

In this paper, we study how the prospect of acquisition shapes the entry decisions of small biotechnology firms in oncology on both the extensive margin – entry versus exit – and the intensive margin – into which cancer market(s) they enter. After developing a structural model of entry that endogenizes the likelihood of acquisition, we quantify how proposed policies that ban overlapping acquisitions would affect entry along these margins and compare entry effects against external estimates of killer acquisitions.

We construct a novel quarterly panel that follows 2,539 oncology compounds from 2006 through 2019, linking each compound's development history to the firm controlling its United States rights. Our model is disciplined by 4 descriptive facts from this panel. First, the set of incumbent acquirers is small and identifiable. The 39 large firms we define as incumbents, between 22 and 26 of them holding an oncology asset in any given quarter, account for 87% of acquisition value and are essentially the only repeat acquirers. A potential entrant (and their venture capital) forming expectations about acquisition is forming expectations about a known set of likely acquirers. Second, demand for acquisition is predictable. An additional $1 billion of revenue within 2 years of losing patent exclusivity in a given market for cancer raises the rate at which incumbents acquire entrants in that market by 48%. Entry into those same markets rises alongside it, suggestive of entrant anticipation of incumbent demand. Third, the sale of development-stage compounds may be considered an auction among few incumbent participants. In the merger chronologies that selling companies are legally required to file, the median process puts 2 parties under a confidentiality agreement when a whole company is sold and 5 when the canvass is for a license, and in both the median number that go on to price the asset is 2. Fourth, consumer (i.e., patient) demand for compounds is price-inelastic, and prices are generally determined as a factor of an observable Medicare average sales price. As new, competing products enter a given cancer market, there is no statistically significant change in price for existing products.

We proceed to estimate a two-stage model of entry. In the first stage, a compound is sold through a first-price auction among incumbents that engage with the sale process. The pool of potential acquirers becomes a small set of engaged bidders, defined by those that elect to conduct costly diligence. The seller initiates the process only when its option value covers the cost. Incumbents holding an overlapping position in the compound's cancer, a Phase II or later development asset or an approved product still on patent, draw their

valuations from a stronger distribution. Following Guerre et al. (2000), we recover valuation distributions by inverting the equilibrium bid function on observable acquisitions. In the second stage, entrants that have completed Phase I trials (which trials are agnostic to specific cancers) choose specific cancer markets for their first Phase II trials, or choose to exit. We estimate this discrete choice model as a conditional logit over 27 cancer indications and the outside option of exit.

Rather than estimate a model of demand, we use the share of patients in a given cancer market to govern how competing products split total revenue available. This is because these products do not appear to compete on price. The payoff in each market is the sum of the expected acquisition value and the expected commercialization value absent acquisition. These payoffs are scaled by a match quality measure built from the compound's mechanism of action (MoA) to account for the fact that not all potential entrants can enter any given cancer indication. For example, a drug with HER2 antagonism MoA is a plausible breast cancer drug, a weaker candidate in gastric cancer, and a sufficiently unlikely opportunity in acute leukemia. Among the indications a compound's mechanism makes feasible, entrants tilt toward the ones where acquisition pays more. A $10 million increase in the match-weighted acquisition payoff raises the odds that a compound enters a given indication by about 38%.

The primary counterfactual we study – the policy most often proposed to remedy killer acquisitions – bars incumbents from acquiring development-stage compounds in cancer indications for which they hold an existing product. Within our auction framework, such a ban is likely to remove the strongest bidders from each market, since overlap is strongly associated with acquisition likelihood. At the working calibration, the expected winning bid falls by 32% on average, and the probability that any engaged party prices the asset falls by 18% on average. As a result, the expected acquisition payoff at entry decreases from $36 million to $18 million on average. Entry responds on both margins. Under the ban, aggregate entry falls by 2.2%. On the intensive margin, 2.9% of entrants substitute their cancer indication and enter a different cancer market than they otherwise would have entered. Non-small cell lung cancer loses 16% of its entrants and non-Hodgkin lymphoma loses 8%, while cancers with fewer overlapping incumbents gain entrants by 7-11%.

While these are entry effects rather than welfare effects, they support a useful accounting. Valuing deterred entrants and prevented killer acquisitions at the same Phase II approval odds, the ban raises expected approvals and increases innovation only if about 22% of the acquisitions it eliminates are killer acquisitions – roughly 3 times the highest estimate of the killer acquisition share in the literature (Cunningham et al.,

2021; European Commission, 2024). Though these findings are preliminary, and extrapolating to welfare is the most valuable quantity to determine, initial estimates suggest that this policy comes at the cost of innovation. Rather than promote mass exit, instead entrants are likely to substitute to markets less affected by this policy, with potential benefits for patients in these markets and costs to patients in other markets.

Our question sits between the literature on entry and innovation. A theoretical literature recognizes that entry can occur in anticipation of buyout, such that the likelihood of acquisition increases the return to entry (Rasmusen, 1988; Gans and Stern, 2000; Gans et al., 2002; Norbäck and Persson, 2009). Simultaneously, recent work has studied the settings in which acquisitions are made only to deter competition (Cabral, 2021; Fumagalli et al., 2023; Letina et al., 2024; Denicolò and Polo, 2026), and Teh and Wang (2025) survey this literature. Evidence from Cunningham et al. (2021) and others moved killer acquisitions to the center of merger policy, and a small set of structural papers now attempts to quantify the entry subsidy from acquisition in several industries (Eisfeld, 2025; Wabiszewski, 2025). In these models, however, the acquisition payoff is either exogenous to the model or a result of a bargaining game with only one acquirer. In both cases, counterfactuals cannot directly address the object of policy debate: the set of firms permitted to tender offers for acquisition. We contribute uniquely to this literature by endogenizing the set of acquirers and likelihood of acquisition in our entry model.

Using novelly-constructed drug-quarter panel of firm ownership, our methods closely follow the empirical auctions literature. Theoretically, we build upon Guerre et al. (2000) to invert the distribution of observable winning bids distribution into a distribution of private valuations. Empirically, our use of merger chronologies as records of potential and actual bidders draws from the takeover-auction work of Boone and Mulherin (2007) and Gorbenko and Malenko (2014). However, because we only observe the distribution of winning bids across heterogeneous assets, we do not directly model the valuations in levels, but instead, in shares. Our approach may transfer to other thin asset markets where only transaction prices are observed (Haile and Tamer, 2003; Krasnokutskaya, 2011). On the entry side, a substantial literature shows that market size, expected profitability, and regulatory design steer pharmaceutical development (Blume-Kohout and Sood, 2013; Budish et al., 2015; Yin, 2008; Dranove et al., 2022; Gupta and Kao, 2025), and that incumbents buy development-stage assets as their own products approach the end of exclusivity (Higgins and Rodriguez, 2006; Krieger et al., 2022). From our investigation, the likelihood of acquisition is just as critical to consider.

The paper proceeds as follows. Section 2 provides background on oncology drug development and fixes the definitions of specific markets within oncology, entrants, and acquisitions. Section 3 describes the data

sources and construction of the panel. Section 4 presents the entry model, and Section 5 provides reduced form evidence for imposed assumptions. Section 6 estimates the model's parameters, and Section 7 applies these estimates to the policy counterfactual. Section 8 concludes with paths for future work.

2 Setting

Oncology drug development is an attractive setting to study how acquisitions affect entry and innovation. First, entry is observable. Phase II trials are a dated event that require regulatory filings and a named market choice. Second, development stage drugs change ownership through a sale process that may be conceptualized as an auction. Third, approved oncology drugs have a market exclusivity that is known to market participants years in advance. The end of these exclusivity periods is known to induce *revenue cliffs*, after which significant revenue loss is expected. This section describes each of these features in turn and fixes definitions of markets, entrants, acquisitions, and patent cliffs.

2.1 Drug Development and Clinical Trials

To become an approved oncology drug, a compound must move through a fixed sequence of stages. Preclinical work and an Investigational New Drug (IND) application are first required to establish that the compound can be tested in humans. Phase I trials then enroll upwards of a few dozen patients to test the toxicity of the compound and determine the dosage. These trials say little about whether the compound actually works, and are not required to determine a specific cancer to trial in. Phase I trials often include patients across several types of cancer which are bucketed together as a trial for "solid tumors" or "hematological malignancies". Phase II trials enroll significantly more patients, often around one hundred, and test for efficacy against a named cancer. Phase III trials are significantly larger and more expensive, requiring evaluating efficacy across hundreds of patients against the standard of care. Finally, if the trial sponsor believes these results are compelling, they can fire a New Drug Application (NDA) or Biologics License Application (BLA) for approval in a specific indication.

Development across these stages is slow and usually ends in failure. Roughly one compound out of every twenty that enters Phase I trials will reach approval, which is the lowest of any therapeutic area (Biotechnology Innovation Organization et al., 2021; Wong et al., 2019). All told, the sequence from first human dose to approval will take roughly a decade, though trial lengths vary significantly by cancer indication (Budish et

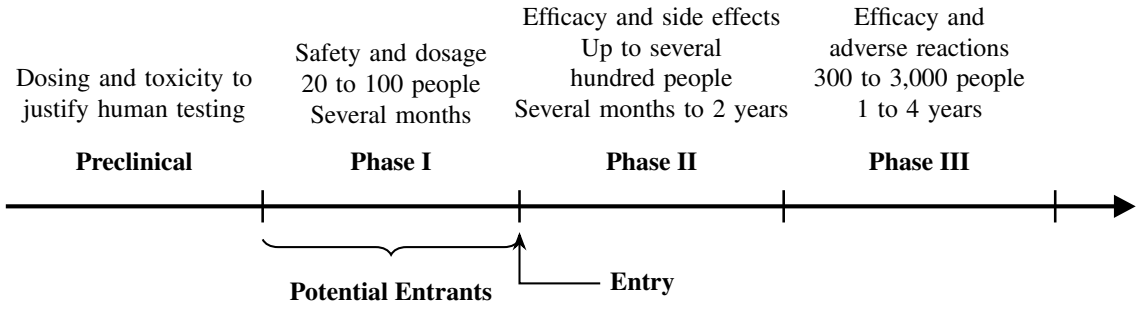
al., 2015). In addition to the significant time investment, trials are also prohibitively expensive and require expertise and resources to pursue. The full cost of bringing compound through the regulatory process to commercialization has been estimated to be between $100 million up to $1 billion (DiMasi and Grabowski, 2007).

Two definitions are required from this process to answer our research question. First, we define a market as a cancer indication. Biomedtracker (discussed in Section 3.1) records development against 64 oncology indications, several of which are clinically close enough that a drug approved in one is used in the others. We collapse these categories into 27 parent indications along the lines of the treatment regimens they share, following the market definitions used in the economics of cancer treatment (Dix and Lensman, 2026) and validated by a board certified oncologist. For example, marginal zone lymphoma and Waldenström macroglobulinemia are both treated with rituximab-based regimens and sit together under non-Hodgkin lymphoma in our categorization. The parent indications also align with the site groupings in which the National Cancer Institute reports incidence, which we use for market size. Biomedtracker also carries two placeholder categories, "Other Solid Tumors" and "Hematologic Cancer", which a record takes when it names no specific cancer. Neither is a market a compound can be approved in, and Section 3.1 describes how we use trial registry records to place these entries in the cancers they test. Markets are listed in Section 3. A compound can enter more than one market, 27% of first Phase II trials are basket trials naming more than one parent indication. We treat such a trial as entry into each market it names, with additional definitions for robustness.

Second, we define market entry as the start of a compound's first Phase II trial in the market that the trial names. Figure 1 places this definition into the regulatory sequence. An additional attractive feature of the market and this definition is that we are then presented with a defined set of potential entrants, or those that have finished Phase I trials but not yet initiated a Phase II trial.

Three features make Phase II the correct definition of entry. It is the first point at which a sponsor must commit to a cancer, since a Phase II protocol tests efficacy in a defined patient population. It is the first stage that is expensive enough to require a new round of financing, representing a commitment of capital as well as of direction. Importantly, the sponsor is not bound by anything from Phase I trials aside from dosing safety. A compound that was dosed in a basket of solid tumors can begin Phase II in any of them, and compounds with the same mechanism routinely begin in different cancers. Industry participants we spoke with described the Phase II start as the point at which a company decides what its drug is for.

Figure 1: The Drug Development Process and Definitions



Note: Enrollment and duration by phase are from the FDA ("The Drug Development Process, Step 3: Clinical Research"). The set of potential entrants is the set of compounds that have completed Phase I. Entry is the start of a compound's first Phase II trial, which names the cancer it tests.

The firms making the entry decision are small. Of the 50 first-in-class cancer drugs approved in the United States between 2010 and 2020, seven originated solely at a large pharmaceutical firm and 38 were owned or licensed by one at launch (Kennedy et al., 2023). The gap is the subject of this paper. The typical originator is a company built around one compound and financed by venture capital, without the capital or the sales force to run Phase III and a launch. In 2024, 39 percent of public biotechs held less than a year of cash (EY, 2025). Licensing revenue from alliances with large firms can exceed all venture capital raised by biotechs in the same year, as it did by a wide margin in 2024 (McLaren, 2025). In our panel, firms defined as Big Pharma initiate 25% of first Phase II trials, large biotech firms initiate 6%, and small firms initiate 69%. The median small firm entrant has only a single compound in development at the time of Phase II trial initiation.

2.2 Pharmaceutical Acquisitions

Most cancer drugs are started by one firm and launched by another. The transfer happens during development, most often between Phase I and Phase III. In our panel, from 2006 through 2019, we record 1,026 transactions in which a firm took exclusive United States or worldwide rights to a development-stage oncology compound, not including so-called "mega-deals" or M&A between two established, large firms. Of these, 293 took place while the compound was in Phase I, 480 while it was in Phase II or III, and 54 after approval[1] Large firms, defined in Section 3.4, were the acquirer in 345 of them, about a third by count, and they paid substantially more per compound than other acquirers, which Section 5.1 documents.

We consider acquisitions take one of two general legal forms. The first is an exclusive license, otherwise

[1]The remaining 185 occurred in preclinical trials.

known as an "alliance" in the industry. In an exclusive license, the biotech which owns the compound grants a partner the exclusive right to develop and sell the compound in a territory in exchange for an upfront payment, payments contingent on development and sales milestones, and a royalty on sales. In this case, the biotech survives as a company and may retain rights in other territories or to other compounds. In terms of dollar value, large firms account for 60% of up-front payments across all licensing deals in our sample. The second form of acquisition we consider is a company acquisition in which the buyer purchases the biotech's equity, usually through a tender offer or a merger vote, and acquires all the firm's assets.

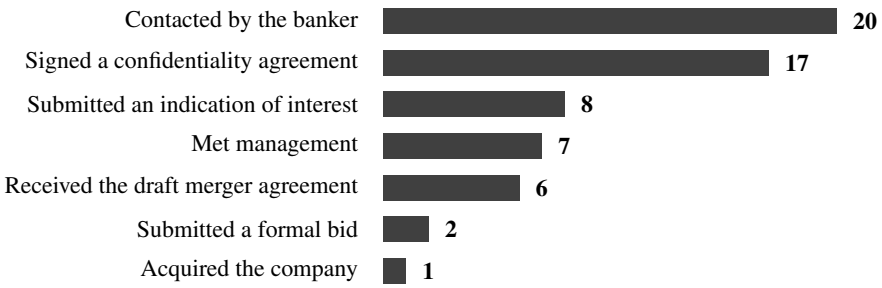
Two thirds of the transfers we consider are license deals with the remaining one third being company acquisitions. Both forms move control of the compound's United States commercialization from the biotech to the acquirer. The allocation of control rights is what the alliance literature finds matters for how the asset is subsequently developed (Higgins, 2007). We therefore treat both as an acquisition of rights and describe the construction in Section 3.2. When we need a price per compound we use licenses together with the whole-company acquisitions whose price can be tied to a single compound; Section 3.3 gives the rule.

The sale is a process the biotech runs, not an offer it waits for. A board that has decided to sell hires an investment bank, which contacts a list of potential buyers, signs confidentiality agreements with those that want to look at internal data, and opens a data room. Interested parties submit non-binding indications of interest with a price range, a subset are invited to meet management and receive a draft agreement, and a smaller subset submit final bids. The board compares the best offer against continuing alone and either signs or stops. A licensing canvass often runs alongside the sale, and boards treat it as the market check for a sale, on the reasoning that any party with an interest in acquiring the company would already have surfaced. The process is close to a sealed-bid auction with a reserve, as formalized in Section 4. Bidders do not observe one another's offers, the seller takes the highest bid that clears its own continuation value, and losing bidders are common.

Figure 2 shows one such process, detailed in SEC Filing SC 14D-9 (discussed in Section 3.7). Cougar Biotechnology was founded in 2003 around one compound, abiraterone, licensed from a British research institute. When its Phase II trial in prostate cancer read out in 2008 the board hired a banker to sell the company. About twenty buyers were contacted, seventeen signed confidentiality agreements, eight submitted indications of interest, seven met management, six received a draft merger agreement, and two submitted formal bids. The highest early bidder withdrew as they wanted to first see Phase III results. Johnson & Johnson won at 43 dollars a share, about one billion dollars. In 2009, J&J finished Phase III and launched

the drug as Zytiga in 2011.

Figure 2: Acquisition Funnel, Cougar Biotechnology



Source: Cougar Biotechnology Schedule 14D-9, June 2009, "Background of the Offer".

Both sides of the transaction have reasons to want a large-firm buyer. The biotech needs a partner that can fund Phase III, manage a regulatory filing, and field a sales force. The biotech's investors also critically need an exit. Venture investors describe choosing which biotechs to fund by asking who the likely buyers would be.[2]

A small firm could in principle carry its own compound to market, and a few have. The reason most do not is that the assets required to commercialize a drug are large and indivisible, and a firm with one product cannot use them efficiently. A United States oncology launch requires manufacturing capacity, a regulatory and pharmacovigilance function, contracting with payers, and a specialty sales force of several hundred representatives. A firm with one compound cannot spread those costs and a firm with forty can. Teece (1986) makes the general form of this argument, that an innovator captures the returns to an innovation only when it controls the complementary assets needed to commercialize it, and otherwise sells to the party that holds them. Oncology sits close to the limiting case, because the complementary assets are concentrated in the same small set of firms that Section 5.1 identifies as the buyers.

The cost of financing what remains also rises sharply at the end of Phase II. A biotech holding one unapproved compound cannot raise the hundreds of millions of dollars a Phase III oncology program requires at a price that reflects what the firm knows about the compound. A large firm funds the same trial from internal cash flow, against a portfolio in which one failure is a single draw. The gap between the two costs of capital is a gain from trade that requires no product market synergy, and it widens at the point where the remaining development bill jumps. That point is also where a Phase II readout has resolved most of what a buyer would price, which is one reason 480 of the 1,026 transfers we record occur while the compound sits

[2] "When he considers investing in a firm, he doesn't eye the public market, but rather looks at whom likely buyers would be." A managing director of a venture firm, quoted in BioWorld Today, 2006.

in Phase II or Phase III.

Large firms have economy of scale, particularly if they already have a significant presence in the same market as the compound they're acquiring. In that case, the large firm will have commercial organization including pre-existing expertise and capital in manufacturing, marketing, and distribution. Thus, the pre-existing presence in the market will result in the compound being worth more to the large firm compared to a firm that would have to invest in these fixed costs, and it faces a schedule of exclusivity losses if it has an approved product that Section 2.3 describes. Acquisition has been the pharmaceutical industry's main channel for replacing internal research since at least the 1990s (Higgins and Rodriguez, 2006), and firms whose products erode respond by buying development-stage assets in the same area (Krieger et al., 2022). Section 4 formalizes these two motives as portfolio overlap and time to exclusivity loss, and Sections 5.2 and 5.3 take them to the data.

The same transactions raise a competition concern. An incumbent that buys a development-stage competitor in its own market may shelve it, and the evidence that this happens is strongest in pharmaceuticals (Cunningham et al., 2021). The remedy under discussion targets exactly the transactions in which the acquirer already sells in the market. The 2023 United States merger guidelines and the European Commission's 2024 review both treat an acquirer's existing position in the target's market as the basis for scrutiny (U.S. Department of Justice and Federal Trade Commission, 2023; European Commission, 2024). The counterfactual in Section 7 is this policy: a firm with a significant, pre-existing presence may not acquire a development-stage drug in that cancer.

2.3 Market Exclusivity and Patents

An approved drug earns most of its revenue while it is the only version of itself on the market. There are two layers of market protection that prevent direct competition of an approved compound. Fist is patent protection. The primary patent covers the active ingredient and is filed years before approval, so its remaining life at launch is typically eight to twelve years after adjustment. Secondary patents cover formulations, methods of use, and combinations, and expire later. The Hatch-Waxman Act restores up to five years of patent life lost to clinical testing and regulatory review, capped at fourteen years of protection after approval, and the sponsor chooses which patent receives the extension, usually the primary patent (Hemphill and Sampat, 2025). The second protection is regulatory exclusivity, which is granted at regulatory approval regardless of the status of the patent protection. A new chemical entity (NCE) is granted five years of exclusivity while

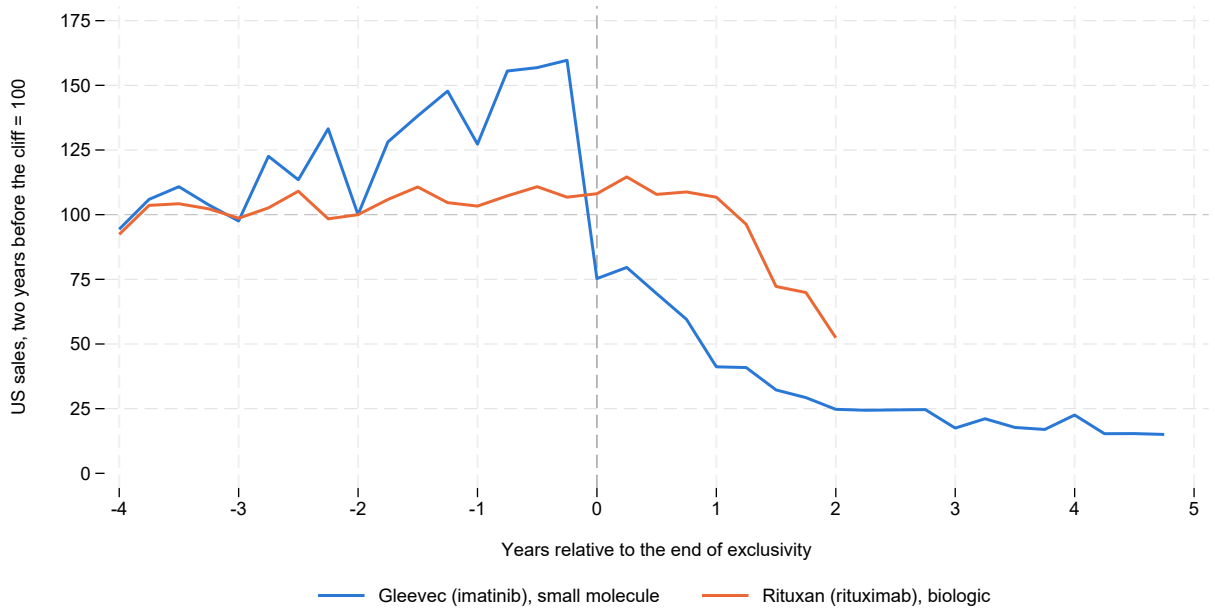
a new approval that is supported by new clinical evidence is given three years. If an NCE is approved for an orphan indication, the firm is rewarded with seven years of exclusivity. Finally, biologics are granted to longest exclusivity for an approval at twelve years. For a firm, monopoly protection for their drug is granted until the later date of either the exclusivity period or the termination of patent protection. Because both are set at or before approval, the date is public knowledge for a decade or more before it arrives to market.

The date of the last patent expiry is not always the termination date of a products patent protection. Generic manufacturers can challenge patents before they expire, even primary patents that cover the active ingredient. The challenges, however, fall disproportionately on the later-expiring secondary patents. Generic manufacturers succeed in 68% of the cases against secondary patents, paving the way for generic entry. The number of secondary patents has increased dramatically over the past twenty years due to a practice to delay generic entry, referred to as "evergreening" or "life-cycle management". Revlimid, for example, carried some thirty patents with different expiry dates. From outside the firm it is difficult to say which patent will be challenged and when, and analysts and the firms themselves work from a consensus expected date rather than the nominal one. Section 3.6 describes how we construct exclusivity end for each drug and market from the Orange Book and from analyst consensus dates, and how we treat the challenge margin. Despite this increased complexity, effective market life of pharmaceuticals has remained remarkably steady since 2000 at 13 years (Hemphill and Sampat, 2011, 2025).

When exclusivity ends, revenue falls, and does so quickly. Figure 3 shows quarterly worldwide sales for two oncology drugs around their loss of exclusivity in the United States. Gleevec exemplifies the life-cycle of a small molecule which typically loses most of its revenue within a year. For small molecules, the manufacturing process is simpler and produces chemically identical compounds. Biologics, on the other hand, are more complicated. Biosimilars require specialty knowledge to produce and do not result in therapies that are identical to their approved counterpart. Do to the higher barrier to entry and potential quality heterogeneity, both manufacturing and prescription uptake are slower for biosimilars, which can be seen by Rituxan. In both cases, a firm's loss is concentrated in one market, since an oncology drug's sales come from the cancers it is approved in.

The fact that exclusivity periods and patents are public knowledge provide predictable pipeline variation for incumbent demand. An incumbent firm with an approved, branded product in a given cancer indication has commercial organization, expended fixed costs, and expertise in that cancer. An upcoming exclusivity cliff places their revenue at risk, and the market knows the date by which they will want to seek a replace-

Figure 3: Revenue after the end of exclusivity



Notes: Quarterly United States sales from Evaluate Pharma. Each series is indexed to its own sales in the quarter two years before the end of exclusivity. End of exclusivity is designated by the vertical dashed line; 2016q1 for Gleevec and 2018q4 for Rituxan. Two years prior to the exclusivity end date, Gleevec sold $420 million a quarter and Rituxan $949 million. Two years after, they sold $104 million and $498 million, respectively. Series are deflated to 2015q1 dollars and end in 2020q4.

ment. A development-stage compound in that cancer is worth more to that firm than the same compound is to a firm with no such product, and the value rises as the date approaches (Higgins and Rodriguez, 2006).

3 Data

We construct a quarterly panel that tracks every oncology compound from its first clinical trial through acquisition, approval, and loss of exclusivity, linked in each quarter to the firm that controls its United States rights. The unit of observation is a drug by indication by quarter, covering 2,539 compounds across the 27 markets defined in Section 2 and the two placeholder categories, from 2006q1 through 2019q4. The link between ownership and development is the data contribution, giving us control right ownership at the quarter level. This enables us to count acquisitions by the acquirer's status at the deal date, price them, and measure the acquirer's portfolio in the drug's market at that moment.

We estimate using three aggregations of this panel. The market-quarter panel of Section 5.2 collapses it to the 25 markets that ever hold an approved drug within eight years of losing exclusivity, 1,400 market-quarters. Bladder cancer and myelofibrosis are the two markets this excludes. The deal-level sample of

Sections 5.1, 5.3 and 7 contains 580 acquisitions of control rights. The entry sample of Section 7 contains 1,195 compound-level entry decisions. All dollar amounts in the paper are deflated to 2015q1 dollars with the CPI-U. This section describes the sources, their uses, and the cleaning decisions with first order considerations for the analysis. Additional information is contained in Appendix A.

3.1 Drug Development, Trials and Classification

To track drugs through their development history, we use Biomedtracker, a Citeline product used extensively in prior work (Blume-Kohout and Sood, 2013; Branstetter et al., 2022; Cunningham et al., 2021). Biomedtracker assigns an identifier to every drug and indication pair and records clinical trials, regulatory events, and analyst assessments. From the drug records we take the mechanism of action and target, whether the compound is a biologic or a small molecule (excluding vaccines), and regulatory designations such as orphan status. From the commercially relevant trial and event records we build the life-cycle of each drug-indication pair. This includes the dates it enters each phase, its approval, and its discontinuation.

The life-cycle of a drug is not always cleanly defined, however, and necessitated two decisions. First, some drugs stop progressing without a publicly disclosed termination. Biomedtracker lists these drug-indication pairs as suspended, but does not assign a date. We assign these a suspension date equal to the last recorded event plus the sample median duration of the final recorded phase. Second, a drug can run several trials in the same indication at once, for instance a combination trial that begins after an approval. We collapse these parallel tracks by taking the running maximum of the development phase within the drug-indication pair, and account for the additional exclusivity a combination approval earns in the patent data rather than in the development panel.

Biomedtracker also supplies the analyst likelihood of approval, a phase-specific probability by indication. In our estimated model, we use the Phase II value, since entry is defined as the first Phase II trial. This probability enters the entrants payoff function as π^{app} and the bid index of Section 6 as a price control.

We supplement the development panel with two additional data sources. The first is ClinicalTrials.gov records, through the AACT database. This is used to place the first Phase II trials whose Biomedtracker indication is designated with the placeholder "Other Solid Tumors". When the AACT registry record names specific cancers in the trial, we assign the entry decision to those cancers. This is the case with 27 of the 57 placeholder entries. The 30 that cannot be placed are tumor-agnostic programs, which leave the entry estimation (Section 6.5). Additionally, the AACT data supplies us with a compound specific therapeutic

class from the EphMRA ATC system at level 4, with 72 classes defined by modality and target family. This forms the basis of the match-quality measure in Section 6.5.

3.2 Ownership, Licenses and Acquisitions

The control right ownership of each compound through the panel is constructed as a chain from two tables. The first is a company identity chain which includes every company that appears anywhere in the data, together with who acquired it and when, so that the controller of any asset at any date resolves by walking the chain. The second is a drug-level ownership event table, which lists the originator of each compound and every subsequent transfer of control, assembled from Biomedtracker alliance records, Evaluate Pharma company M&A, Evaluate product-level deals, and four hand-collected files.

The object of interest that we track is United States control rights. The control-rights definition we use throughout covers company acquisitions, asset purchases, and exclusive worldwide or US licenses. Examples of transactions excluded from this chain include non-exclusive co-development deals, co-commercialization deals, and reverse mergers. Additionally, for the analysis sample, we remove "mega-mergers", or whole-company M&A between large, established, diversified firms. These amount to 14 deals, identified by hand, detailed in Appendix A. These are excluded as we are unable to assign individual values to the different compounds to any one development compound, and the acquisitions are motivated by different market forces than those we consider here. Finally, licensing deals and whole-company M&A occasionally cover several assets. In these cases, we assign the deal to the compound the deal description identifies as the lead, or to the compound furthest in development. If we are unable to determine a clear lead, the deal is removed from the analysis sample. We report robustness to dropping multi-drug deals in Appendix A.

3.3 Deal Terms and Prices

We leverage both Biomedtracker and Evaluate Pharma for deal terms, serving as validation against one another. For product level alliances, we primarily use Biomedtracker records, which report upfront payments, milestone schedules and royalty provisions for licenses along with text fields containing descriptions of the deals. Evaluate Pharma provides an additional check on these alliances with the product-deal file, and supplements for business line and direct asset purchases. Finally, Evaluate Pharma provides the terms for company M&A. From these terms, we carry through two main measures of deal value: total potential

deal value and the upfront payment. The total potential deal value is the total amount committed if all development and commercialization milestones are realized, which is often not the case. The upfront value represents a more immediate and guaranteed payment. For robustness checks, we also employ a measure which captures the upfront payment plus 50% of the milestone payments in an attempt to capture the total value expectation at point of sale.

To avoid concerns about correlation of deal structure and drug quality, our primary price metric used in Section 6 is the total potential deal value, as it is the whole of what the acquirer commits to. Additionally, it is the object on which the acquisition route is comparable to self-commercialization. The milestones are not risk-adjusted at the deal level, because the data are unable to decompose the payments by trigger, and the upfront payment, the only component paid with certainty, is carried as an alternative metric throughout for robustness. Since most milestones never pay, the realistic transfer sits between the two, and Section 7.2 also reports a risk-adjusted price, the upfront payment plus the milestone schedule at a fixed discount, on the deals reporting both components.

A price is disclosed for 246 of the 410 exclusive licenses on total value and 211 on the upfront, and for most whole-company acquisitions. A whole-company price covers everything the target owns, so it enters the priced-deal analyses only when it can be tied to one compound. This is done either when target holds a single drug or when a clear lead drug is identifiable. This adds 91 whole-company deals on total value and 35 on the upfront, for a priced-deal sample of 337 and 246. One limitation of the data is that deal term disclosure is not legally compelled and subject to selection. Conditional on phase, scope, orphan status, drugs per deal and year, a large-firm acquirer is 37 percentage points more likely to report a total value (see Appendix A, Table 14). Because the missing prices are concentrated in deals between smaller firms, the large-firm premia in Section 5.1 are best read as lower bounds.

3.4 Firms and the Acquirer Pool

We define the incumbent characteristic as a state, not a fixed attribute of a firm. Firms my enter or leave the incumbent set through acquisitions and divestitures, and is thus defined at the quarter level. We define two types of incumbents. The first are Big Pharma firms, encompassing roughly twenty diversified firms meeting size and scope thresholds on market capitalization, revenue, research intensity, and geographic reach. The second set are Large Biotech firms that are smaller in size, scope, and diversification that Big Pharma firms, but still hold significant commercialized assets. Our primary measure is the superset, which we refer to as

"large firms". Appendix B lists both sets with their inclusion dates.

The acquirer pool of the structural model, K_t, is every incumbent holding an oncology asset in the quarter, between 22 and 26 firms over the sample. The pool is deliberately not indication-specific, since incumbents routinely buy into cancers where they hold nothing, and Section 6.1 measures its composition by market. Table 2 additionally reports a residual category of other large firms, defined by an oncology presence in the panel together with either $500 million of revenue and three control-rights acquisitions or $250 million and five.

3.5 Sales, Market Revenue and Indication Attribution

Quarterly drug sales combine Evaluate Pharma and Biomedtracker revenue series, at the worldwide and United States level. About 22 percent of on-patent approved drug-quarters have no sales row, all of which are products that fall below disclosure thresholds. These are set to zero and the count of covered drugs is carried alongside.

A drug approved in several cancers reports one sales figure, and the paper needs revenue by cancer. Attributing the full amount to each cancer double counts, and an equal split is arbitrary. We attribute sales with indication shares built from MarketScan claims. For molecule d, cancer c and quarter t, the share is patient-months on d in c over patient-months on d in all cancers, with boundary quarters carried outward beyond the 2009 to 2019 claims window. Where the claims data cannot see the market, shares are assigned by hand from labels and analyst sources. This is done where the claims data do not identify specific site groups, such as CML, GIST, neuroendocrine tumors, sarcoma, and thyroid cancer. Claims shares carry about 71 percent of attributed sales, hand shares about 11 percent, and single-indication drugs, which need no attribution, the remaining 18 percent. Revenue at risk, the paper's demand shifter, is built from this attributed series and defined in Section 5.2.

3.6 Exclusivity and Patents

An approved drug's protection ends at the later of patent expiry and regulatory exclusivity. The regulatory side we construct from the approval record, with three years of protection from an approval in the market, seven from an orphan approval, and a floor of five years from first approval for a new chemical entity. The patent side is harder to observe ex ante. As Section 2.3 describes, secondary patents are challenged often

and successfully, so the nominal last-to-expire date overstates protection and market participants work from an expected date instead.

We follow that practice. The expiry date is Evaluate Pharma's consensus analyst forecast of the first commercially significant patent expiry in the drug's key market, which incorporates term extensions and the thicket of secondary patents. Where Evaluate has no record, we fall back to the Orange Book active-ingredient patent with its term extension, taken from the NBER-maintained database (Durvasula et al., 2023), and a biologic with neither is assigned fifteen years from first approval, the shorter end of realized biosimilar entry in oncology. A small number of drugs whose litigation outcome or settlement date is public take that date. We convert the expiry to a patent term at the drug level and extend it with market-level regulatory exclusivity, so a combination approval earns its additional protection in the market it is approved in. We also construct a realized exclusivity end, the later of the expected date and the Orange Book's first generic approval, which marks the cliffs that the expected date understated. The expected date is the object an entrant planning years ahead weighs, and it enters the quarters-to-expiry measures behind revenue at risk and the deal-price regressions. The realized date defines whether a drug is on patent in a given quarter.

3.7 Merger Disclosures

A public US target that sells itself must explain the process to its shareholders. The Schedule 14D-9 filed for a tender offer and the DEFM14A filed for a merger vote both contain a "Background of the Merger" chronology, written by deal counsel from the banker's log. Boone and Mulherin (2007) establish these chronologies as the standard source on how corporate sales actually run, and Gorbenko and Malenko (2014) use the bidding records they contain to estimate what acquirers were willing to pay. The filings are legally compelled, contemporaneous, and name or consistently alias ("Company A") the parties that lost, which no transaction database does. Of 171 Evaluate M&A targets that could plausibly have filed, 135 have a filing on EDGAR, 95 yield a usable chronology, and 88 support the full coding, covering 2000 to 2024.

With these data, we are able to identify distributions of parties that are contacted in the event of a sale, the parties that signed a confidentiality agreement to receive information on the compound, and parties that eventually submit bids. This process is what we leverage in Section 6 to identify the process which governs which firms actually engage with a sale process. Appendix C holds the full codebook and audit protocol. Section 5.4 defines the engaged-bidder count and reports the results.

3.8 Claims Data

MarketScan Commercial Claims and Encounters and Medicare Supplemental data, 2009 to 2019, are used to construct the indication shares of Section 3.5 and the quantity outcome for the competitor-approval event study of Section 5.5. The population is the commercially insured under 65 plus retirees with employer supplemental coverage. The Medicare supplement is critical as cancer incidence is concentrated over 65. A patient is assigned to a cancer by any inpatient or outpatient claim carrying a cancer diagnosis. Drug exposure comes from outpatient J-codes and pharmacy NDCs in the antineoplastic classes, 266 molecules after crosswalking to our drug identifiers. The working panel is an enrollee by cancer by month file with 10.4 million patient-months across 21 primary-site groups.

There are two well known limitations to this approach and data. First, inpatient drug administration is under-captured, which is why the acute leukemias are excluded from claims-based shares. Second is that the contributing employer base shrinks over the window, so patient counts in levels require year effects while shares do not. Every exported table suppresses cells under 11 patients.

3.9 Samples and Descriptive Statistics

Table 1 describes the 25 markets of the market-quarter panel. Market size in the descriptive table is annual cancer incidence from the SEER program of the National Cancer Institute, whose site groupings align closely with the 27 parent indications. The estimating equations control for total market revenue and likelihood of approval. Development activity is broad. Across the 25 markets, 1,716 distinct compounds ran 3,966 drug-market development programs from 2006 through 2019, since a compound developed for several cancers counts once in each. Attrition is consequential across the development cycle, with only 20% of programs reaching Phase III and 7% reaching approval. Revenue is concentrated by market. The top seven markets carry $15.0 billion of the $19.1 billion in mean quarterly sales. Entry and deal activity track development rather than revenue. Non-small cell lung cancer, the largest development market, has 103 small-firm entries and 103 deals against mean quarterly revenue of $1.9 billion, while breast cancer, the largest revenue market, has 79 entries. Development intensity and commercial size are correlated, but far from perfectly so. For example, the acute leukemias have 70 entries and 105 deals on $0.1 billion of quarterly revenue. To capture this divergence, we employ a measure of the match-quality and crowding controls in Section 6.

Figure 4 plots small-firm entries and deals by year. Entry rises through the sample, roughly doubling

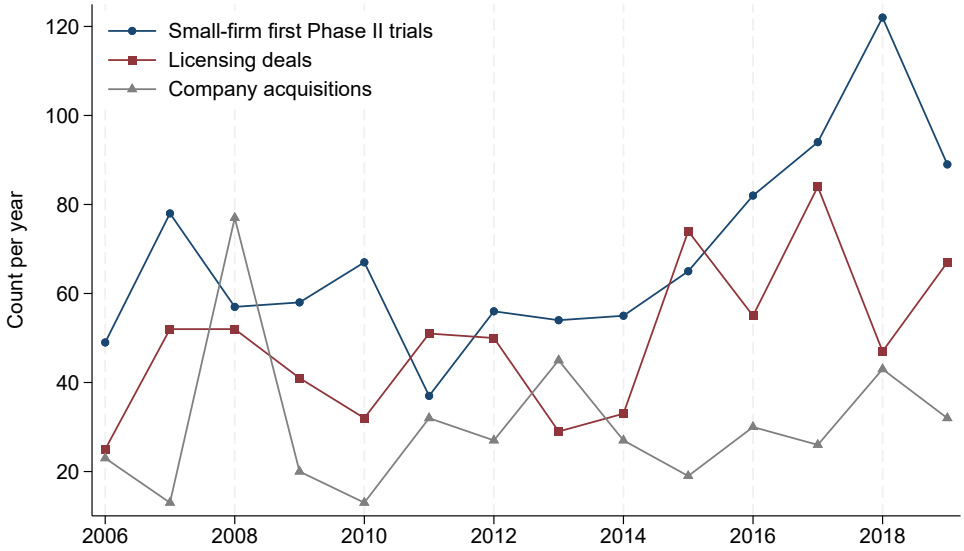
Table 1: The 25 markets: development, entry, deals and revenue, 2006–2019

Market	Drugs in development				Entries	Deals		
	All	Ph. II	Ph. III	Appr.	Small firm	Lic.	M&A	Sales (USD bn)
Breast Cancer	353	286	72	30	79	47	32	3.78
Multiple myeloma	208	131	27	11	44	32	26	2.29
Non-small cell lung cancer	410	362	94	28	103	65	38	1.89
Colorectal cancer	244	194	34	14	52	37	31	1.87
Non-Hodgkin's lymphoma	252	188	43	23	67	52	36	1.84
Chronic leukemias	151	121	36	18	34	34	19	1.73
Prostate Cancer	243	193	47	13	53	43	26	1.57
Neuroendocrine tumors	56	39	15	8	13	15	6	0.83
Melanoma	215	166	41	12	57	30	20	0.63
Renal cell cancer	125	111	27	14	33	21	10	0.59
Brain cancer	170	126	25	2	44	38	11	0.34
Female reproductive cancers	258	207	43	9	60	46	37	0.25
Sarcoma	93	79	17	6	25	19	10	0.24
Gastrointestinal stromal tumor	24	21	9	3	2	3	3	0.23
Head and neck cancer	145	123	28	6	49	18	11	0.20
Hepatobiliary cancers	147	124	35	8	31	19	12	0.17
Acute leukemias	279	180	52	20	70	63	42	0.13
Pancreatic cancer	198	160	38	3	48	33	12	0.11
Small cell lung cancer	93	80	15	4	19	15	10	0.09
Gastro-esophageal cancers	120	108	34	6	23	21	9	0.09
Hodgkin's lymphoma	47	40	5	3	17	7	7	0.07
Non-melanomatous skin cancers	32	25	7	6	7	3	7	0.05
Cutaneous lymphomas	41	37	13	8	14	18	5	0.05
Thyroid cancer	27	26	11	8	11	6	4	0.03
Mesothelioma	35	30	11	1	8	7	3	0.02
Total	3,966	3,157	779	264	963	692	427	19.11

Notes: The 25 markets of the market-quarter panel, 2006q1 to 2019q4, sorted by revenue. Drugs in development are distinct compounds in the analysis sample holding a Phase I or later record in the market during the window; the phase columns count those reaching at least Phase II, at least Phase III, and approval in that market by 2019q4. Entries are first Phase II trials by small firms, with a trial naming several markets counted in each. Lic. is control-transferring licensing deals and M&A company acquisitions of drugs with a record in the market; a deal for a drug developed in several markets appears in each. Sales are mean quarterly US revenue of approved drugs in the market, indication-weighted, USD billions, 2015q1 dollars. A drug and a deal can appear in more than one row, so the total row exceeds the count of distinct drugs and deals.

from the early 2010s to the late 2010s. Licensing deals are the more common form of acquisition throughout and move with entry.

Figure 4: Small-firm entry and acquisitions of control rights by year



Notes: Annual counts over the 25 markets of the market-quarter panel. Entries are first Phase II trials by small firms, each named market counted. Licensing deals are control-transferring exclusive licenses; company acquisitions are whole-company M&A of firms with a drug in the market. A deal for a drug developed in several markets is counted in each.

The remaining samples are as follows. The deal sample holds 580 control-rights acquisitions reachable by at least one analysis, 410 exclusive licenses and 170 whole-company acquisitions, covering 659 distinct drugs and 330 acquiring firms; 385 of the 580 enter an estimation sample. The entry sample holds 1,195 cases, 633 compound by named-market entries and 562 compounds that completed Phase I and never started Phase II, choosing over 27 indications and the outside option - discontinuation.

4 Model

In this section, we develop a model of entry in which the prospect of acquisition is priced by an auction among incumbents. For every market it could enter, a potential entrant compares two payoffs in expectation. The first is the value of developing and commercializing the compound alone. The second is the price an acquirer would pay for it, weighted by the probability that a sale occurs. The purpose of the model is to express the second payoff as a function of the set of incumbents permitted to bid, so that a policy restricting that set has a defined route to the entry decision.

Three decisions are made by the entrants. First, at $t = 0$ the entrant chooses the market in which to begin Phase II or chooses to not proceed. Second, at each later date the entrant chooses whether to initiate a sale process, which is costly. Third, conditional on a process, the incumbents that have engaged with the entrant submit sealed bids, and the entrant accepts the highest if it clears the value of continuing alone. We solve the model backward. Section 4.1 fixes the primitives. Section 4.2 takes a process as given and derives the price. Section 4.3 derives when a process runs. Section 4.4 returns to the entry decision, and Section 4.5 states the three results the counterfactual relies on.

4.1 Definitions and Setting

4.1.1 Agents, Markets, and Timing

Firms are partitioned into two mutually exclusive sets. The first set comprises potential entrants, or small biotechnology firms holding a compound that has completed Phase I trials. These firms are indexed by $i \in \mathcal{I}$.[3] The second set comprises incumbents, the large firms of Section 3.4, indexed by $k \in \mathcal{K}$. Incumbents are capable of acquiring development-stage assets and hold the means for commercialization.

Markets are defined as cancer indications, indexed by $j \in \mathcal{J} = \{1, \ldots, J\}$. The outside option is denoted $j = 0$, which is equivalent to not entering any market. Time is discrete and indexed by $t \in \{0, 1, \ldots, T\}$, where $t = 0$ designates the entry decision period, which is the commitment to begin Phase II trials in a chosen indication, and T is the expected date of regulatory decision. Let d_{it} denote the development phase of compound i at t. Firms share a common discount factor $\delta \in (0, 1)$.

Market j at date t is characterized by a vector ψ_{jt}, known to all firms. It collects the market's revenue M_{jt}, the stock of compounds in development by phase and owner, and the indication's scientific promise. Entry into market j carries a fixed cost F_j, common across potential entrants. Development from t to $t + 1$ carries flow cost c_{it}.

4.1.2 Asset Match Quality

Each compound has quality θ_{ij} that is specific to the drug-indication pair. The quality parameter is a match term that reflects how well the compound's target and mechanism of action satisfy the biology of indication j. For example, the same compound may have potential in pancreatic cancer but not be indicated in

[3] We index potential entrants at the drug level, as the overwhelming majority of these firms are single-product. Multi-product firms are treated as multiple firms, each of which holds one compound.

leukemia. The originating biotechnology firm and, upon disclosure, any counterparty assess a noisy signal of θ_{ij} from the compound's mechanism and the clinical evidence accumulated through date t, and expectations $\mathbb{E}_t[\cdot]$ are taken with respect to that information. The probability of eventual regulatory approval is $\pi_j^{\text{app}}(\theta_{ij})$, and conditional on approval, the firm that retains the asset earns V_{ij}^{own}, the discounted value of self-commercialization. Section 6.5 constructs the empirical counterpart of θ_{ij}.

4.1.3 The Acquirer Pool and Its Types

We restrict attention to acquisitions by incumbents.

Assumption 1 *Only incumbents may acquire a biotechnology compound, and the identity of the incumbent set is common knowledge.*

Incumbency is time-specific and every incumbent can bid in every market. Let

$$\mathcal{K}_t = \{k : k \text{ is an incumbent active in oncology at date } t\}, \qquad K_t = |\mathcal{K}_t|, \tag{1}$$

denote the set of potential acquirers. The pool is deliberately not restricted to firms already present in market j, since incumbents acquire compounds in indications where they hold no assets. What varies by market is the pool's composition. Let

$$\mathcal{O}_{jt} \subseteq \mathcal{K}_t \tag{2}$$

denote the incumbents holding an overlapping position in indication j, meaning a Phase II, Phase III or NDA-stage development asset, or an approved product still on patent. Let $H_{jt} = |\mathcal{O}_{jt}|$, and let $L_{jt} = K_t - H_{jt}$ be the number of incumbents without such an overlap. This distinction is central to the counterfactual policy. The firms in $\mathcal{O}_{jt}$ are the ones that could conduct a killer acquisition, and they are the firms prohibited from bidding in indication j under the policy we study.

We refer to the incumbents in $\mathcal{O}_{jt}$ as high types and to the others as low types. An incumbent's valuation of compound i is private information, and the two types draw from different distributions.

Assumption 2 *Conditional on $(\theta_{ij}, \psi_{jt}, d_{it})$, incumbent valuations are independent across incumbents. A high type draws its valuation from $F_H(\cdot \mid \psi_{jt})$ and a low type from $F_L(\cdot \mid \psi_{jt})$, where F_H first-order*

stochastically dominates F_L.

Two motives place the overlapping incumbents above the others. The first is the value of a compound to a firm with an established commercial and clinical presence in the indication, which can spread the fixed costs of a launch over an existing organization. The second is the incentive to replace revenue from a product approaching loss of exclusivity, a mechanism documented by Higgins and Rodriguez (2006). Both motives attach to the position the counterfactual restricts, and the model carries them together as the type. Section 5.2 provides evidence for the exclusivity motive and Section 5.3 for the overlap premium in prices.

4.1.4 The Sale Process

A sale process is costly, as it requires engaging an advisor, preparing diligence materials, and disclosing clinical information to potential rivals (Section 2.2). We model this as a market-invariant fixed cost $\kappa > 0$, which is incurred whether or not a transaction eventually results. The entrant decides whether to bear this cost, and when. Conditional on a process at date t, the sale proceeds in four steps.

First, incumbents engage. Evaluating a clinical-stage asset requires costly diligence, which an incumbent undertakes only when the expected return exceeds the cost. Let $\mathcal{E}_{it} \subseteq \mathcal{K}_t$ denote the set of engaged incumbents, those that open diligence on compound i at date t, with $m_{it} = |\mathcal{E}_{it}|$.

Assumption 3 *Each pool member engages independently with probability* $p \in (0, 1]$*, common across types and invariant to the pool.*

The engaged count and its composition are therefore

$$m_{it} = m_{it}^H + m_{it}^L, \qquad m_{it}^H \sim \text{Bin}(H_{jt},\, p)\,, \qquad m_{it}^L \sim \text{Bin}(L_{jt},\, p)\,. \tag{3}$$

Engagement weighs the cost of diligence against the option value of learning a valuation, and we treat the rate as a primitive. Equation (3) is the object through which the counterfactual reaches the price. A restriction removing $\mathcal{O}_{jt}$ from the set of acquirers reduces the expected number of engaged parties by pH_{jt} and, separately, removes the high-type draws from the upper tail of the value distribution.

Second, an engaged party may exit before forming a valuation. Diligence proceeds in rounds, and a party that drops out early is observed to do so by the others through the banker. We call this exit the diligence

screen, and let each engaged party pass it independently with probability $\sigma \in (0, 1]$. Let $n_{it} \leq m_{it}$ denote the number of parties past the screen.

Third, each party past the screen learns its valuation v, drawn as in Assumption 2, and decides whether to bid. The entrant compares any offer against the value of holding the asset one more period. Let W_{it} denote the value of holding compound i at date t with the option to initiate a process, and write

$$R_{it} = -c_{it} + \delta \, \mathbb{E}_t\big[W_{i,t+1}\big], \tag{4}$$

the net cost of one more period of development plus the discounted value of holding the asset into the next period. An offer must clear R_{it} to be accepted, so a party bids only when $v \geq R_{it}$. A party that draws a value below R_{it} walks away, and this exit, unlike the screen, is not observed by its rivals.

Fourth, the parties that bid submit sealed offers b_{kit}, and the entrant observes the highest, $b_{it}^{(1)}$. The entrant sells at $b_{it}^{(1)}$ if that offer exceeds R_{it}, and retains the asset otherwise. We assume that R_{it} and the number of parties past the screen, n_{it}, are common knowledge among those parties, while bids and the bid-stage exits are not.

The probability that an engaged party prices the asset is $\omega_{it} = \sigma\,(1 - F(R_{it}))$, the product of the two pass rates. The merger filings of Section 5.4 measure this product.

4.2 Bidding

Throughout this subsection we take as given that a process has been initiated at date t, that m_{it} parties engaged, and that n_{it} of them passed the screen.

Consider first the case of a single acquirer type, so that every party past the screen draws from a common distribution F. A bidder holding valuation $v \geq R_{it}$ wins when every rival either abstains or bids less, and both events are contained in $F(v)$. Expected surplus is $(v - b)\,F(v)^{n_{it}-1}$, and the symmetric equilibrium bid solves

$$b(v) = v - \frac{1}{F(v)^{n_{it}-1}} \int_{R_{it}}^{v} F(x)^{n_{it}-1}\, dx. \tag{5}$$

Shading is governed by n_{it}, the number of parties past the screen, rather than by the number that ultimately submit an offer. A bidder does not observe which rivals drew valuations below R_{it}, so it shades

against every party that remained past the screen, and a party that is alone past the screen is not told so. The screen's exits, by contrast, are observed and thin the field directly. Shading falls as n_{it} rises, so a thinner field widens the gap between valuations and the prices we observe.

With two types of potential bidders, the bid function is no longer symmetric, since high-type and low-type incumbents draw from F_H and F_L and best respond to the composition of their rivals. Let n_{it}^H and n_{it}^L denote the numbers of high and low types past the screen, with $n_{it} = n_{it}^H + n_{it}^L$, and let f_H and f_L be the densities of F_H and F_L. Each type bids by a strictly increasing function, and the equilibrium is characterized by the inverse bid functions $\phi_H(b)$ and $\phi_L(b)$, the valuation of each type that bids b. A bidder's first-order condition equates the marginal cost of a higher bid to the marginal gain from beating each rival it faces, so the pair solves

$$\begin{aligned}
\frac{1}{\phi_H(b) - b} &= \left(n_{it}^H - 1\right) \frac{f_H\left(\phi_H(b)\right) \phi_H'(b)}{F_H\left(\phi_H(b)\right)} + n_{it}^L \frac{f_L\left(\phi_L(b)\right) \phi_L'(b)}{F_L\left(\phi_L(b)\right)}, \\
\frac{1}{\phi_L(b) - b} &= n_{it}^H \frac{f_H\left(\phi_H(b)\right) \phi_H'(b)}{F_H\left(\phi_H(b)\right)} + \left(n_{it}^L - 1\right) \frac{f_L\left(\phi_L(b)\right) \phi_L'(b)}{F_L\left(\phi_L(b)\right)},
\end{aligned} \tag{6}$$

with boundary conditions $\phi_H(R_{it}) = \phi_L(R_{it}) = R_{it}$ at the reserve and a common maximum bid at the top of the two supports. Setting $F_H = F_L$ collapses both lines to the symmetric first-order condition whose solution is equation (5). The system has no closed form, but the solution exists, is unique, and is solved numerically (Lebrun, 1999; Maskin and Riley, 2000, 2003; Hubbard and Paarsch, 2014). Because each type shades against the realized composition (n_{it}^H, n_{it}^L), we assume that composition, and not only the count, is common knowledge among the parties past the screen.

Two properties of the equilibrium carry the counterfactual. First, shading falls as the number of parties past the screen rises. Second, the expected winning bid rises when a high type replaces a low type among them. In the symmetric treatment that the estimation uses, both follow from the results in Appendix D. In the asymmetric equilibrium of (6) the first holds generally, while the second is the typical case rather than a theorem, since a low type facing stronger rivals bids more aggressively than a high type with the same valuation (Maskin and Riley, 2000).

An entrant sells when at least one engaged party prices the asset at or above R_{it}, and retains it otherwise.

Conditional on a process with m_{it} engaged parties, the probability of sale is

$$\Pr\left(\text{sale} \mid \text{process}, m_{it}\right) = 1 - \left(1 - \omega_{it}\right)^{m_{it}}, \tag{7}$$

which reduces to $1 - F(R_{it})^{m_{it}}$ when there is no screen. Equation (7) also states the selection the estimation sample faces, since a transaction price is observed only for a process that produced a sale. The derivation of this correction is provided in Appendix F.

4.3 Initiating a Sale Process

The biotech chooses when to bear the cost κ, if at all. Running a process at t yields the best offer when it clears R_{it} and the asset otherwise, so the value of holding the compound satisfies

$$W_{it} = \max\Big\{ \underbrace{\mathbb{E}_t\big[\max\{b_{it}^{(1)},\, R_{it}\}\big] - \kappa}_{\text{initiate at } t},\;\; \underbrace{R_{it}}_{\text{wait}} \Big\}, \tag{8}$$

with R_{it} given by equation (4). At $t = T$ the compound has reached a regulatory decision and no option remains, so $W_{iT} = \pi_j^{\text{app}}(\theta_{ij})\, V_{ij}^{\text{own}}$.

We define the option value of running a process,

$$S_{it} \;\equiv\; \mathbb{E}_t\big[\max\{b_{it}^{(1)},\, R_{it}\}\big] - R_{it} \;=\; \mathbb{E}_t\big[(b_{it}^{(1)} - R_{it})^{+}\big]. \tag{9}$$

Because the biotech keeps the asset whenever the best offer falls short, $S_{it} \geq 0$ always. Comparing the two branches of (8) gives the condition under which a process occurs.

Proposition 4.1 *The biotech initiates a sale process at date t if and only if $S_{it} \geq \kappa$.*

Proposition 4.1 says that a process runs when the expected upside over holding the asset covers the cost of soliciting one. We use three implications of this proposition in our analysis.

First, the static auction is a special case. Setting $\kappa = 0$ makes the condition hold everywhere, since $S_{it} \geq 0$, so every asset is marketed at every date and the only source of non-sale is a low realization of $b_{it}^{(1)}$. A first-price auction with a reserve price and no initiation decision is therefore nested inside this model at $\kappa = 0$, and the initiation channel is economically meaningful only to the extent that κ is large.

Second, the acquirer pool enters the initiation decision. S_{it} is an integral over the upper tail of the winning-bid distribution,

$$S_{it} = \int_{R_{it}}^{\infty} \left(1 - G_{it}^{(1)}(x)\right) dx, \tag{10}$$

where $G_{it}^{(1)}$ is the distribution of the maximum bid. Removing high-type incumbents compresses that tail, so a restriction on the strongest bidders lowers the option value of a process and, through Proposition 4.1, the set of states in which one runs the process at all.

Third, the probability of acquisition factors into two pieces:

$$\pi_{it}^{A} = \underbrace{\Pr\left(S_{it} \geq \kappa\right)}_{\text{a process runs}} \times \underbrace{\Pr\left(\text{sale} \mid \text{process}\right)}_{\text{the best offer clears } R_{it}}. \tag{11}$$

4.4 Entry and Market Choice

At $t = 0$, an entrant holds a compound that has completed Phase I and assesses its quality from the Phase I evidence. The entrant chooses an indication in which to begin Phase II, or chooses not to proceed and exits. Entering market j costs F_j and delivers the value of holding a compound in that market. Expected profit is

$$\Pi_{ij} = W_{i0}\big(\theta_{ij},\, \psi_{j0},\, \mathcal{O}_{j0};\, \mathcal{K}_0\big) - F_j. \tag{12}$$

The entrant enters market $j^* = \arg\max_{j \in \mathcal{J}} \Pi_{ij}$ when $\Pi_{ij^*} > 0$, and selects $j = 0$ otherwise. The match θ_{ij} varies across indications for the same compound.

Solving (8) forward and substituting into (12) separates the two routes by which a compound generates value,

$$\Pi_{ij} = \mathbb{E}_0\Big[\underbrace{\pi_{ij}^{A}\, A_{ij}}_{\text{acquisition}} + \underbrace{\left(1 - \pi_{ij}^{A}\right) \pi_{j}^{\text{app}}(\theta_{ij})\, \delta^{T} V_{ij}^{\text{own}}}_{\text{self-commercialization}}\Big] - F_j, \tag{13}$$

where π_{ij}^{A} is the cumulative probability that the compound is acquired over its development horizon and A_{ij} is the expected transaction price conditional on a sale. Both terms depend on the pool's composition in j, and this is where a restriction on the acquirer pool reaches the entry decision. A policy that removes high-type incumbents from market j lowers A_{ij} through the bidding stage of Section 4.2, and lowers π_{ij}^{A}

through both factors of (11). Neither channel changes π_j^{app} or V_{ij}^{own}, so a restriction on the acquirer pool moves the acquisition payoff without moving the value of developing the compound independently.

Because a restriction binds in some markets and not in others, the model predicts reallocation as well as deterrence. A restriction that binds in market j lowers Π_{ij} without lowering it elsewhere, so compounds at the margin move toward indications where the acquirer pool is unaffected. Whether aggregate entry falls depends on how many compounds sit near $\Pi_{ij^*} = 0$ rather than near the boundary between two markets.

4.5 How a Restriction on the Pool Reaches Entry

The following are three results that are relevant to the counterfactual policy of restricting the incumbents in $\mathcal{O}_{jt}$ from bidding. Proofs are in Appendix D. They treat the parties past the screen as symmetric, each drawing from the mixture of F_H and F_L implied by the pool, which is the case that we estimate and simulate in Section 6.

Proposition 4.2 *Entry value is nondecreasing in the acquirer pool. Adding an incumbent to $\mathcal{K}_t$ weakly raises Π_{ij}, and the increase is weakly larger when the added incumbent is a high type.*

The first part is standard. The second part is what the counterfactual effect runs through, since the policy we study does not remove a randomly chosen incumbent. It removes the incumbents holding an overlapping position in the indication, which Assumption 2 identifies as the firms with the highest valuations. Removing them truncates the value distribution from above rather than thinning it uniformly.

Proposition 4.3 *Consider a restriction that removes $\mathcal{O}_{jt}$ from the pool in market j. Under Assumption 3, the expected number of engaged parties falls by pH_{jt}, and Π_{ij} weakly falls in every market where $H_{jt} > 0$ and is unchanged elsewhere.*

Proposition 4.3 holds the engagement rate at its baseline value under the restriction. If the rate instead adjusted so that the remaining low types earn zero surplus from engaging, a banned high type would be replaced by more than one low type and the engaged count would fall by less than pH_{jt}. Appendix E derives that replacement ratio. The fixed rate modeling assumption bounds the participation channel from above.

Proposition 4.4 *Let $\Pi_{ij}^{auction}$ denote entry value when $\kappa = 0$, so that every asset is marketed at every date. Then $\Pi_{ij}^{auction} \geq \Pi_{ij}$. Holding R_{it} fixed, a restriction that removes $\mathcal{O}_{jt}$ weakly lowers S_{it} at every date, so*

the set of states in which a process runs weakly shrinks. Under $\kappa = 0$ a process runs in every state, so $\Pi_{ij}^{auction}$ responds to the restriction only through the sale stage, and a counterfactual that holds the initiation factor of (11) *at its baseline understates the fall in the acquisition payoff.*

Proposition 4.4 is the basis for how the counterfactual is implemented. The measured hazard is the product in (11), and the counterfactual moves only its second factor and the price. The proposition signs the omission, so the entry effects we report are conservative on the initiation channel.

5 Descriptive and Reduced Form Evidence

In this section, we provide evidence for the assumptions underpinning the model in Section 4. These assumptions are required for the estimation in Section 6. The first, Assumption 1, is that only incumbents acquire development-stage compounds and that the identity of the incumbent set is known to potential entrants. Section 5.1 measures that set and shows that it is small. The second, Assumption 2, is that incumbents holding an overlapping position in an indication draw their valuations from a distribution that first-order stochastically dominates the one other incumbents draw from. We address this in two parts, with Section 5.2 taking up the exclusivity motive behind the ordering and Section 5.3 taking up the ordering itself. The third, Assumption 3, is that each member of the pool engages with a given sale independently and with a probability common across types. Section 5.4 measures that engagement rate from merger chronologies.

We then provide reduced form evidence in Section 5.5 which estimates the share a newly approved product takes from the incumbents already selling in a cancer. The share is the quantity from which the self-commercialization payoff V^{own} is built in Section 6.4.

5.1 Who Acquires Development-Stage Oncology Drugs

The first assumption the model makes is Assumption 1: all potential entrants know the set of incumbent firms K_t capable of acquisition. These firms must have the resources to make acquisitions and the expertise to gain from acquisitions. Table 2 summarizes the deal sample of Section 3.2 by acquirer type: 580 control-rights acquisitions between 2006 and 2019, comprised of 410 product-level acquisitions and 170 company acquisitions, covering 659 unique compounds, with some compounds acquired multiple times. There are 19 Big Pharma firms and 20 Large Biotech firms in our sample. In contrast, the remaining set of firms that make at least one acquisition in our sample is comprised of 294 distinct entities, as shown in Figure 5.

Table 2: Acquisitions of rights to development-stage oncology drugs, by acquirer type

		Large firms			Difference from others	
	All	Big Pharma	Large Biotech	Others	Big Pharma	Large Biotech
Acquisitions of control rights, 2006q1 to 2019q4						
Distinct drugs	659	220	140	361		
Distinct deals	580	149	71	362		
Exclusive licenses	410	92	33	287		
Whole-company acquisitions	170	57	38	75		
Distinct acquiring firms	330	19	20	294		
Deals entering an estimation sample	385	149	60	178		
Share of Phase II+ events	1.00	0.25	0.19	0.57		
Sample drugs per deal (mean)						
Exclusive licenses	1.2	1.3	1.3	1.2	0.1	0.1
					(0.1)	(0.1)
Whole-company acquisitions	2.6	3.2	3.7	1.7	1.6***	2.1***
					(0.5)	(0.5)
Phase at the deal (shares)						
Preclinical or IND	0.14	0.18	0.14	0.11	0.07***	0.03
					(0.03)	(0.03)
Phase I	0.32	0.35	0.31	0.30	0.05	0.01
					(0.04)	(0.04)
Phase II or III	0.49	0.40	0.46	0.56	-0.15***	-0.09*
					(0.04)	(0.05)
NDA review or approved	0.05	0.06	0.09	0.04	0.02	0.05**
					(0.02)	(0.02)
Prices, exclusive licenses (2015q1 USD m)						
Upfront payment, mean	66.7	123.0	78.8	27.9	95.1***	50.9**
					(21.4)	(19.8)
Licences with an upfront reported	211	70	30	111		
Total potential value, mean	425.3	929.8	445.8	139.1	790.6***	306.7***
					(86.6)	(52.4)
Licences with a total value reported	246	77	31	138		
Prices, whole-company acquisitions (2015q1 USD m)						
Total value, mean	1,896.0	3,404.0	1,804.5	124.1	3,279.9***	1,680.4***
					(861.4)	(350.0)
Acquisitions with a value reported	135	55	35	45		

[1] Acquisitions that transfer control rights to a development-stage oncology drug from 2006q1 to 2019q4, in the form of either a whole-company acquisition or an exclusive license or asset purchase conveying US or worldwide rights. Co-development, co-commercialization, reverse mergers and shell transactions are excluded, as are mega-mergers of two large, established, multi-product firms. A deal enters if the drug is between its first Phase I trial and NDA review, or if it is an exclusive license reporting a price.

[2] Acquirer type is attributed to the ultimate parent company in the quarter of the deal. Big Pharma firms have about $40 billion of market capitalization, $20 billion of annual revenue, research spending near 15 percent of revenue, and approvals and commercial operations across the United States, Europe, Japan and China. Large Biotech corresponds to at least $15 billion of market capitalization and $2 billion of revenue, a pipeline-driven business, independence from any Big Pharma parent, and commercial presence in more than one major market. The Large Biotech column also holds large firms outside both segments that have an oncology presence and either $500 million of revenue with at least 3 acquisitions here, or $250 million with at least 5.

[3] Prices are deflated to 2015q1 prices and reported in millions, and only where terms were disclosed, so each price row carries its own count. Prices for whole-company acquisitions represent the full acquisition of the target firm. Disclosure is more common among large-firm acquirers, by 37 percentage points (SE 5) conditional on phase, scope, orphan status, drugs per deal and year, so the price gaps below are lower bounds.

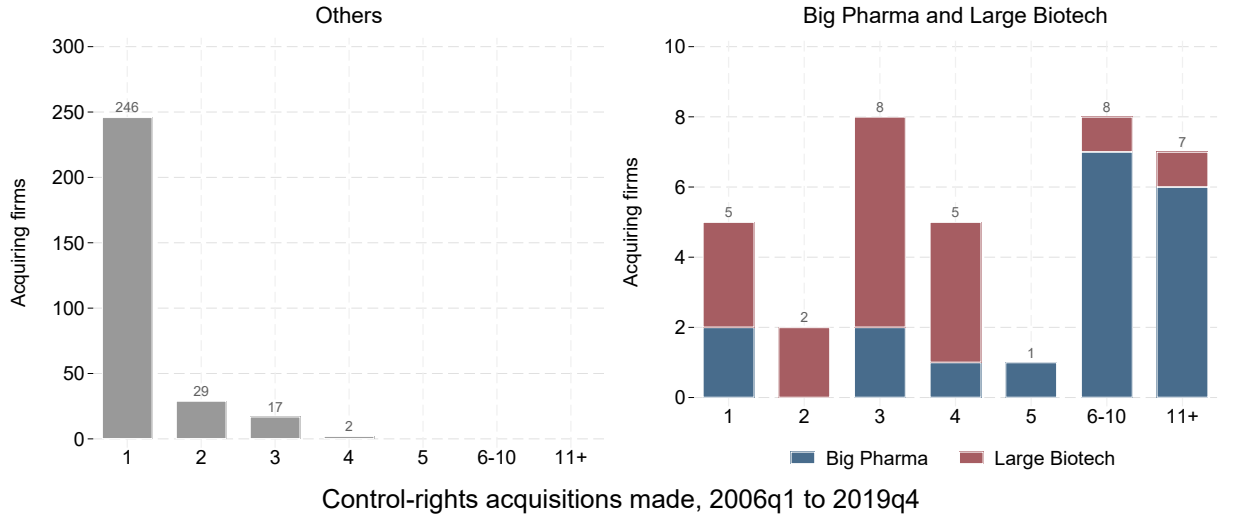
[4] The last two columns are the difference between each large-firm group and the residual group, with the standard error of the difference in parentheses. Counts of drugs, deals and firms are taken within each column, so a drug or deal involving two types of acquirer appears in more than one column.

$^{*}\ p < 0.10,\ ^{**}\ p < 0.05,\ ^{***}\ p < 0.01.$

Table 2 establishes that the set of Big Pharma and Large Biotech firms, which we refer to as *large firms*, feasibly constitute the set of potential acquirers. While the pure count of the number of deals presents a less compelling metric with large firms comprising 220 of 580 (37.9%) of all deals, the dollar value share presents a different picture. Large firms constitute 72% and 79% of the upfront value and total value, respectively, of exclusive licensing deals, 89% of company acquisition value, and 87% of deal value across both categories. Looking at mean deal values, this gap is also evident, with mean whole-company deal values $3.3 billion and $1.7 billion higher for Big Pharma and Large Biotech firms, respectively, compared to other deals.

Big Pharma firms are less likely to acquire Phase II/III assets than the residual firms, consistent with a story that the residual firms are not investing as speculation on promising technology, but due to financial pressures to continue with existing research and development. This is then driven by the high level of deals at these stages from the residual firms, given that the plurality of deals for Big Pharma firms still occurs at this stage. Given the differences in prevelance, volume, and frequency, the typical entrant will have reasonable expectations that if an acquisition were to occur, one of the large firms are by far the most likely to do so.

Figure 5: Only large firms are likely to be repeat buyers



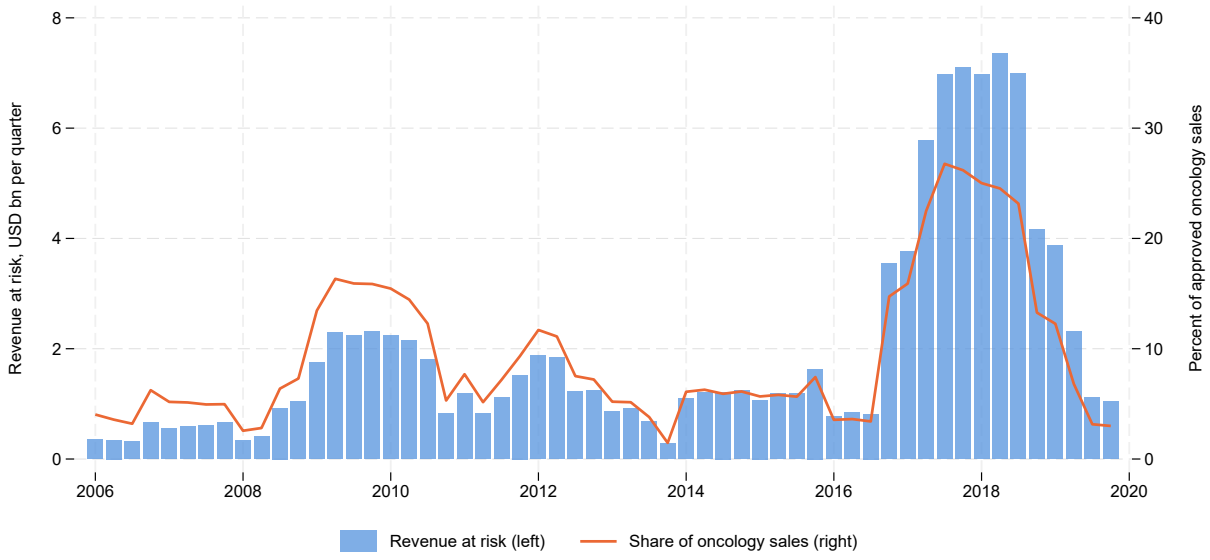
Notes: Each firm that acquired control rights to a development-stage oncology drug between 2006q1 and 2019q4 is counted once, in the bin given by how many it acquired. Deals are those in Table 2. A firm that crosses an incumbency threshold during the window is assigned the highest type it attains, so it appears in one panel only. The left panel is the 290 firms outside both large-firm segments, of which 243 acquired once and none more than four times. The right panel is the 36 Big Pharma and Large Biotech firms. The vertical scales differ between the panels, so heights are not comparable across them. *Source: Biomedtracker and Evaluate Pharma.*

5.2 Revenue at Risk, Acquisitions, and Entry

We next provide evidence suggestive of the exclusivity motive behind Assumption 2. We show that incumbents value a compound in an given indication more when there is revenue at risk in that indication due to an exclusivity cliff. This effect is pooled across all incumbents in the indication, indicative of market-level incentives. Expiry opens the therapeutic space that the expiring product defended to whichever incumbent refreshes it (Caves et al., 1991; Magazzini et al., 2004; Ellison and Ellison, 2011).

Section 3.6 defines revenue at risk (RAR) in market j and quarter t, RAR^8_{jt}, as the sales of approved drugs in j whose exclusivity ends within eight quarters, in billions of 2015q1 dollars per quarter[4]. Figure 6 shows how RAR moves in aggregate and Figure 7 shows where RAR falls. The revenue cliffs provide rich variation across out sample. The cliffs are large, lumpy, and staggered across cancers. RAR averages $1.95 billion a quarter across all markets and reaches $7.4 billion at the 2017 peak. On the other end of the spectrum, there are three quarters which have zero RAR across all cancers and there are four markets that never carry any revenue risk at all.

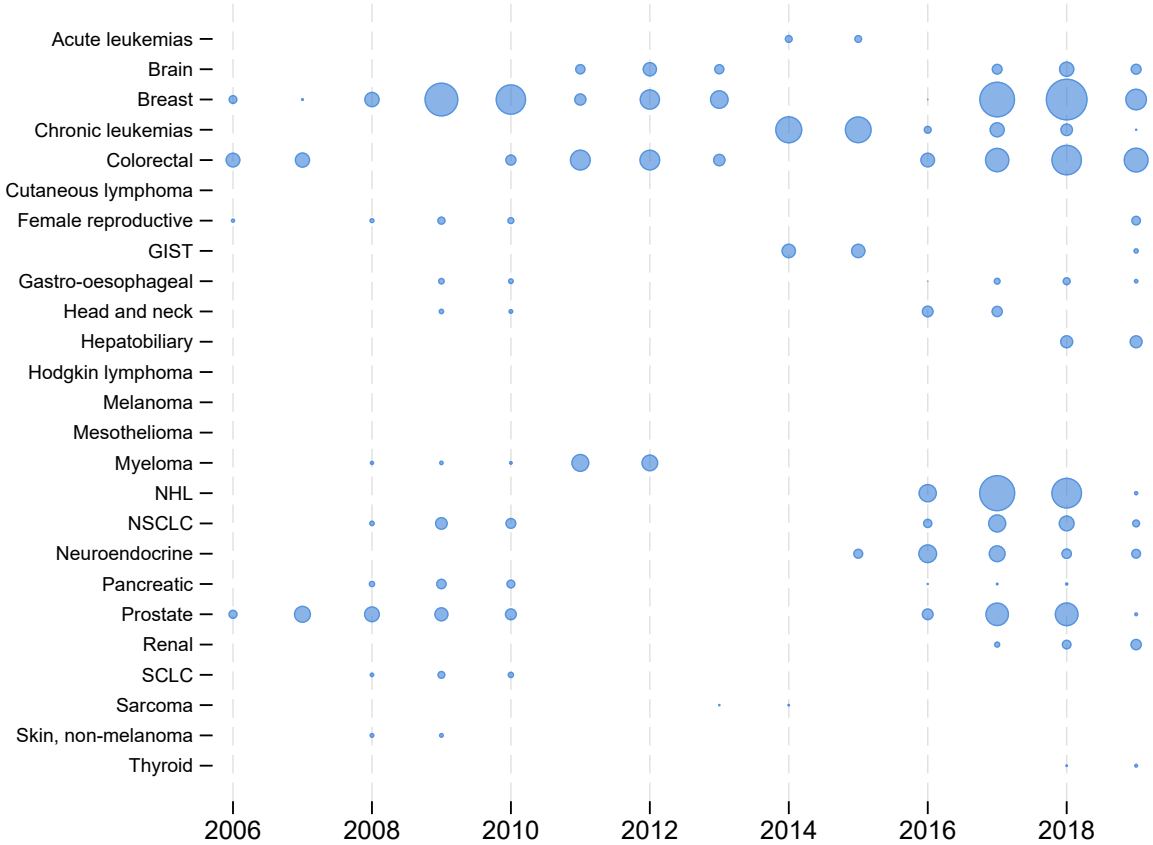
Figure 6: Revenue at risk across all 25 markets, by quarter



Notes: Revenue at risk in a market is the sales of approved drugs in that market whose exclusivity ends within eight quarters, with a drug approved in several cancers having its sales split across them as described in Section 3.5. Bars sum revenue at risk over the 25 markets in each quarter, on the left axis. The line is that total as a percentage of the sales of all approved drugs in the 25 markets, on the right axis.

[4]Appendix G Table 23 reports robustness checks of the analysis using RAR measures with different horizons.

Figure 7: Revenue at risk by cancer and year



Notes: The same quantity as Figure 6, plotted by cancer and year with circle area proportional to revenue at risk. Cancers are listed alphabetically. Blank space is a cancer-year with no approved drug within eight quarters of losing exclusivity, which is most of the panel: 76.5 percent of the 1,400 market-quarters carry no revenue at risk, 21 of the 25 cancers ever carry any, and 104 of the 350 cancer-years do. Conditional on being positive the mean is $0.33 billion a quarter against an unconditional mean of $0.08 billion..

We estimate

$$\mathbb{E}[N_{jt} \mid \cdot] \;=\; \exp\big(\beta\, RAR^{8}_{jt} + Z'_{jt}\gamma + \alpha_j + \delta_t\big)\,, \tag{14}$$

where N_{jt} is either N^{acq}_{jt}, the number of large firm acquisitions in market j and quarter t, or N^{ent}_{jt}, the number of entries into market j in quarter t. The controls Z_{jt} are market revenue M_{jt} and the market's Phase II likelihood of approval π^{app}_j, α_j and δ_t are market and year effects, and standard errors are clustered by market. Because the mean is in log terms, β is a proportional change in the count per billion dollars at risk rather than a number of events.[5]

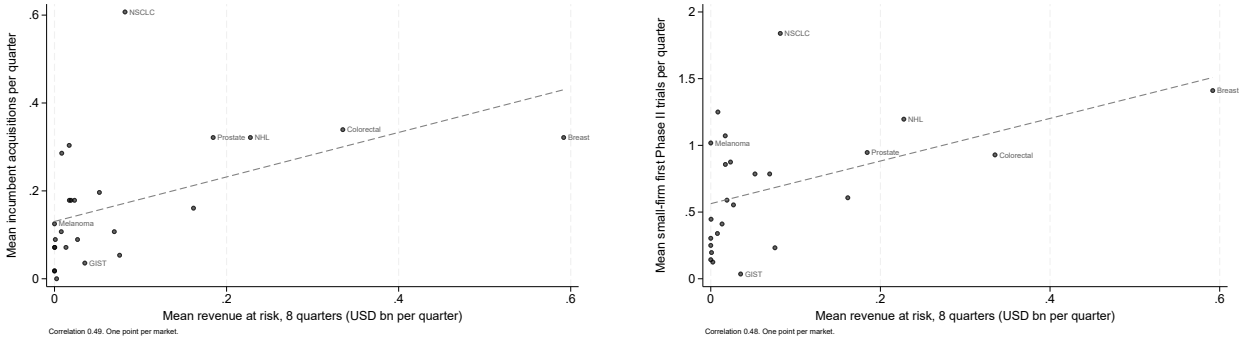
Figure 8 visualizes the cross-sectional evidence that is estimated in Table 3, Column (1). Without the includsion of market fixed effects, however, this specification runs the risk of simply measuring market density. That is, markets that are more profitable will persistently have more entry and revenue at risk will drive the relationship. If entry is only responding to market profitability, we would be unable to separately identify the entry incentives of acquisition and market profitability. To address this, Column (2) includes market fixed effects to isolate within-market variation.

Table 3, Column (2) reports regression output with acquisition as the outcome of interest. With market and year effects, \$1 billion in revenue at risk increases the incumbent acquisition rate by 0.394 log points, or by 48% over baseline. Column (3) includes a linear trend term to rule out the effect being driven by secular trend of increased entry or acquisition. Inclusion of this trend term lowers the point estimate and precision slightly, but does not meaningfully alter the interpretation. In levels, the marginal effect of a \$1 billion increase in revenue at risk is an increase of 0.067 acquisitions per market-quarter against a mean count of 0.17. This measure is difficult to interpret, and most compounds never face \$1 billion at risk. A more meaningful measure is that a market moving from no revenue at risk to the average positive amount (\$332 million/quarter) sees its acquisition rate rise by 14%.

Table 4 reports analogous regression output with entry as the outcome of interest. Entry by non-incumbents increases by 0.232 log points per \$1 billion at risk, which only counts trials dedicated to at most three cancers, and by 0.179 when we count every named cancer in a basket trial. Importantly, incumbents' own entry into Phase II trials does not move, with a point estimate of -0.147 statistically indistinguishable from zero. This heterogeneity is informative of several narratives consistent with our model. The first is that

[5] We estimate by Poisson rather than by ordinary least squares in levels for two reasons. First, acquisitions average 0.17 per market-quarter and their base rate varies by an order of magnitude across cancers, so the response has to scale with the base rate. Second, the counterfactual in Section 7 lowers RAR, where a linear model predicts negative counts. Least squares already puts 199 of the 1,400 fitted values below zero for acquisitions.

Figure 8: Markets with more revenue at risk have more acquisitions and more entry



(a) Acquisitions (b) Entrants

Notes: One point per market, 25 markets, averaging over 2006q1 to 2019q4. Revenue at risk is the sales of approved drugs in the market whose exclusivity ends within eight quarters, in USD billions per quarter, with a multi-market drug's sales split across its markets as described in Section 3.6. The dashed line is an unweighted least-squares fit. This is the cross-sectional variation that the market fixed effects in the regressions below absorb; it motivates the design and does not identify it.

Table 3: Revenue at risk and incumbent acquisitions of development-stage drugs

	(1)	(2)	(3)
Revenue at risk, 8 quarters (USD bn)	0.272	0.394***	0.303**
	(0.218)	(0.136)	(0.127)
Controls			
Market sales (USD bn)	0.293***	-0.073	0.091
	(0.101)	(0.182)	(0.235)
Likelihood of approval	-0.371**	0.011	-0.174
	(0.173)	(0.176)	(0.225)
Mean of dependent variable	0.170	0.170	0.170
Observations	1,400	1,400	1,400
Year FE	Yes	Yes	Yes
Market FE	No	Yes	Yes
Market linear trend	No	No	Yes

[1] Parent indication by quarter, 2006q1 to 2019q4, 25 markets. Poisson regressions; coefficients are the proportional change in the rate per USD billion at risk. Standard errors clustered by market in parentheses. Controls are market sales (the sales of approved drugs in the market, USD billions per quarter) and the likelihood of approval. Column (3) adds a market-specific linear time trend.

[2] The dependent variable counts acquisitions by a Big Pharma or Large Biotech incumbent of drugs in Phase II, Phase III or NDA review in the market that quarter: company acquisitions, asset purchases and exclusive worldwide or US licenses, including co-development agreements that carry exclusive rights; one event per acquirer, drug and quarter. The definition is the control-rights definition of the acquirer-types table: non-exclusive co-development and co-commercialization agreements transfer no control and are not counted, and neither are reverse takeovers, in which a private company merges into a listed shell to obtain the listing. Whole-company mergers between diversified firms are also excluded (14 deals, listed in the appendix).

[3] Revenue at risk is the sales of approved drugs in the market whose exclusivity ends within 8 quarters, in USD billions per quarter. A drug approved in several markets has its sales split across them by MarketScan patient-month shares, or by label and analyst shares where the claims data do not identify the market (appendix).

$^{*}\ p < 0.10,\ ^{**}\ p < 0.05,\ ^{***}\ p < 0.01.$

incumbent firms are not driven by acquisition incentives in the same way that entrants are motivated. We expect this is because they are not resource-constrained and do not require additional support to profitably commercialize their products. Second, the compounds that appear where a cliff is coming are the ones held by incumbent firms that would have to sell to realize their value.

Table 4: Revenue at risk and Entry

	Small-firm first Phase II trials		Incumbent first Phase II trials
	(1) Dedicated	(2) All	(3) Incumbent
Revenue at risk, 8 quarters (USD bn)	0.232***	0.179**	-0.147
	(0.059)	(0.074)	(0.142)
Controls			
Market sales (USD bn)	-0.043	-0.045	0.175
	(0.080)	(0.116)	(0.130)
Likelihood of approval	-0.116	-0.116	0.354**
	(0.101)	(0.104)	(0.145)
Mean of dependent variable	0.541	0.688	0.296
Observations	1,400	1,400	1,400
Year FE	Yes	Yes	Yes
Market FE	Yes	Yes	Yes

[1] Parent indication by quarter, 2006q1 to 2019q4, 25 markets. Poisson regressions; coefficients are the proportional change in the rate per USD billion at risk. Standard errors clustered by market in parentheses. Controls are market sales (USD bn) and the likelihood of approval.
[2] Entry is a drug's first Phase II trial in any indication, assigned to the market(s) of that trial. Column (1) excludes trials that name four or more markets at once; column (2) counts each named market in full. Small firms are neither Big Pharma nor Large Biotech.
[3] Revenue at risk as in the acquisitions table, 8-quarter horizon.
$^{*}\ p < 0.10,\ ^{**}\ p < 0.05,\ ^{***}\ p < 0.01.$

There are several limitations of this reduced-form design. First, we are unable to separate the acquisition channel from other mechanisms associated with a cliff. For instance, a cliff arrives fifteen to twenty years after a therapeutic class's first approval, which is also when the follow-on science in that class matures, and generic availability lowers the cost of developing a combination therapy around it. Market and year effects absorb the level and the common trend, but not the within-market correlation. In our structural model, Section 6 separates these channels. Second, this regression approach cannot estimate aggregate entry effects. Because we condition on year effects, we are measuring where entry goes across cancers within a year and not how much entry there is in total. Our structural model is able to recover the aggregate margin, since it places the outside option of developing nothing, which a count regression never observes, in the entrant's choice set. This is shown in Section 7.3.

We provide further robustness checks in Appendix G, all of which are consistent. First, we show that the response is concentrated in the two years before a cliff rather than after it. Revenue that has already

lost exclusivity does not move acquisitions. If anything, there is suggestive evidence of a "kill-zone" effect (Kamepalli et al., 2020), where small firms are less likely to enter after a cliff (and presumably an acquisition) occur, since they now have decreased expectations of acquisition. We also provide consistent results using alternative estimators (ordinary least squares and zero-inflated negative binomials).

5.3 Acquirer Portfolios and Deal Prices

Assumption 2 argues that incumbents with a drug in development for a given cancer indication value compounds in that market more than other incumbents. This assumption separates high type and low type incumbents.

To test this, we study the association between portfolio overlap and acquisition price (i.e., winning bid). Index a deal by d, its drug by i, its acquirer by a and its lead market by j, in the quarter t of the deal, and let P_d be the price in 2015q1 dollars. Let o_{ajt} be the number of drugs the acquirer already holds in the drug's market from Phase II through approval, counting an approved drug only while it remains on patent, and measured the quarter before the deal takes place. Because a drug can be developed for several indications at once, we aggregate o_{ajt} in 2 ways: the sum over the drug's markets, and the largest position the incumbent holds in any one market. We estimate:

$$P_d \;=\; \beta' o_{ajt} + \theta\,\text{LARGE}_{at} + Z_d'\gamma + \alpha_{j(d)} + \delta_t, \tag{15}$$

where LARGE_{at} is an indicator variable for whether the acquirer is a large firm, α_j and δ_t are lead-market and year fixed effects, and $Z_d$is a vector of deal controls. Included in the vector are the phase of the compound at the time of deal, if the deal is for US or worldwide rights, an indicator variable for the presence of a royalty to the licensee, an indicator for orphan designation, the number of sample drugs in the deal, the acquirers oncology sales, total sales in the market, the market total count of Phase II to NDA pipeline, and the number of markets the drug is currently in.

In these regressions, we use the sample of deals with observable prices as discussed in Section 3.3. Omitted deals (i.e., those that do not report prices) are likely between smaller non-incumbent firms, so the premia below may be interpreted as lower bounds.

Table 5 reports both prices. On total potential value, each overlapping drug that a high-type incumbent has adds \$74.1 million when the holdings are summed over the drug's markets and \$79.5 million when

Table 5: The acquirer's portfolio in the drug's market and the deal price

	Upfront payment		Total potential value	
	(1) Summed Across Markets	(2) Max Across Markets	(3) Summed Across Markets	(4) Max Across Markets
Portfolio overlap, count	11.057*	7.704	74.080**	79.501**
	(5.958)	(7.720)	(31.046)	(35.673)
Whole-company deal	186.516***	187.814***	122.340	131.770
	(45.586)	(47.188)	(132.191)	(136.007)
Acquirer is a large firm	96.695***	115.317***	497.859***	546.194***
	(31.604)	(29.418)	(127.739)	(117.100)
Market sales (USD bn/qtr)	5.580	7.229	-3.286	8.908
	(6.739)	(8.233)	(18.677)	(26.405)
Mean of dependent variable	95.8	95.8	486.4	486.4
Observations	270	270	366	366
R^2	0.50	0.49	0.40	0.38
Year FE	Yes	Yes	Yes	Yes
Lead-market FE	Yes	Yes	Yes	Yes

Dependent variable: the deal price in 2015q1 USD m, the upfront payment in columns 1 and 2 and the total potential value including milestones in columns 3 and 4. One observation per acquirer parent, drug and deal quarter. The sample is the exclusive licenses with a reported price, plus the whole-company acquisitions whose price can be tied to one compound: a single sample drug, or two to three sample drugs with a unique most-advanced drug still in development and a company value under USD 5bn; the whole-company indicator absorbs the deal-form level. The regressor throughout is the number of drugs the acquirer already holds in the drug's market from Phase II through approval, counting an approved drug only while it is still on patent, and measured the quarter before the deal. An expired product is not a franchise the acquirer can defend, which is the position the counterfactual restricts. A drug can be developed for several parent indications, so the acquirer's holdings have to be aggregated across them. The first column of each pair sums the holdings over the drug's markets; the second takes the largest position the acquirer holds in any one of them. Counting approved drugs whether or not they remain on patent gives 10.3 and 67.3 in place of the first and third columns. On the 223 deals for a drug in a single indication (164 on the upfront), where the aggregation choice does not arise, the coefficient is not distinguishable from zero on either price. Estimated with no holdings measure at all the large-firm indicator is 132.6 with a standard error of 26.0 on the upfront and 713.4 with a standard error of 101.2 on total value. Controls: phase at the deal, worldwide rights, royalty in the deal, orphan designation, sample drugs in the deal, acquirer oncology sales (USD bn), market sales (USD bn), Phase II-NDA drugs in the market, number of markets, year and lead-market fixed effects. Standard errors clustered by acquirer parent.
$^{*}\ p < 0.10,\ ^{**}\ p < 0.05,\ ^{***}\ p < 0.01.$

the largest position in any one market is used instead. On the upfront payment, the summed count gives $11.1 million and the largest-position count $7.7 million. The whole-company indicator is $122 million on total value and $187 million on the upfront; the upfront premium is mechanical, since a company price has no milestone schedule to strip out, and the total-value premium is modest once the drug and market are controlled.

A high-type incumbent has a sales force, formulary access and prescriber relationships there, and it has a franchise with an expiry date, which Section 5.2 shows is when incumbents buy. Counting only its on-patent approved drugs and nothing in development gives $53.4 million on the upfront, a standard error of 27.1, and $165.7 million on total value, a standard error of 94.1: a larger premium per drug than the broader count, on far fewer deals. Whether approved products are counted only while on patent makes little difference to the broader measure, $11.1 million against $10.3 million on the upfront, because an acquirer with an approved drug in a cancer usually has one still on patent.

5.4 Engaged Bidders in Merger Disclosures

This subsection provides the evidence for Assumption 3, that each member of the acquirer pool engages with a given sale independently and with a probability common across types, so that the parties bidding in an auction are a subset of the pool rather than the pool itself.

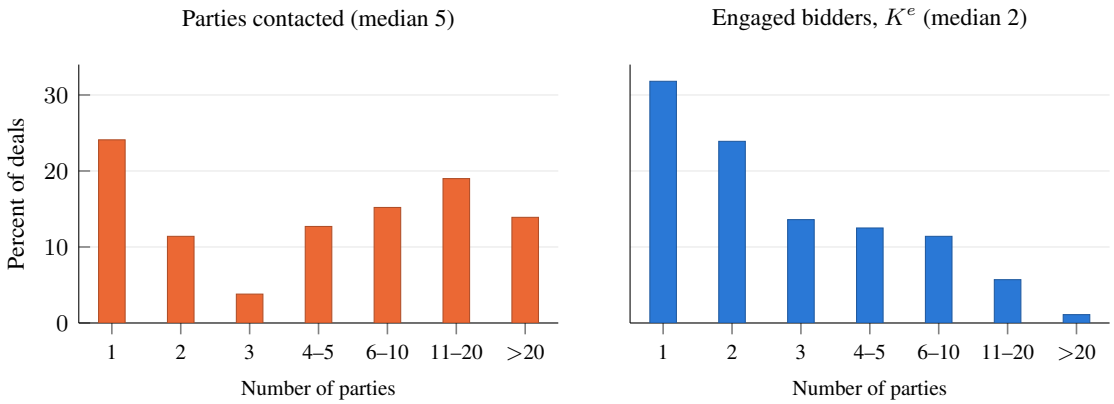
Throughout the model, the sale price that an entrant expects for its own product is a function of the number of incumbents that bid in a first-price auction and the distribution of incumbent private valuations. Indeed, we are explicitly modeling the price formation process with these dynamics. While we have thus far defined the set of large firms, allowing each to be a bidder would result in auctions in which firms bid their private value, and any counterfactual would be hard pressed to increase the bid shading in an economically meaningful capacity. Thus, the model defines a process in Section 4.1.4 in which the full set of potential acquirers, defined as large incumbent firms, is transformed to a set of potential bidders (i.e., engaged) drawing from their valuation and submitting bids. We micro-found that step with data rather than leaving it to assumption.

The merger chronologies of Section 3.7 record every party the seller's banker approached and every party that went on to receive confidential information or to price the asset. Because the losing parties are tracked through the process, we are able to measure participation after engagement and identify the firms that undertook costly diligence but ultimately chose not to bid. We define the engaged bidders K^e in a

sale as the parties that received confidential information under a confidentiality agreement executed for the transaction, or that made a proposal carrying a price, with the acquirer always counted. The number of parties the banker approached bounds the set from below.

Figure 9 and Table 6 report the distribution of the sales processes. The number of banker-contacted parties is large, with a median of five parties and a mean of 11.9 contacted. The funnel below this to engaged bidders is thinner. The median sale has two engaged bidders with a mean of 3.8, 75th percentile of five, a 90th percentile of eight, and a maximum of 22. The modal value is one, in 28 of the 88 sales, and 82 percent have five or fewer. From this, a median of two parties bid at all and a median of one submits a final bid[6].

Figure 9: How many parties engage in a sale



Notes: 88 sales of US-listed oncology targets, 2000 to 2024, coded from the "Background of the Merger" chronology in the target's Schedule 14D-9 or DEFM14A. Left: parties the seller's banker contacted, where the filing states it (79 deals, median 5). Right: engaged bidders, parties that received confidential information under a confidentiality agreement for the transaction or made a price-bearing proposal (median 2, modal value 1, 82 percent at five or fewer).

These filings are reported only by public companies undergoing a whole-company merger or acquisition. The auction sample of Section 6 contains deals of exactly this form, the 91 whole-company acquisitions whose price can be tied to one compound, so the funnel evidence bears directly on part of the estimation sample. The larger part of that sample remains exclusive licenses, and licensing deals are less costly on average for the acquiring firm, so they may have a lower threshold for engagement and more bidders per process. Two measurements inside the same filings address this possibility. First, twenty-two chronologies in the data describe a partnering canvass running alongside the sale, with a median of 12 parties contacted, 5 informed and 2 pricing the asset. Second, eighteen processes describe a parallel exploration of licensing

[6]Fifteen of the 88 counts are floors, where the filing describes informed parties without enumerating them; we flag these rather than dropping them, which pushes the measured count down.

Table 6: Parties at each stage of the sale and of the parallel licensing canvass

	N	Mean	p25	Median	p75	p90	Max
Sale of the company							
Parties contacted by the banker	79	11.9	2	5	14	30	100
Engaged bidders, K^e	88	3.8	1	2	5	8	22
Parallel licensing canvass, same filings							
Parties contacted	17	18.9		12			100
Informed (confidentiality agreement and information)	19	7.5		5			33
Priced (submitted a term sheet)	18	2.6		2			6

[1] Sales of US-listed oncology targets in the Evaluate M&A sample, 2000 to 2024, with a usable "Background of the Merger" chronology in the Schedule 14D-9 or DEFM14A (88 of 171 candidate targets). An engaged bidder received confidential information under a confidentiality agreement executed for the transaction or made a price-bearing proposal; the acquirer always counts. Fifteen of the 88 are lower bounds where the filing describes informed parties without counting them. Parties contacted is reported where the filing states it.
[2] The licensing rows code the partnering canvass that ran alongside the sale in the same chronologies, where the filing states the stage counts (22 of 95 sections). Six canvasses solicited a sale and a license together; excluding them, the informed maximum is 13.

or asset purchase with a median of 62 contacted, 8 informed, and 2 priced. From these data, we see that licensing draws a wider field at the top of the funnel but the same field at the bottom, and no single-asset process in either measurement has more than eight parties price the asset. Boards themselves treat the partnering canvass as the market check for a sale, which is the micro-foundation for using partnering counts to bound the set.

Section 6.1 keeps the set and participation as separate objects and calibrates the engagement rate to these counts. About a third of informed parties go on to price the asset, 47 of 134 on the sale side, which is the total walk-away the model takes as given (Section 6.2).

5.5 Competition Among Approved Drugs

Unlike the subsections above, this one does not provide evidence for an assumption of the model. It estimates an input the model takes from the data, the share a newly approved product takes from the incumbents already selling in a cancer, which Section 6.4 uses to build V^{own}.

Beyond expected value of acquisition, entrants must also calculate their expected value of self-commercialization. That value is the market's revenue times the share that a newly approved product takes from existing drugs. We measure the share in reduced-form rather than from a structurally-estimated demand system.

The choice to estimate expected revenue from self-commercialization using a reduced-form approach is informed by two observations. First, the relevant object that is needed in the structural model is entrant revenue expectations over market entry, not contemporaneous realized entry. In speaking with market

participants, the reduced-form approach we describe is how entrants describe their own internal estimates. Second, the oncology market has idiosyncrasies which result in difficult-to-interpret parameters from traditional demand estimation. In the oncology market, physicians mediate patient demand, with price-inelastic preferences. Additionally, oncology treatments are structured as lines of therapy, in which there are formal regimens defined for a given indication that physicians follow. Newly-approved drugs are almost always approved to be the final treatment in the line of therapy, and drugs added to the end of the line may be complements to prior treatment rather than direct substitutes, which is difficult to reconcile in traditional models of demand. Finally, prices are generally determined as a percent over the Medicare Average Sales Price (ASP), meaning they do not directly negotiate drug prices, and the decision-relevant price, net of rebates, is unobserved.

Empirically, price-mediated substitution appears to be absent. In distributed-lag regressions of log price on the number of competitor approvals, the average sales price of infused drugs moves by -0.007 over the twelve quarters after an approval, statistically indistinguishable from zero, and oral list prices keep rising throughout, about one point a year more slowly than they otherwise would (Appendix I). No incumbent cuts price after a competitor product is approved and enters the market.

There is, however, a modest but measurable quantity displacement from entry. Figure 10 and Table 7 report a stacked event study in the construction of Cengiz et al. (2019), one stack per competitor approval in a cancer over a window of eight quarters before to twelve after, so that each event carries its own clean controls and its own clock (Goodman-Bacon, 2021; Sun and Abraham, 2021). Treated incumbents are drugs approved and selling in the cancer at least eight quarters before the event and not sold by the entrant's seller, and controls are incumbents in other cancers over the same quarters. We measure outcomes in logs because the parameter that the entry model needs is the proportional effect: the entrant's share s of Section 6.4 is the incumbents' percentage loss, and the post-by-treated coefficient on a log outcome measures that loss directly. For drug d in the stack built around event e, observed in quarter t, we estimate:

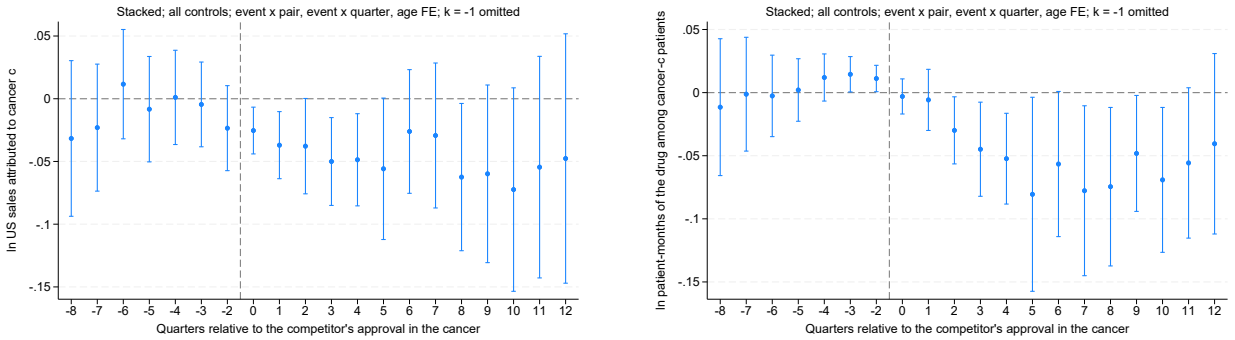
$$y_{det} \;=\; \beta\left(\text{Post}_{et} \times \text{Treated}_{de}\right) + \alpha_{de} + \lambda_{et} + \gamma_{a(d,t)} + \varepsilon_{det}, \tag{16}$$

where y_{det} is the log of attributed sales, of patient-months, or of the drug's share of its cancer's treated patients; Post_{et} marks quarters at or after the event; Treated_{de} marks the incumbents in the event's cancer; α_{de} are event-by-drug effects, λ_{et} event-by-quarter effects, and $\gamma_{a(d,t)}$ age-in-market effects that absorb the

launch curve. Standard errors are clustered by drug.

The event-time paths in Figure 10 replace the single post term with indicators for each quarter relative to the event, -8 to 12 with -1 omitted. Important for identification, there is no statistically significant trend in the quarters before the event. With the full control group, patient-months in the cancer fall by 0.048 log points, the incumbent's share of treated patients falls by 0.031, the total number of treated patients in the cancer does not move at all, and there is no statistically significant decrease in sales. The same regressions in levels agree in sign and implied magnitude, with attributed patient-months falling significantly and the dollar and point-share estimates noisier, because levels weight the largest incumbents (Appendix H).

Figure 10: Incumbent revenue and volume in a cancer around a competitor's approval



(a) Log US sales allocated to the cancer

(b) Log patient-months in the cancer

Notes: Stacked event study, one stack per competitor approval in a cancer, window -8 to $+12$ quarters, 118 events. Treated: incumbents approved and selling in the cancer at least eight quarters before the event and not sold by the entrant's seller. Controls: incumbents in all other cancers over the same quarters. Event by pair, event by quarter, and age-in-cancer fixed effects; standard errors clustered by drug. Sales in the cancer are US sales times the drug's MarketScan share of patient-months in that cancer; patient-months are MarketScan counts. Later approvals in the same cancer are not netted out, so the post-period average is the effect of about two net approvals.

Table 7: Incumbent outcomes after a competitor's approval in the cancer

	ln sales in c		ln patient-months in c		ln share of c		ln total treated in c	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Post x treated	-0.118	-0.035	-0.122**	-0.048**	-0.106*	-0.031	0.012	-0.003
	(0.088)	(0.026)	(0.060)	(0.022)	(0.057)	(0.020)	(0.049)	(0.021)
Control cancers	clean	all	clean	all	clean	all	clean	all
Observations	31,394	151,243	31,927	154,556	31,927	154,556	12,043	38,137
Clusters	81	84	80	83	80	83	18	18

One stack per competitor approval in a cancer (event date: approval), window -8 to +12 quarters. Treated: incumbents approved and selling in the cancer at least 8 quarters before the event, not sold by the entrant's seller. Controls: incumbents in other cancers over the same quarters; clean = no approval in the control's cancer within 4 quarters of the event. Event x pair, event x quarter, and age fixed effects; SEs clustered by drug (columns 1-6) or cancer (7-8; 20 clusters). Sales in c = US sales x Marketscan share of the drug's patient-months in cancer c. * p<0.10, ** p<0.05, *** p<0.01.

Taken together, a competitor's approval reduces an incumbent's use in a cancer by two to three percent, with the reduction coming from the incumbent's share rather than from the number of patients treated. Given that revenue falls by less than volume, and this change in revenue is indistinguishable from zero, a competitor count is an adequate summary of the competitive effects of an additional entrant.

6 Estimating the Structural Model

This section takes the model of Section 4 to the data, naming the empirical counterpart of each model object and the choices made to measure it. Section 7 then defines the counterfactual, estimates the two stages and reports what they find. Estimation imposes one restriction on the model, namely the symmetric-within-type inversion, and we state where this is implemented. The acquisition hazard is measured as the unseparated product in equation (11), which doesn't impose any requirements on the process cost, κ, and the counterfactual holds the initiation factor fixed in the conservative direction (Proposition 4.4).

6.1 The Pool and Engagement

The pool of Section 4.1.3 is measured as $\mathcal{K}_t$, the set of large incumbents, comobining the Big Pharma and Large Biotech firms as defined in Section 5.1. These firms must hold an oncology asset in quarter t, with $K_t = |\mathcal{K}_t|$ between 22 and 26 over the sample. The overlap set $\mathcal{O}_{jt}$ is measured using an overlap definition of development assets in Phase II or later or an approved product still on patent, consistent with Assumption 2. $H_{jt} = |\mathcal{O}_{jt}|$ averages nine, and the high-type share H_{jt}/K_t averages 0.36 with wide variation, from around 0.7 in non-small cell lung and breast cancer to near 0.1 in the smallest markets. $L_{jt} = K_t - H_{jt}$ is the permitted pool under the policy ban, with a mean of fifteen and a minimum of three in the sample.

The engaged count follows equation (3). The filings require at least one engaged party, providing us with

$$m_{it} \sim \text{Bin}(K_t, p) \text{ truncated at } m \geq 1, \qquad m_{it}^{\text{cf}} \sim \text{Bin}(L_{jt}, p) \text{ truncated at } m \geq 1, \tag{17}$$

in the baseline and under the ban. We calibrate p to the merger filings of Section 5.4 such that the mean engaged count reproduces the filings' mean of 3.8, which gives $p = 0.16$ at $K_t = 24$. For robustness, we report every result over a grid of p that includes the value reproducing the filings' median of 2 ($p = 0.055$). Equation (17) holds the rate at its baseline value under the ban, as Assumption 3 and Proposition 4.3 do.

Allowing for p to be determined in equilibrium is an area of open work. Such a change would increase the engagement probability for low-type firms, increasing the number of bidders on average in the counterfactual (Appendix E). Thus, the fixed probability is the case in which the counterfactual has the largest effect.

6.2 Valuations, the Reserve, and Bidding

In this subsection, we describe how we measure three objects from the auction stage of Section 4.2 with the data: the seller's reserve, which is measured and deliberately slack; the walk-away, whose total the filings fix and whose split between its two margins we vary over a grid; and the type-specific valuation distributions, recovered by inverting the observed bids.

The reserve value of a given small biotech is the continuation value of developing and commercializing the product themselves, given by equation (4). This is measured deal-by-deal as

$$R_i = \phi \, \pi^{\text{app}}_{ij(i)} \, V^{\text{own}}_{j(i)t(i)}. \tag{18}$$

where $\pi^{\text{app}}_{ij(i)}$ is the compound's analyst likelihood of approval in the deal's indication at the deal date, a drug-indication object that proxies for the quality term, θ_{ij}. This is measured as the drug-indication pair's own value where the analysts cover it and the indication mean otherwise. V^{own}_{jt} is the discounted revenue of one more approved product in its market and year (detailed in Section 6.4), and $\phi \in (0, 1]$ is the discount a small firm applies to commercializing alone, the reason it sells at all. We are unable to separately identify ϕ from the entry coefficient on V^{own}, so we do not estimate it. Instead we use the requirement that an observed sale means the winning bid beat the seller's accept threshold, so the estimated reserve should rarely, if ever, exceed the price actually paid. We set ϕ to the largest value at which the winning bid clears the reserve in 95 percent of the deals by incumbent pool members. The requirement constrains ϕ from one side only, and Appendix K.6 reports where it binds and shows that nothing in the counterfactual turns on the choice.

The measured reserve is small relative to the prices we observe, which is an intentional model feature. Among deals by the incumbent pool members, the median price is about thirty times $\pi^{\text{app}} V^{\text{own}}$ on total deal value, nineteen times at the risk-adjusted price, and six times on the upfront payment alone (Appendix K.6), so even the only component paid with certainty sits well above the seller's continuation value. The reserve is intentionally slack because its only job is the seller's accept threshold. It is not asked to explain why engaged parties fail to bid, which is instead determined by the walk-away margins.

The walk-away is measured in two pieces. Recall from Section 4.1.4 that an engaged party that never prices the target exits either at the diligence screen, before forming a valuation and observed by its rivals, or at the bid stage, after forming a value. There might be multiple, unobservable, reasons that drive the choice to disengage after forming a value: internal operating frictions, the opportunity cost in the acquisition budget, or shocks to firm expectations. Let F denote this bid-stage share, or the probability that a party which passes the screen nevertheless never bids. The filings fix the two margins' product, since 65 percent of informed parties never price, but the deal prices alone cannot separate the split between the two channels, so we carry F as a grid. A bid-stage share of F leaves a screen exit rate of $1 - 0.35/(1 - F)$, and the grid runs from all of the walk-away at the screen ($F = 0$) to all of it at the bid stage ($F = 0.65$, the one-margin model), inverting and simulating at the same split in each cell, with $F = 0.35$ as the working value. The bids themselves discipline the split through a bid-fit check, where the recovered valuations pushed back through the bid function should return the observed bids, and a large F implies a pile-up of winning bids just above the no-bid threshold that the smooth lower tail of observed bids does not show. A simulation confirms the check has power, since the fit gap stays small when bids are generated under a true F of any size on the grid, and Section 7.2 reports the gap by cell.

The inversion itself works on surplus bids. A bidder's valuation is the reserve plus a private surplus, so the surplus bid $s_i = b_i - R_i$ is the object we invert, and the bid function is equation (5), with the bid-stage share F entering as a point mass at the bottom of the distribution each bidder shades against:

$$b(v) = v - \frac{\int_R^v F(x)^{m-1}\,dx}{F(v)^{m-1}}, \qquad F(x) = F(R) + \big(1 - F(R)\big)\,\tilde{F}(x), \tag{19}$$

with $\tilde{F}$ the distribution of surplus above the reserve and $F \equiv F(R)$ that point mass. Here and in the rest of this subsection, m is the number of parties past the screen, the count n_{it} of Section 4.2, since those are the parties a bidder shades against. The screen's exits enter through that count, not through F, because rivals observe them. Observing only winning bids, and only for completed sales, the distribution of a representative engaged party's bid is recovered from the winning-bid distribution by the order-statistic identity $\tilde{G}(s)^m = F(R)^m + (1 - F(R)^m)\,G_{\text{win}}(s)$, and valuations follow from the first-order condition (Guerre et al., 2000),

$$v(s) = s + \frac{\tilde{G}(s)}{(m-1)\,\tilde{g}(s)}. \tag{20}$$

Please see Appendix F for the derivation of this correction for the reserve.

The bids entering the inversion are residuals from a bid index rather than raw surplus bids, so that deals for different drugs in different markets are comparable. The index regresses the surplus bid on the drug's phase and promise, the deal's form and terms, the market's sales and pipeline, and year and indication effects, with the covariates listed in Appendix J. The acquirer's own position in the market is deliberately excluded, as it is the margin the counterfactual acts on. Because the index is additive, a markup recovered on the residual scale can be added back to the fitted bid, which is how simulated prices return to dollars.

These inversion has two limitations. First, equation (20) is valid only if the residual dispersion is the private-value dispersion. The bid index explains 39 percent of the variance in surplus bids, and a symmetric independent-private-values model reads the rest as dispersion within an auction, leading us to use estimate the scale factor instead of level of valuations. This is described in more depth in Section 7.2. Instead, the inversion provides the shape of the valuation distribution above the reserve, by type, which is all the two counterfactual ratios use. Second, the types are recovered by applying equation (20) to each type's winning bids and splitting by the acquirer's type, a symmetric shortcut we take due to data limitations. We only have 25 acquisitions in the inversion sample where the winning bids are by low types. The asymmetric inversion of Campo et al. (2003), in which each type best responds to the other's distribution, is the estimator a larger sample would support, and a place we highlight for future work.

6.3 The Acquisition Payoff

The expected payoff from the acquisition route to a compound entering indication j in year t is the price an acquirer pays, times the discounted probability that the compound is acquired at some point over the years it stays in development there:

$$\Pi^{A}_{jt} = A_{jt} \sum_{\tau=0}^{T_j-1} \delta^{\tau} (1 - \pi^{A}_{jt})^{\tau} \, \pi^{A}_{jt}, \tag{21}$$

This definition ensures that the payoff is on a comparable scale to self-commercialization. π^{A}_{jt} is the annual acquisition hazard, defined as the incumbent acquisitions of small-firm compounds in Phase I through NDA review in j during t, over the stock of relevant compounds. This measure is first smoothed via a Poisson regression on the pool mix H_{jt}/K_t, market revenue, the incumbents' revenue at risk from Section 5.2, and indication and year fixed effects. T_j is the median number of years a small-firm program stays in the indication after its first Phase II, 4.3 on average, and δ is the annual discount factor. A_{jt} is a hedonic price,

predicted from the deal's and the market's characteristics[7]. For this, we use the total potential deal value for the measure to capture the total commitment, with the upfront payment and the risk-adjusted price of Section 7.2 carried as alternatives.

We define two ratios that carry the auction's counterfactual adjustments. The first, η^{A}_{jt}, measures what the ban does to the price a compound fetches when it sells. The second, η^{π}_{jt}, measures what it does to the chance that anyone prices it at all. Both compare the restricted market to the baseline in the same indication and year, so each is a number close to one that scales a measured payoff rather than replacing it.

$$\eta^{A}_{jt} = \frac{\mathbb{E}\big[\, b(v^{(1)}) \mid m^{\mathrm{cf}},\, F_L \,\big]}{\mathbb{E}\big[\, b(v^{(1)}) \mid m,\, F_{\mathrm{mix}}(H_{jt}/K_t) \,\big]}, \qquad \eta^{\pi}_{jt} = \frac{\Pr\big(m^{\mathrm{cf}} \geq 1 \text{ party bids}\big)}{\Pr\big(m \geq 1 \text{ party bids}\big)}, \tag{22}$$

Both ratios are simulated from the recovered type distributions, drawing the engaged count from equation (17): in the baseline a party is a high type with probability H_{jt}/K_t, and under the ban every party is a low type drawn from the permitted pool (Appendix K.4 gives the simulation). The ratios are how the counterfactual policy interacts with the auction model. η^{A}_{jt} carries the composition effect, since the remaining bidders draw weaker valuations and shade differently, together with the effect of a thinner field on shading. η^{π}_{jt} carries the effect of a thinner field on whether anyone prices the compound at all. These ratios scale payoffs that we measure in the data, giving the counterfactual price $\eta^{A}_{jt} A_{jt}$ and the hazard $\eta^{\pi}_{jt} \pi^{A}_{jt}$.

6.4 Self-Commercialization

If the compound is not acquired and is approved, the biotech earns

$$V^{\mathrm{own}}_{jt} = \sum_{\tau=T}^{T+40} \delta^{\tau}\, s\, M_{jt}, \tag{23}$$

where M_{jt} is US revenue of approved drugs in indication j in year t, s is the share of the market a newly approved product takes, $T = 24$ quarters is the lag from a first Phase II to approval, forty quarters of exclusivity follow, and δ is the quarterly discount factor.

The entrant's share s comes from the stacked event study of Section 5.5, where a competitor's approval lowers each incumbent's share of treated patients by 3.1 percent, does not change the number of patients

[7]Every priced deal in the Section 3.3 sample is regressed on the pool mix, market sales, and the deal's own terms, then predicted at each indication-year's pool mix and market sales with the deal characteristics at their sample means (Appendix J gives the specification).

treated, and does not move incumbent prices, so the entrant's revenue is the incumbents' loss, about 3 percent of the market. We also carry two alternative measures throughout the analysis: the clean-control estimate of 10.6 percent and the equal-share benchmark $1/(n_{jt}+1)$ (Appendix J). The approval probability that multiplies V^{own} in the payoff is the realized rate at which Phase II programs in the indication reach approval, empirically measured in our data and reported in Table 8.

Table 8: Realized approval rates by indication

Indication	Phase II pairs	Approved (%)
Non-Small Cell Lung Cancer (NSCLC)	233	12.4
Breast Cancer	216	15.7
Prostate Cancer	169	11.8
Female reproductive system cancers	148	8.1
Colorectal Cancer (CRC)	142	9.2
Acute leukemias	134	22.4
Non-Hodgkin's lymphoma	130	20.0
Melanoma	117	12.0
Pancreatic cancer	111	4.5
Multiple Myeloma (MM)	96	18.8
Chronic leukemias	94	23.4
Brain Cancer (Malignant Glioma; AA and gliob	91	4.4
Hepatobiliary cancers	87	9.2
Renal Cell Cancer (RCC)	80	16.2
Gastro-esophageal cancers	70	5.7
Head and neck cancer	56	8.9
Sarcoma	55	23.6
Small Cell Lung Cancer (SCLC)	52	7.7
Bladder Cancer	42	14.3
Hodgkin's lymphoma	33	18.2
Neuroendocrine tumors	31	32.3
Myelofibrosis (MF)	29	6.9
Cutaneous lymphomas	28	39.3
Mesothelioma	20	5.0
Gastrointestinal Stromal Tumor (GIST)	16	18.8
Thyroid cancer	15	33.3
Non-melanomatous skin cancers	8	50.0

Notes: Every drug-indication pair in the development panel that reached Phase II in the indication and whose first record is 2012 or earlier, so each pair has a decade of runway before the panel ends; all firms and all of a drug's indications, not only the first. Approved means the pair ever reaches approval there. The approval share is the probability the entry model's self-commercialization payoff uses (Section 6.4). No unconditional rate is reported, because the sample conditions on a clinical record. The registry placeholders ("other solid tumors" and "hematologic cancer") name no approvable indication and are excluded here as in the estimation. Across all indications the rate is 13.3 percent, 329 of 2,478 pairs, above the 6.7 percent that Wong et al. (2019) estimate from Phase II for oncology and the 10.8 percent in Biotechnology Innovation Organization et al. (2021); the cohort restriction accounts for the difference, since conditioning on a first record in 2012 or earlier both resolves the censoring and selects a period before oncology approval rates fell. The ranking is the published one, with the hematologic cancers above the large solid tumors and pancreatic and brain cancer at the bottom. Six indications rest on fewer than 30 pairs, from myelofibrosis and cutaneous lymphomas at 29 and 28 down to non-melanomatous skin cancers at 8, and their rates are correspondingly imprecise.

6.5 Entry

A compound i that has completed Phase I chooses an indication for its first Phase II trial, or does not proceed. Its payoff from indication j in decision year t is

$$U_{ijt} = \lambda_1 \, q_{ij} \, \Pi^A_{jt} + \lambda_2 \, q_{ij} \, \pi^{\text{app}}_j V^{\text{own}}_{jt} + \gamma \ln q_{ij} + Z'_{jt}\beta + \alpha_j + \varepsilon_{ijt}, \tag{24}$$

with ε_{ijt} type-I extreme value. Equation (24) is the econometric counterpart of the entry value in equation (13), where the entrant has two possible payoffs: acquisition or self-commercialization. The entry cost F_j of equation (12) is absorbed into the alternative constants α_j. Z_{jt} includes the indication's scientific promise, defined as the number of trials in the prior year run by non-industry sponsors, and a measure of crowding which captures the number of small-firm programs already in Phase II or later in the indication the prior year. A higher crowding measure results in lower expected payoffs of both acquisition and self-commercialization, and is lagged so that the year's own entrants are not in it. α_j are alternative-specific constants, and they absorb everything permanent about an indication, so λ_1 and λ_2 are identified by variation across the alternatives one compound faces rather than by differences in level between markets.

In measuring the possible payoffs, we scale them by a match quality measure, q_{ij}, which supplies most of that variation, since the indication-year terms move little once the constants are in (indication and year effects explain 88 percent of the fitted acquisition hazard) while a compound's mechanism makes some markets feasible for it and others not. This is the empirical analog to the quality measure, θ_{ij} in Section 4. The measure also serves to discipline the entrant's choice set. For example, the quality measure we use captures the fact that a HER2 antagonist has the highest match in treating breast cancer, a lower probability of treating gastric cancers, and no probability of treating an acute leukemia. The metric is constructed as the probability that a compound with i's mechanism of action is developed in indication j. It is the leave-one-out share of same-mechanism compounds in the development panel with a Phase II or later record in j, shrunk through three nested classifications of the compound's mechanism, from its specific action to its target family to its modality class, so that a compound whose exact mechanism is rare borrows from mechanisms aimed at the same target. Using the example, this metric becomes very small for an acute leukemia, which shrinks the possible payouts from entering that market toward zero. It stays strictly positive by construction, so that a compound whose mechanism has no precedent anywhere is not recorded as infeasible everywhere. See Appendix J.1 for additional details on construction.

The data also require two final adjustments. First, we must right-censor the outside option, as the majority of the potential entrants (Phase I compounds) are still feasibly running their Phase I trials by the end of the analysis window. Second, we require the outside option in the choice set, so the model is estimated at the compound level and the counterfactual's aggregate-entry effect is the change in the predicted no-entry probability. Aggregate-share versions, the multinomial logit over indications estimated as a Poisson regression on indication-year counts and the Berry log-odds form (Berry, 1994; Guimarães et al., 2003), are reported in Appendix K.2 as a reference, since every payoff term other than q_{ij} varies at the indication-year level and the constants leave them little variation to work with.

7 Counterfactual and Estimation

The counterfactual policy bars incumbents holding an overlapping position in an indication (Phase II through approved, on-patent product) from acquiring development-stage compounds in the same indication, with the portfolio fixed at the entry date. We apply the two ratios in equation (22) to the measured payoff, hold V^{own}, π^{app}, Z, q and the constants fixed, and re-solve the choice probabilities, including the outside option of no entry.

7.1 Measured Inputs

Table 9 reports the measured components of the acquisition payoff by indication, averaged over 2006 to 2019. The pool is the same 24 to 25 incumbents for every indication. There is variation, however, in the number of high-type firms in the pool, ranging from 10% to 70%. The annual acquisition hazard averages 3 percent, a program stays in its indication for a median of 4.3 years after its first Phase II, and the predicted price averages \$366 million on total deal value, so the discounted probability of being acquired over the horizon is 10 percent and the expected acquisition payoff of equation (21) averages \$35 million with variation across markets. Appendix J.3 reports the hazard and price regressions behind those averages and how each loads on the pool mix. The own payoff, the Phase II approval probability times V^{own}, averages \$41 million, so the two routes are close to parity for the average entrant: \$36 million from acquisition against \$41 million from commercializing alone. Both are expectations formed at the entry decision, and both are small relative to the prices that change hands when a sale happens, because a compound reaches either outcome with probability around one in ten.

Table 9: The measured payoff components by indication, 2006 to 2019

Indication	Entries	K	H	H/K	π^A (%)	T_j	A (USD m)	Π^A (USD m)	V^{own} (USD m)
Non-Small Cell Lung Cancer	39	24.6	17.4	0.71	3.9	4.0	451.0	52.8	677
Breast Cancer	44	24.6	17.3	0.71	2.6	4.0	511.8	41.9	1353
Non-Hodgkin's Lymphoma	39	24.6	16.3	0.67	4.7	3.9	444.8	61.2	657
Female Reproductive Cancer	28	24.6	16.2	0.66	3.0	4.8	393.4	41.1	91
Prostate Cancer	29	24.6	15.5	0.63	3.0	4.8	432.8	44.9	561
Colorectal Cancer	28	24.6	15.4	0.63	4.2	3.5	442.2	51.4	670
Acute Leukemias	44	24.6	14.6	0.60	2.6	4.8	382.7	35.6	48
Pancreatic Cancer	26	24.6	13.2	0.54	1.7	4.2	376.1	20.6	40
Chronic Leukemias	19	24.6	12.8	0.52	3.6	4.0	426.3	45.2	619
Renal Cell Cancer	16	24.6	12.2	0.50	3.0	4.0	387.2	34.7	210
Melanoma	26	24.6	12.0	0.49	2.0	4.4	387.8	25.0	226
Multiple Myeloma	24	24.6	11.7	0.48	3.6	4.8	439.8	54.1	818
Brain Cancer	21	24.6	11.3	0.46	1.7	4.8	375.3	22.7	123
Gastroesophageal Cancers	12	24.6	10.5	0.43	2.4	3.4	364.0	24.0	32
Head and Neck Cancer	17	24.6	10.5	0.43	3.4	3.4	367.5	33.9	72
Hepatobiliary Cancer	17	24.6	10.3	0.42	1.3	3.8	365.9	13.8	68
Small Cell Lung Cancer	9	24.6	10.0	0.41	3.9	4.0	361.6	42.8	34
Sarcoma	11	24.6	9.3	0.38	3.0	4.1	363.7	33.3	85
Bladder Cancer	10	24.6	6.9	0.28	2.6	3.0	347.5	23.0	17
Myelofibrosis	8	24.6	6.1	0.25	5.9	4.8	348.5	66.0	107
Neuroendocrine Tumors	6	24.6	5.8	0.24	2.2	6.2	367.6	32.4	295
Hodgkin's Lymphoma	13	24.6	5.2	0.21	2.9	3.1	341.0	24.6	25
Gastrointestinal Stromal Tumor	2	24.6	4.9	0.20	3.7	3.8	344.8	36.1	83
Thyroid Cancer	4	24.6	4.6	0.19	4.9	4.5	336.9	52.3	11
Mesothelioma	7	24.6	4.3	0.17	1.6	7.5	335.1	24.8	5
Cutaneous Lymphomas	10	24.6	2.9	0.12	1.4	5.8	330.4	18.8	18
Non-Melanomatous Skin Cancers	2	24.6	2.8	0.11	2.2	2.6	330.0	16.0	18

Notes: Means over 2006 to 2019 by parent indication, sorted by the high-type share of the pool. Entries are first Phase II trials by small firms directed at the indication, every named indication counted. K is the number of Big Pharma and Large Biotech incumbents holding an oncology asset in the quarter; H the number holding an overlapping position in the indication, a Phase II or later development asset or an on-patent approved product. π^A is the fitted annual acquisition hazard of a small-firm Phase I to NDA compound (Table 26, column 2); T_j the median years a small-firm program stays in the indication after its first Phase II; A the predicted total deal value (Table 27, column 1), 2015q1 $m; Π^A the expected acquisition payoff over the horizon, equation (21). V^{own} is the discounted revenue of one more approved product in the indication, equation (23), at the event-study share.

V_{jt}^{own} is built on the entrant's share s, which is the 3.1 percent loss of treated patients that a competitor's approval imposes on each incumbent in the stacked event study of Section 5.5. At this share V_{jt}^{own} averages \$242 million conditional on approval, with 83 percent of its variance across indications, and the \$41 million own payoff above is that value times the indication's approval probability. The reserve of equation (18) applies the same risked value deal by deal and averages \$34 million across deals by pool members, with the calibration putting the commercialization discount at its bound of one on total deal value and at 0.40 on the upfront payment alone (Appendix K.6).

7.2 The Auction Stage

The bid index is fitted on all 337 priced deals, the 246 exclusive licenses with a reported total deal value and the 91 eligible whole-company acquisitions with clearly assignable lead assets, from 2006 to 2019, with the deal type included in the index covariates. For the inversion, we restrict the deals made by incumbents, which totals 140 of the 337 priced deals. Additionally, we drop the top and bottom 5% of deals, leaving us with a total of 126 in the inversion. Valuations are recovered at $m = 3$ and the counterfactual is simulated with the engaged count drawn from equation (17). The low-type distribution is identified on 28 deals, which is thin, and prohibits us from using the preferred asymmetric estimator. We discuss this limitation further at the end of the section.

The estimand of this stage is the proportional effect of the ban, defined as the ratio of the expected winning bid with and without the barred bidders, and the ratio of the probability that anyone bids. Our current proportional approach is required as we only observe the winning bid of a heterogeneous asset, and we cannot separate dispersion of values within an auction from differences in asset quality across auctions. Bidder shading increases with this dispersion, leaving us with large implied markups. At the working cell the mean recovered valuation is about four times the mean winning bid, where Gorbenko and Malenko (2014), estimating the same object on takeover auctions, find winning slack of 7 to 15 percent of willingness to pay. Thus, as a validation of our estimation strategy, we run a Monte Carlo simulation that generates the predicted winning bids from a known distribution at our sample size and inverts them. This process recovers the truth to within 10 to 25 percent of the prediction, and the same Monte Carlo shows that unobserved auction-level heterogeneity of a plausible size roughly doubles recovered valuations (Krasnokutskaya, 2011). Instead, we estimate the ratios which are far less exposed. A common inflation of all valuations cancels in a ratio of expected bids. We also run this stage using the alternative price measures as a robustness check in

Appendix K.5, which supports the use of the proportional estimand and demonstrates that the choice of price measure does not meaningfully change the estimates.

Two thirds of the parties that sign a confidentiality agreement and receive information never price the asset. Section 4.1.4 assigns that walk-away to two margins with different footprints in the price, the diligence screen that rivals observe and the bid-stage exit they do not, and Section 6.2 carries the bid-stage share F as a grid because the deal prices alone cannot separate the split.

Table 10 reports what each value of F implies for the two margins and how well the bids support it. The check first inverts the observed bids at a given F, then pushes the recovered valuations back through the bid function and compare the result to the bids we started from. A large bid-stage share puts a large mass of valuations below the reserve, so explaining the bids we observe requires valuations far above them. The gap is \$18 million at $F = 0$ against a mean bid of \$761 million, \$125 million at 0.35, and \$981 million at 0.65, where it exceeds the mean bid itself. The bids therefore reject the story in which most of the walk-away happens after a valuation is formed, and place most of it at the diligence screen.

The best-fitting cell is the corner, $F = 0$, but we do not use it. It asserts that no firm ever values an oncology asset and declines to bid, which the data does not support. We take $F = 0.35$ instead, which divides the walk-away about evenly, with 46 percent of engaged parties exiting at the screen and 35 percent of those that pass declining to bid. The choice is conservative for what follows, as the price effect of the restriction is larger at the cells that fit better, and we report every cell rather than this one alone.

Table 10: Where the walk-away sits, and whether the bids support it

	Where the walk-away sits (%)		Bid-fit check	
F (bid stage)	Screen exit	Bid stage	Gap (USD m)	Rank corr.
0.00	65.0	0.0	17.9	1.000
0.25	53.3	25.0	46.2	1.000
0.35*	46.2	35.0	125.2	1.000
0.50	30.0	50.0	422.6	0.996
0.65	0.0	65.0	981.3	0.910

Notes: F is the bid-stage share, the probability that a party which has formed a valuation stays out of the bidding. The filings fix the total walk-away at 65 percent of informed parties, so the probability that an engaged party prices is 0.35 at every F and the screen-pass rate is the residual $0.35/(1-F)$; the first two columns report the resulting exit rates at each margin, as a percent of engaged parties for the screen and of screen-passers for the bid stage. Bid-fit check: the surplus bids are inverted at the stated F, the recovered valuations are drawn as the per-party primitive, the equilibrium winning bid is simulated at $m = 3$, and the gap is its mean less the observed mean surplus-bid residual, in 2015q1 \$m, against a mean bid of \$761m. Rank corr. is the Spearman correlation between the bid and its recovered valuation.
* Working value.

A separate exercise isolates the composition effect by holding the engaged count fixed and drawing every

party from the low-type distribution instead of the sample's mix (Appendix K.4, Table 30). The expected winning bid falls by 14 percent at $m = 2$, 18 percent at $m = 3$ and 21 percent at $m = 5$, so the composition of the field moves the price even with its size held fixed. The expected maximum valuation falls by about 15 percent at every m, with the price falling by less than the valuations in the smallest field and by more in larger ones, since equilibrium shading depends on the composition of rivals and not only on their number.

7.3 Entry Estimates

In this section, we report the estimates from the conditional logit of equation (24). A compound that has completed Phase I trials chooses among the 27 indications, or chooses to develop nothing. In each market it weighs two expected payoffs, acquisition and self-commercialization, and each is the indication-year payoff scaled by the compound's match quality q_{ij}. The scaling factor serves to alter the probability of both approval and acquisition by taking the ex-ante indication-specific quality match into the expectation. For example, a predicted price of $366 million is worth little to a compound whose mechanism cannot reasonably be expected to serve the given market. Match quality also enters on its own, in logs, to govern the indication feasibility separately from how it scales expected payoffs. For instance, a drug with few plausible markets is more likely to exit in the counterfactual, independent of the scaled price. It is what lets a poor match push a compound out of development entirely. The alternative constants absorb everything permanent about an indication, including the entry cost. Thus, λ_1 governs whether, among the markets a compound's mechanism fits, it tilts toward the ones where acquisition pays more. The sample holds 633 compounds with a named indication and 562 that never start a Phase II trial, giving 1,195 cases in all.

Table 11 reports the estimates. In Column 1, the model makes no attempt to determine the ex-ante quality of indication match for a given drug. In other words, each compound is equally likely to be approved in any indication - an unrealistic assumption. As a result, this specification finds nothing. Columns 2 and 3 report the same specification but using different samples. Column 2 utilizes the full data, and the coefficient on the match-weighted acquisition payoff is significant with a point estimate of 0.032 per $1 million. Column 3 is our preferred estimate, as it corrects for a right censoring issue in the data. This specification uses the sample of potential entrants that we are certain have made a choice to not enter any market to identify the extensive margin. This correction prevents us from erroneously assigning the "no indication" choice to a drug that is actually still in Phase I development. Reassuringly, the coefficient of interest is not substantially altered from Column 2 to Column 3, indicating a stable parameter. What does change is that the own payoff

parameter becomes marginally significant under the right censoring correction, at 0.009 per $1 million. The feasibility term $\ln q_{ij}$ is the strongest single predictor of where a compound goes through all specifications. We report additional robustness checks in Appendix K.1 varying how we measure the feasibility term as well as using different deal value metrics. Across every price measure and every match-quality tier the response is the same sign and the same economic size.

Table 11: Entry at the compound level, with match quality

	(1) Without q	(2) With q	(3) With q, right censor correction
Acquisition payoff Π^A	-0.0004		
	(0.0074)		
Own payoff $\pi^{app} V^{own}$	-0.0023		
	(0.0023)		
Crowding (log)	-0.0143	-0.1055	-0.2086
	(0.4324)	(0.4368)	(0.1702)
Promise (log)	0.1032	0.0950	-0.3425***
	(0.2054)	(0.2246)	(0.0769)
$q\times$ acquisition payoff		0.0316***	0.0322***
		(0.0091)	(0.0096)
$q\times$ own payoff		0.0035	0.0086*
		(0.0059)	(0.0046)
$\ln q$		0.3791***	0.3355***
		(0.0615)	(0.0624)
Cases	1,195	1,195	835
Choice rows	33,460	33,460	23,380
Clusters (years)	25	25	24

Conditional logit with alternative constants (not shown) over 27 indications and the outside option, so one coefficient governs both the choice of market and the choice to enter at all. A case is a compound and a named market (a first Phase II naming k markets is k cases, with compounds naming more than four markets excluded), or a compound that never starts Phase II. Payoffs in USD m. Match quality q is the compound's specific mechanism of action's leave-one-out share of compounds developed in the indication, shrunk toward its target family, its EphMRA ATC level-4 class, and the indication's overall share; it multiplies both payoff terms and enters in logs. Crowding is log(1 + small-firm Phase II to NDA programs in the indication the prior year), and promise is log(1 + non-industry trials started in the prior year). Columns 1 and 2 are estimated on the same full sample and differ only in whether q is in the model. Column 3 is the headline and applies the right-censoring correction described in the text, dropping the non-entrants whose imputed decision date falls within seven years of the panel's end. Standard errors clustered by decision year. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

7.4 Counterfactual Results

At the auction stage, the ban lowers the expected winning bid by 32 percent and the probability that any engaged party prices by 18 percent at the working cell, for a combined fall of 42 percent in the price times the pricing probability. Table 12 and Figure 11 report the ban's effect on entry, both margins from the headline model of Column 3. Under the ban the acquisition payoff falls from $36 million to $18 million on average, falling more in markets with larger high-type shares. Entry moves out of the indications where incumbents hold overlapping positions and into the indications where they do not: non-small cell lung

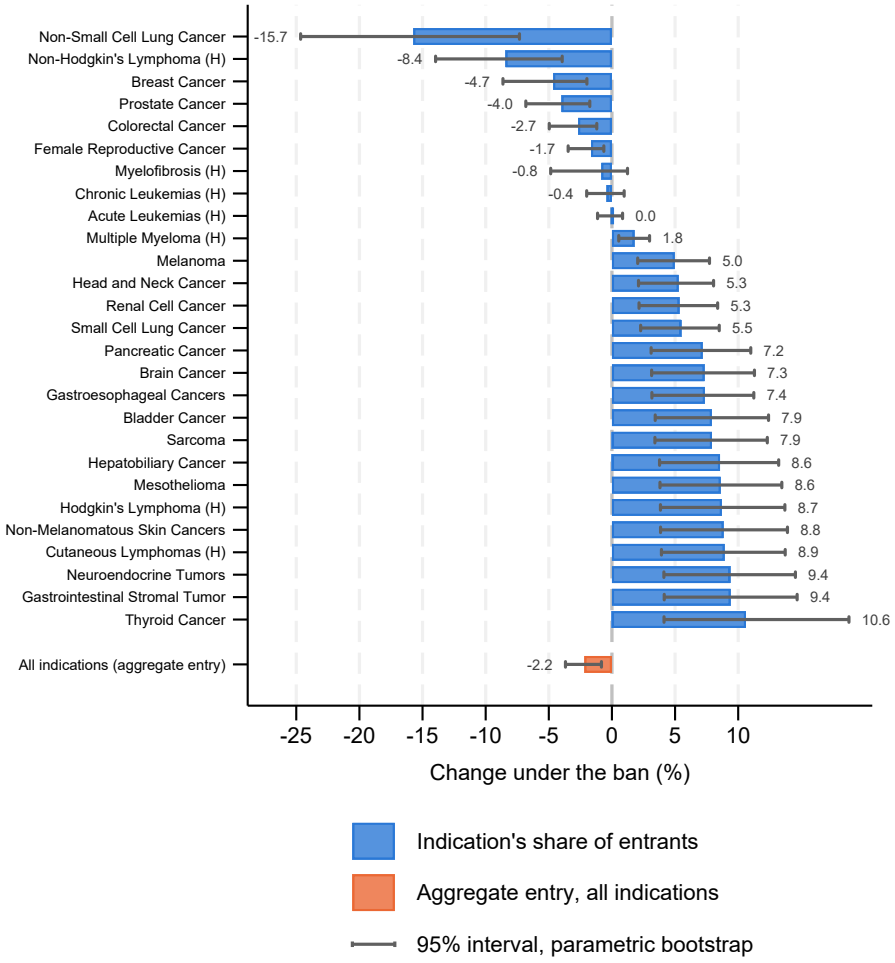
cancer loses 16 percent of its entrants, non-Hodgkin's lymphoma 8 percent, breast cancer 5 percent and prostate cancer 4 percent. The indications with few overlapping incumbents gain 7 to 11 percent each, and 2.9 percent of entrants change indication in total. On the extensive margin the predicted no-entry share rises from 24.2 to 25.8 percent of decisions, a fall in aggregate entry of 2.2 percent of its baseline. Priced at the upfront payment instead, the model reallocates 3.2 percent of entrants and the same indications move, and at the risk-adjusted prices it reallocates 3.4 percent. At the coarse ATC tier it reallocates 0.9 percent of entrants, the same pattern at a smaller size, the attenuation of Appendix K.1 again.

Table 12: The counterfactual: entry shares by indication under the overlap ban

Indication	Baseline share (%)	Restricted share (%)	Change (%)
Acute Leukemias	7.76	7.77	0.0
Bladder Cancer	1.88	2.03	7.9
Brain Cancer	4.13	4.44	7.3
Breast Cancer	8.44	8.05	-4.7
Chronic Leukemias	4.65	4.63	-0.4
Colorectal Cancer	5.28	5.14	-2.7
Cutaneous Lymphomas	1.96	2.14	8.9
Female Reproductive Cancer	4.77	4.69	-1.7
Gastroesophageal Cancers	2.47	2.65	7.4
Gastrointestinal Stromal Tumor	0.32	0.35	9.4
Head and Neck Cancer	3.27	3.44	5.3
Hepatobiliary Cancer	3.15	3.42	8.6
Hodgkin's Lymphoma	2.17	2.36	8.7
Melanoma	5.46	5.73	5.0
Mesothelioma	1.10	1.20	8.6
Multiple Myeloma	5.28	5.38	1.8
Myelofibrosis	1.57	1.56	-0.8
Neuroendocrine Tumors	1.07	1.17	9.4
Non-Hodgkin's Lymphoma	7.32	6.70	-8.4
Non-Small Cell Lung Cancer	9.06	7.63	-15.7
Non-Melanomatous Skin Cancers	0.32	0.35	8.8
Pancreatic Cancer	4.60	4.93	7.2
Prostate Cancer	5.87	5.64	-4.0
Renal Cell Cancer	3.14	3.31	5.3
Sarcoma	1.98	2.14	7.9
Small Cell Lung Cancer	2.15	2.27	5.5
Thyroid Cancer	0.81	0.90	10.6
All indications: entrants changing indication (%)			2.9
All indications: aggregate entry (%)			-2.2

Notes: Predicted share of small-firm first Phase II entries directed at each indication, mean over entrant cases 2006 to 2019, baseline and under the ban, from the compound-level conditional logit with match quality (Table 11, column 3), which carries both margins. Shares sum to 100 by construction, so the two aggregate rows report different objects: the share of entrants that change indication under the ban (half the sum of absolute share changes), and the change in the number of entrants. The ban lowers the acquisition payoff in each indication-year by the product of the two ratios in Appendix Table 31 at the working cell. V^{own}, approval probabilities, promise and the constants are held fixed.

Figure 11: Change in each indication's share of entrants under the ban



Notes: Blue bars: percent change in the indication's share of small-firm first Phase II entries, mean over 2006 to 2019, from the compound-level conditional logit with match quality (Table 11, column 3) at the working cell, bid-stage share $F = 0.35$, $p = 0.16$. (H) marks hematologic indications. The last bar is a different object: the percent change in the number of entrants across all indications, from the same model and the same coefficient. Whiskers are 95 percent intervals from a parametric bootstrap of the entry coefficients, 200 draws from the estimated coefficient distribution of each column with the counterfactual re-solved per draw. The auction-stage ratios are held fixed in every draw; their variation is reported across the grid of Appendix Table 31, not as a standard error. The share of entrants changing indication carries an interval of 1.3 to 4.6 percent around the 2.9 point, and aggregate entry -3.7 to -0.8 around -2.2.

The reallocation is not neutral in terms of expected approvals. Table 8 reports the empirical approval rate estimates by indication conditional on a Phase II trial, which vary substantially by market. For instance, averaged across indications among Phase II pairs with a decade of runway, compounds in the hematologic cancers reach approval at 18 to 23 percent, the large solid tumors sit near 9 to 16, and pancreatic and brain cancer approve under 5. A compound the ban moves out of non-small cell lung cancer therefore lands somewhere with a different conversion rate from entry to approved drug.

The auction-stage effect moves with the walk-away split, and Appendix K.4 reports it across the grid of Section 7.2. The combined ratio runs from 0.49 to 0.65 over the cells, with 0.58 at the working value, and the entry counterfactual uses the working cell.

8 Conclusion

Small biotech firms do not choose where to develop cancer drugs in isolation from the market for their assets. We model the sale of a development-stage compound as a first-price auction among the incumbent firms that engage, measure every payoff the entry decision weighs, and estimate where entry goes when the acquisition payoff moves. Barring overlapping incumbents from acquiring, the policy debated as the remedy for killer acquisitions, cuts the expected acquisition payoff by 50 percent, redirects about 3 percent of small-firm Phase II starts away from the most contested cancers, and lowers the number of entering compounds by about 2 percent.

These headline figures that we report are entry effects, not welfare effects. They do, however, provide insight into the potential welfare effects, and particularly for expected approvals. Per 100 entering compounds, the ban deters 2.2 entrants. Over the years a compound stays in development, about 12 of the 100 are acquired, roughly 10 of them by the overlapping incumbents the ban bars. If some of those banned deals are killer acquisitions, each killer acquisition prevented restores the compound's chance of approval. Valuing both sides of the ledger at the Phase II likelihood of approval, which assumes a killed compound and a deterred marginal entrant carry the same approval odds, the ban pays for itself only if about 22 percent of the acquisitions it eliminates are killers. This would require that 18 percent of all acquisitions are killer acquisitions, far above measured rates. For example, Cunningham et al. (2021) put the killer share at 5.3 to 7.4 percent of pharmaceutical acquisitions, and the European Commission's ex-post evaluation finds a discontinuation of an overlapping project without a technical or safety explanation after 89 of 3,193 trans-

actions, under 3 percent (European Commission, 2024). The Commission's conditional rate is higher, 37 percent of the 240 transactions whose projects overlapped narrowly, but narrow molecular overlap describes a small subset of the broad indication-level class this ban bars. On this margin the policy forgoes three to seven times the approvals it saves.

Approvals are not welfare, and that gap is left for future work. An approval's value depends on it's market. The ban moves entry out of crowded cancers and toward markets with fewer approved options, where variety plausibly matters most. The move also has an approval cost of its own. The entry model's match-quality gradient says a compound pushed from its best-fit indication toward its second choice is a worse scientific fit there, so some of the 3 percent of compounds that reallocate arrive with lower approval odds.

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A Data Appendix

This appendix collects the construction details behind Section 3, in the order the sources appear there. It records the decisions a replicator needs and the checks behind the one-line summaries in the text; the code paths are in the replication package.

A.1 Development Histories

Two cleaning decisions shape the drug-indication life-cycles. First, some drugs stop progressing without a disclosed termination and Biomedtracker lists them as suspended. We assign these a suspension date equal to the last recorded event plus the sample median duration of the final recorded phase: six quarters from preclinical, two from an IND filing, eight from Phase I, twelve from Phase II or Phase III, and two from NDA review. The imputation creates 126 artificial suspension events, 63 where the recorded history ends with a trial start and 63 where development goes quiet with no further event of any kind, and a further 186 drug-indication pairs are marked suspended at their last recorded event with no added time. Officially disclosed suspensions, 2,376 events, carry their reported dates. Second, a drug can run several trials in the same indication at once, most often a combination trial that begins after an approval. We collapse these parallel tracks by taking the running maximum of the development phase within the drug-indication pair, so a drug's phase never falls while any program in the indication is alive, and the additional exclusivity a combination approval earns enters through the patent data rather than the development panel. The maximum rarely binds. Recorded phases move backward in only 10 of 5,670 drug-indication event histories, so the collapse is bookkeeping rather than measurement.

A.2 Ownership and Control-Rights Deals

The ownership build has two tables. Table A is the company identity chain. It lists every company appearing anywhere in the data, with its acquirer and the transfer date, so that the ultimate parent of any firm at any date resolves by walking the chain. Table B is the drug-level ownership event table, the originator and every control transfer, assembled from Biomedtracker alliances, Evaluate company M&A, Evaluate product deals and four hand files. Validation is against Biomedtracker's own lead-company field on its pull date: the chain matches for 2,459 of 2,539 drugs (96.8 percent), and 99.1 percent at Phase II or later.

Exclusions from the deal analysis, with counts: BMT Marker rows (non-exclusive co-development or

co-commercialization, which transfer no control), reverse takeovers (a listing transaction in which the operating company merges into a listed shell, 22 drug-rows), and the 14 whole-company mergers between diversified firms identified by hand (going concerns with marketed portfolios across therapeutic areas). Option deals, which involve no immediate change of control, are dropped. Table 13 reproduces the hand list.

Table 13: The 14 excluded mergers between diversified firms

Acquirer	Target	Quarter	USD bn	Reason
Pfizer	Wyeth	2009q4	68.0	Diversified pharma target (305 products, 137 marketed)
Bristol Myers Squibb	Celgene	2019q4	74.0	Going concern with a marketed portfolio (167 products; Revlimid)
Roche	Genentech	2009q1	46.8	Buy-out of the minority in a majority-owned subsidiary
Merck & Co	Schering-Plough	2009q4	41.1	Diversified pharma target (243 products, 144 marketed)
Shire	Baxalta	2016q2	32.0	Diversified target (hematology and immunology)
Bayer	Schering AG	2006q3	21.3	Diversified pharma target (148 products, 95 marketed)
Sanofi	Genzyme	2011q2	20.1	Diversified target (rare disease)
Pfizer	Hospira	2015q1	17.0	Diversified generics and injectables target
Novartis	GSK oncology business	2015q1	16.0	Portfolio swap between two incumbents
Merck KGaA	Serono	2007q1	13.8	Diversified target (fertility, multiple sclerosis)
Teva	IVAX	2006q1	7.4	Diversified generics target
Teva	Cephalon	2011q4	6.8	Diversified pharma target (127 products, 87 marketed)
Novartis	Chiron	2006q2	5.7	Diversified target (vaccines and diagnostics)
Servier	Shire oncology business	2018q3	2.4	Business-unit purchase of a marketed franchise (Oncaspar)

Announced deal values in USD billions. A deal is on this list when the target is a going concern with a marketed portfolio across therapeutic areas, or the transaction is between two incumbents, rather than the purchase of a development-stage compound. The full list is in the replication package.

A licensing deal covering several assets is assigned to the compound the deal description identifies as the lead, or to the compound furthest in development; 42 of the 246 licenses priced on total value cover more than one sample drug. Restricting the Table 5 regressions to deals covering exactly one sample drug, which also removes the clear-lead whole-company tier, gives an overlap premium of $8.8 million (SE 3.4) on the upfront and $80.5 million (SE 25.7) on total value, on 203 and 267 deals, so the assignment rule is not what carries the result.

Whole-company acquisitions enter the priced-deal analyses only when the price can be tied to one compound. The rule has two tiers: the target holds a single sample drug (95 deals, 61 with a reported value), or two to three sample drugs with a unique most-advanced drug that is still in development (Phase I to NDA review) and a company value under $5 billion (45 clear-lead deals, 39 priced, 30 surviving the screen). The screen is not cosmetic. The unscreened tier is dominated by multi-billion dollar takeovers of firms with marketed products, and including them roughly halves the fit of the bid index and eliminates the overlap

premium of Section 5.3, because those prices reflect the acquired firm's full commercial assets rather than the lead compound. A whole-company indicator absorbs the deal-form level in every pooled specification.

A.3 Prices and Deflation

All dollar amounts are deflated to 2015q1 dollars with the CPI-U (series CPIAUCSL). License terms come from Biomedtracker alliance records (upfront payment, milestone schedule, royalty provisions); company deal values from Evaluate Pharma; the Evaluate product-deal file is the spine linking deals to drugs. Milestones enter total potential deal value at their nominal amounts because the data do not break them down by trigger. Selective disclosure is measured by a linear probability model of whether a deal reports a price, on acquirer type, phase, scope, orphan status, drugs per deal and year effects, one observation per exclusive-license deal. Table 14 reports it: a large-firm acquirer is 37 percentage points (SE 5) more likely to report a total value and 40 points (SE 5) more likely to report an upfront.

Table 14: Selective disclosure of deal terms

	(1) Total value reported	(2) Upfront reported
Large-firm acquirer	0.370^{***}	0.398^{***}
	(0.046)	(0.048)
Mean of dependent variable	0.53	0.46
Observations	462	462
R^2	0.18	0.17

Linear probability that an exclusive-license deal reports a price, one observation per deal, 2006q1 to 2019q4. Controls: phase at the deal, worldwide rights, orphan designation, sample drugs in the deal, and year fixed effects. Standard errors clustered by acquirer parent. Missing prices are concentrated among smaller acquirers, so the large-firm price premia in the paper are lower bounds.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

A.4 Sales and Indication Attribution

Quarterly drug sales combine Evaluate Pharma and Biomedtracker revenue series. About 22 percent of on-patent approved drug-quarters have no sales row, nearly all products below disclosure thresholds; these are set to zero and the count of covered drugs is carried alongside every market aggregate. A drug approved in several cancers has its sales split by MarketScan indication shares: for molecule d, cancer c and quarter t, patient-months on d in c over patient-months on d in all cancers, with boundary quarters carried outward beyond the 2009 to 2019 claims window. Where the claims data cannot identify the cancer (CML, GIST,

neuroendocrine tumors, sarcoma, thyroid), shares are assigned by hand from labels and analyst sources; the hand file is in the replication package. Claims shares carry about 71 percent of attributed sales, hand shares about 11 percent, and single-indication drugs the remaining 18 percent.

A.5 Exclusivity End Dates

The exclusivity end is the later of expected patent expiry and regulatory exclusivity. The patent side is Evaluate Pharma's consensus analyst forecast of the first commercially significant expiry in the drug's key market, converted to a term at the drug level. Where Evaluate has no record, the Orange Book active-ingredient patent with its term extension, from the NBER-maintained cleaned version, serves as the fallback, and a biologic with neither is assigned fifteen years from first approval and flagged. Expiry is never set earlier than five years after first approval, the new chemical entity exclusivity. Drugs with a public litigation outcome or settlement date take that date. The regulatory side is applied at the market level, three years for an approval in the market and seven for an orphan approval. The realized exclusivity end is the later of the expected date and the Orange Book first generic approval through 2024, holding the expected date where an override or a pre-1991 legacy product applies. The expected date enters quarters to expiry and revenue at risk. The realized date defines the on-patent indicator.

B Potential Acquirer Definitions

This appendix lists the firms behind the incumbent definitions of Section 3.4, with the quarters in which each qualifies inside the analysis window, 2006q1 to 2019q4. Table 15 holds the Big Pharma incumbents and Table 16 the Large Biotech incumbents. A firm acquired by an incumbent rolls up to its parent from the acquisition quarter, and a firm that crosses the Big Pharma thresholds moves from one table to the other in the quarter it crosses. Both tables are generated from the quarterly incumbent panel in the replication package.

Table 15: Big Pharma Incumbents

Incumbent	Member Firm	First Quarter	Last Quarter
Abbott Laboratories	Abbott Laboratories	2006Q1	2012Q4
AbbVie	AbbVie	2006Q1	2019Q4
	Abbott Laboratories†	2013Q1	2019Q4
Amgen‡	Amgen	2011Q1	2019Q4
AstraZeneca	AstraZeneca	2006Q1	2019Q4
	MedImmune†	2007Q2	2019Q4
Bayer	Bayer	2006Q1	2019Q4
Biogen‡	Biogen	2012Q1	2019Q4
Boehringer Ingelheim	Boehringer Ingelheim	2006Q1	2019Q4
Bristol Myers Squibb	Bristol Myers Squibb	2006Q1	2019Q4
	Celgene†	2019Q4	2019Q4
Eli Lilly	Eli Lilly	2006Q1	2019Q4
Gilead Sciences‡	Gilead Sciences	2013Q1	2019Q4
	Kite Pharma†	2017Q4	2019Q4
GSK	GSK	2006Q1	2019Q4
Johnson & Johnson	Johnson & Johnson	2006Q1	2019Q4
	Janssen Pharmaceutica	2006Q1	2019Q4
Merck & Co	Merck & Co	2006Q1	2019Q4
	Schering-Plough†	2009Q4	2019Q4
Novartis	Novartis	2006Q1	2019Q4
Pfizer	Pfizer	2006Q1	2019Q4
	Wyeth†	2009Q4	2019Q4
Regeneron Pharmaceuticals‡	Regeneron Pharmaceuticals	2018Q3	2019Q4
Roche	Roche	2006Q1	2019Q4
	Genentech†	2006Q1	2019Q4
	Chugai Pharmaceutical†	2006Q1	2019Q4
Sanofi	Sanofi	2006Q1	2019Q4
	Aventis†	2006Q1	2019Q4
Schering-Plough	Schering-Plough	2006Q1	2009Q3
Takeda‡	Takeda	2011Q3	2019Q4
	Shire†	2019Q1	2019Q4
	Baxalta†	2019Q1	2019Q4
Wyeth	Wyeth	2006Q1	2009Q3

To qualify as Big Pharma, in a given year the firm must have over $40 billion in market capitalization, $20 billion in annual revenue, 15% of revenue invested in research and development, and significant commercial operations in the US, Europe, China, and Japan.
†Member firm mapped to the incumbent through the company identity chain from the quarter shown. Wyeth and Schering-Plough qualify independently until their 2009 acquisitions; Abbott Laboratories qualifies independently until the AbbVie separation in 2013; Genentech and Chugai map to Roche through majority ownership, with Genentech also qualifying independently as a Large Biotech until 2009Q2 (Table 16).
‡Qualified as a Large Biotech incumbent before the first quarter shown (Table 16).

Table 16: Large Biotech Incumbents

Company Name	First Quarter	Last Quarter
Amgen‡	2006Q1	2010Q4
Baxalta†	2015Q3	2016Q1
Biogen‡	2006Q1	2011Q4
Celgene†	2006Q1	2019Q3
Daiichi Sankyo	2006Q1	2019Q4
Genentech†	2006Q1	2009Q2
Genmab	2006Q1	2013Q1
Genzyme†	2006Q1	2011Q1
Gilead Sciences‡	2006Q1	2012Q4
Incyte	2014Q1	2019Q4
Kite Pharma†	2014Q3	2017Q3
MedImmune†	2006Q1	2007Q1
Merck KGaA	2006Q1	2019Q4
Regeneron Pharmaceuticals‡	2016Q1	2018Q2
Servier	2010Q1	2019Q4
Shire†	2010Q1	2015Q3
Takeda‡	2006Q1	2011Q2
Teva Pharmaceutical Industries	2006Q1	2019Q4

To qualify as a Large Biotech, in a given year the firm must have over $15 billion in market capitalization, $2 billion in annual revenue, 20% of revenue invested in research and development, be independent from any Big Pharma firm, and have significant commercial operations in the US, Europe, China, and Japan.

†Acquired by an incumbent at the end of the span shown and mapped to its acquirer thereafter (Table 15): Baxalta by Shire (2016) and then Takeda (2019), Celgene by Bristol Myers Squibb (2019), Genentech by Roche (2009), Genzyme by Sanofi (2011), Kite Pharma by Gilead Sciences (2017), MedImmune by AstraZeneca (2007), Shire by Takeda (2019).

‡Crossed the Big Pharma thresholds after the span shown and appears in Table 15 from that quarter.

C Merger Disclosure Coding

This appendix is the codebook behind Sections 3.7 and 5.4 and the paired-price spreads of Section 7.2.

C.1 Retrieval and Extraction

The candidate set is every Evaluate M&A target in our deal universe that could plausibly have filed a merger disclosure: a US-listed target acquired in a change-of-control transaction, 171 deals. Matching targets to EDGAR by CIK and searching their filings around the announcement date finds a Schedule 14D-9 or DEFM14A for 135. The "Background of the Merger" (or "Background of the Offer") section is extracted automatically and validated; 95 filings yield a chronology complete enough to code, and 7 of those are excluded from the reported distribution because the document describes a different transaction, the process was a related-party or otherwise unusable one, leaving the 88 processes of Figure 9. The pipeline is a numbered sequence of scripts in the replication package.

C.2 Coding Rules

The unit of coding is the party. Each chronology is read party by party, producing one row per named or aliased participant, 351 rows over the 95 coded processes, and every flag on a row must be supported by verbatim sentences copied from the filing, which is what makes the coding auditable. A party counts as engaged, and so enters K^e, if it either (a) received confidential information under a confidentiality agreement executed for this transaction, or (b) made a price-bearing proposal, with or without an agreement. The acquirer always counts. The rules that decide the hard cases: a confidentiality agreement signed for a partnering or licensing discussion does not qualify, one amended to cover an acquisition does; an unsolicited approach counts only if it named a price; and parties in a parallel licensing or asset canvass are excluded from K^e unless they crossed over into the acquisition process. Where a filing describes informed parties without enumerating them ("several other parties conducted due diligence"), the count is recorded as a floor and flagged; fifteen of the 88 counts are floors, which biases the measured K^e downward. Alongside K^e we record the number of parties the banker contacted, the number that bid at all, and the number submitting final bids.

C.3 Audit

Every counted party is reproduced in an audit sheet, one block per deal, carrying the count, its basis (an enumerated list or a stated count), the funnel numbers, the source URL, an adjudication note explaining any judgment call, and the verbatim evidence for each party's flags. The machine-assisted reading was checked by hand against this sheet, and ambiguous processes carry their resolution in the note. One caveat travels with every use of these counts: the filings describe change-of-control sales of public companies. The auction sample itself contains 91 deals of this form, so the funnel evidence bears directly on part of the estimation sample. For the license majority, the partnering and licensing canvasses inside the same filings (Section 5.4) are the bridge.

C.4 Paired Prices

The paired prices used in Appendix K.5 are built from the same chronologies. For each process we record the winner's first offer, its final price at signing, and the best rival offer stated in the same unit (per share, or total consideration), keeping a flag for whether the rival's price was a firm proposal rather than an indication of interest that was never formalized. A hand adjudication resolves unit mismatches and stale offers, deal by deal, with notes. The result is 22 processes with both prices in one unit, 15 of them with a firm rival proposal, and 37 processes in which the winner's own offer path is observed from first offer to signing.

D Proofs

This appendix provides the proofs for Propositions 4.2, 4.3 and 4.4.

Throughout the proofs, the parties past the screen bid in the symmetric equilibrium of equation (5) (restated here):

$$b(v) = v - \frac{1}{F(v)^{n_{it}-1}} \int_{R_{it}}^{v} F(x)^{n_{it}-1}\, dx.$$

with each party's valuation drawn independently from the mixture $F_{\text{mix}} = (H_{jt}/K_t)\, F_H + (L_{jt}/K_t)\, F_L$ in the baseline and from F_L under the restriction. This is the treatment estimated in Section 6.2 and simulated in Appendix K.4. The proof of Proposition 4.2 establishes three steps that the other two proofs reuse.

Proposition 4.2. *Entry value is nondecreasing in the acquirer pool. Adding an incumbent to $\mathcal{K}_t$ weakly raises Π_{ij}, and the increase is weakly larger when the added incumbent is a high type.*

Proof. The proof proceeds in three steps. Step 1 writes the option value of a process in terms of valuations rather than bids. Step 2 shows that the option value is nondecreasing in the pool at a fixed reserve. Step 3 carries the ordering through the Bellman equation by backward induction.

Step 1: The option value in valuations. Fix a process in which n parties past the screen draw valuations independently from F and the reserve is R. Let $v^{(1:n)}$ and $v^{(2:n)}$ denote the highest and second-highest of the n valuations. For simplicity, we define the convention that $v^{(2:1)} = -\infty$, the value of the second highest valuation drawn from a process with two members. We claim

$$S = \mathbb{E}\big[(v^{(2:n)} - R)^{+}\big] \tag{25}$$

In the symmetric equilibrium a party with $v \geq R$ bids $b(v)$ from equation (5) and wins when every rival has a lower valuation or abstains. The allocation is therefore the same as in a second-price auction with reserve R, and the lowest type that bids, $v = R$, earns zero surplus in both formats. By the revenue equivalence theorem (Myerson, 1981; Riley and Samuelson, 1981), the seller's expected revenue is the same in the two formats. In the second-price format the price is $\max\{v^{(2:n)}, R\}$ when $v^{(1:n)} \geq R$, so

$$\mathbb{E}\big[b^{(1)}\, \mathbf{1}\{v^{(1:n)} \geq R\}\big] = \mathbb{E}\big[\max\{v^{(2:n)}, R\}\, \mathbf{1}\{v^{(1:n)} \geq R\}\big].$$

The entrant retains R when no party bids, so

$$\mathbb{E}\big[\max\{b^{(1)}, R\}\big] = R\,\Pr\big(v^{(1:n)} < R\big) + \mathbb{E}\big[\max\{v^{(2:n)}, R\}\,\mathbf{1}\{v^{(1:n)} \geq R\}\big] = R + \mathbb{E}\big[(v^{(2:n)} - R)^+\big],$$

where the last equality uses that $(v^{(2:n)} - R)^+ > 0$ only when $v^{(1:n)} \geq v^{(2:n)} > R$. Subtracting R gives (25). When n is random, as it is when the engaged count and the screen are drawn, the identity holds conditional on n and therefore in expectation.

Step 2: The option value is nondecreasing in the pool at a fixed reserve. Let two pools be given, with engaged counts drawn as in equation (3) and screen passes drawn independently at rate σ. Construct both pools on one probability space by assigning to each incumbent k three independent uniform draws (u_k^e, u_k^s, u_k^v). Incumbent k engages when $u_k^e \leq p$, passes the screen when $u_k^s \leq \sigma$, and draws valuation $F_H^{-1}(u_k^v)$ if a high type and $F_L^{-1}(u_k^v)$ if a low type. We then need two comparisons:

(a) If $\mathcal{K}' \supseteq \mathcal{K}$ with the same types for the common members, the set of parties past the screen under $\mathcal{K}'$ contains the set under $\mathcal{K}$ realization by realization, and the second-highest element of a superset weakly exceeds the second-highest element of the subset.

(b) If $\mathcal{K}'$ and $\mathcal{K}$ have the same members and one member is a high type in $\mathcal{K}'$ and a low type in $\mathcal{K}$, the sets of parties past the screen coincide realization by realization. First-order stochastic dominance gives $F_H^{-1}(u) \geq F_L^{-1}(u)$ for every u, so the switched member's valuation is weakly larger and every other valuation is unchanged.

In both cases $v^{(2)}$ is weakly larger under $\mathcal{K}'$ realization by realization. Since $x \mapsto (x-R)^+$ is nondecreasing, (25) gives $S_{it}(R;\mathcal{K}') \geq S_{it}(R;\mathcal{K})$ for every R.

Step 3: The ordering passes through the Bellman equation. Write the option value as a function of the reserve and the pool, $S_{it}(R;\mathcal{K})$, and define

$$\Phi(R;\mathcal{K}) = R + \big(S_{it}(R;\mathcal{K}) - \kappa\big)^+, \tag{26}$$

so that equation (8) reads $W_{it} = \Phi(R_{it};\mathcal{K}_t)$. By Step 2, Φ is nondecreasing in the pool at every R. Φ is also nondecreasing in R. By (25), $S_{it}(R)$ is nonincreasing in R with slope $-\Pr(v^{(2)} > R) \in [-1, 0]$ almost everywhere, so for $R' \geq R$, $S_{it}(R) - S_{it}(R') \leq R' - R$. Since $x \mapsto (x-\kappa)^+$ is nondecreasing with slope

at most one,

$$\Phi(R') - \Phi(R) = (R' - R) + \left[(S_{it}(R') - \kappa)^+ - (S_{it}(R) - \kappa)^+\right] \geq (R' - R) - \left(S_{it}(R) - S_{it}(R')\right) \geq 0.$$

The result now follows by backward induction on t. At $t = T$, $W_{iT} = \pi_j^{\text{app}}(\theta_{ij})\, V_{ij}^{\text{own}}$ does not depend on the pool. Suppose $W_{i,t+1}$ is nondecreasing in the pool in every state. Then $R_{it} = -c_{it} + \delta\, \mathbb{E}_t[W_{i,t+1}]$ is nondecreasing in the pool, because expectation preserves the ordering. Writing $R'_{it} \geq R_{it}$ for the reserves under $\mathcal{K} \cup \{k\}$ and $\mathcal{K}$,

$$W'_{it} = \Phi(R'_{it}; \mathcal{K} \cup \{k\}) \geq \Phi(R_{it}; \mathcal{K} \cup \{k\}) \geq \Phi(R_{it}; \mathcal{K}) = W_{it},$$

using first that Φ is nondecreasing in R and then that it is nondecreasing in the pool. Hence W_{i0}, and with it $\Pi_{ij} = W_{i0} - F_j$, is nondecreasing in the pool.

For the second statement, compare $\mathcal{K} \cup \{h\}$ with $\mathcal{K} \cup \{l\}$, where h is a high type and l a low type. Step 2(b) gives $S_{it}(R; \mathcal{K} \cup \{h\}) \geq S_{it}(R; \mathcal{K} \cup \{l\})$ for every R and every date, and the same induction yields $\Pi_{ij}(\mathcal{K} \cup \{h\}) \geq \Pi_{ij}(\mathcal{K} \cup \{l\}) \geq \Pi_{ij}(\mathcal{K})$. □

Proposition 4.3. *Consider a restriction that removes $\mathcal{O}_{jt}$ from the pool in market j. Under Assumption 3, the expected number of engaged parties falls by pH_{jt}, and Π_{ij} weakly falls in every market where $H_{jt} > 0$ and is unchanged elsewhere.*

Proof. Under Assumption 3, the engaged count under the restriction is $m_{it}^L \sim \text{Bin}(L_{jt}, p)$, with mean $pL_{jt} = pK_t - pH_{jt}$. Removing the H_{jt} members of $\mathcal{O}_{jt}$ one at a time and applying Proposition 4.2 at each step gives Π_{ij} weakly lower under the restriction. In a market with $H_{jt} = 0$ the restriction removes no one, so Π_{ij} is unchanged. □

Proposition 4.4. *Let $\Pi_{ij}^{auction}$ denote entry value when $\kappa = 0$, so that every asset is marketed at every date. Then $\Pi_{ij}^{auction} \geq \Pi_{ij}$. Holding R_{it} fixed, a restriction that removes $\mathcal{O}_{jt}$ weakly lowers S_{it} at every date, so the set of states in which a process runs weakly shrinks. Under $\kappa = 0$ a process runs in every state, so $\Pi_{ij}^{auction}$ responds to the restriction only through the sale stage, and a counterfactual that holds the initiation factor of (11) at its baseline understates the fall in the acquisition payoff.*

Proof. For the first statement, write Φ_κ for the map in (26) at cost κ, and let W_{it}^0 and R_{it}^0 denote the value and reserve at $\kappa = 0$. At $t = T$ the two models coincide. Suppose $W_{i,t+1}^0 \geq W_{i,t+1}$ in every state. Then

$R^0_{it} \geq R_{it}$, and

$$W^0_{it} = \Phi_0(R^0_{it}) \geq \Phi_0(R_{it}) \geq \Phi_\kappa(R_{it}) = W_{it},$$

where the first inequality uses that Φ_0 is nondecreasing in R (Step 3 of the proof of Proposition 4.2) and the second uses $(S-0)^+ \geq (S-\kappa)^+$. Hence $W^0_{i0} \geq W_{i0}$ and $\Pi^{\text{auction}}_{ij} \geq \Pi_{ij}$.

For the second statement, fix R_{it}. The restriction replaces the pool $\mathcal{K}_t$ by $\mathcal{K}_t \setminus \mathcal{O}_{jt}$, which by Step 2(a) of the proof of Proposition 4.2, applied H_{jt} times, weakly lowers S_{it} in every state. By Proposition 4.1, a process runs when $S_{it} \geq \kappa$, so the set of states in which one runs weakly shrinks, and the first factor of (11) weakly falls. At $\kappa = 0$ the condition $S_{it} \geq 0$ holds in every state before and after the restriction, so that factor equals one in both and $\Pi^{\text{auction}}_{ij}$ responds only through the second factor and the price. A counterfactual that holds the first factor at its baseline therefore attributes to the restriction a weakly smaller fall in π^A_{ij} than the model implies, and with A_{ij} moving in the same direction, a weakly smaller fall in the acquisition term of (13). □

E The Free-Entry Engagement Regime

This appendix derives the replacement ratio for the alternative to Assumption 3 discussed after Proposition 4.3, in which the engagement rate is an equilibrium object rather than the model primitive the estimation carries.

By the envelope theorem, the ex-ante surplus of a low type bidder that incurs the diligence cost is given by

$$\Pi_{it}^{L} \;=\; \mathbb{E}\left[\int_{R_{it}}^{v} F_H(x)^{m_{it}^{H}}\, F_L(x)^{m_{it}^{L}-1}\, dx\right], \qquad v \sim F_L, \tag{27}$$

where the expectation is taken over the value draw and over the engagement counts in (3), and the integrand is the probability that every rival sits below x. Free entry sets $\Pi_{it}^{L} = c$, which determines the rate at which the remaining low types engage.

Removing $\mathcal{O}_{jt}$ sets $m_{it}^{H} = 0$ and raises the ex-ante low type surplus at every x. If we are assuming that the engagement probability adjusts in equilibrium to maintain the same level of surplus, then the equality requires more low types. Equating the integrand before and after gives a replacement ratio

$$\tilde{m}_{it}^{L} - m_{it}^{L} \;=\; m_{it}^{H}\,\lambda(x), \qquad \lambda(x) \;\equiv\; \frac{\ln F_H(x)}{\ln F_L(x)} \;>\; 1, \tag{28}$$

where $\lambda > 1$ because $F_H \succ_{\text{FOSD}} F_L$ implies $F_H(x) < F_L(x)$. Each banned high type must be replaced by more than one low type. In terms of the engagement probability,

$$\tilde{p} \;\approx\; p\left(1 + \frac{H_{jt}}{L_{jt}}\,\bar{\lambda}\right), \tag{29}$$

where $\tilde{p}$ is the post-ban rate, $\bar{\lambda}$ averages $\lambda(x)$ over the relevant range of valuations, and the approximation holds when the rate is small enough that the engagement counts enter the surplus through their means. Under free entry the ban raises the remaining firms' engagement rate by the banned share scaled by the replacement ratio, which is what leaves the engaged count, and with it the participation channel, nearly unchanged.

F Truncation of the Winning-Bid Distribution

This appendix derives the correction referred to in Section 4.2.

We observe a transaction price only when a sale occurred, so the winning bids in our data are draws from the distribution of the maximum bid conditional on that maximum having cleared the reserve. This requires a selection correction term, derived here.

Fix a sale process at date t with m_{it} engaged parties drawing valuations independently from F, and let R_{it} be the reserve price. Let $\tilde{G}$ denote the marginal bid distribution of a representative engaged party, which includes the convention that a party drawing a valuation below R_{it} abstains and is assigned a notional bid below the support, such that it counts toward $\tilde{G}(b)$ at every b. Then

$$\tilde{G}(b) = \begin{cases} F(R_{it}), & b < R_{it}, \\ F\big(v(b)\big), & b \geq R_{it}, \end{cases} \tag{30}$$

where $v(\cdot)$ inverts the bid function:

$$v(b) = b + \frac{\tilde{G}(b)}{(m_{it} - 1)\,\tilde{g}(b)}. \tag{31}$$

$\tilde{G}(b)$ is continuous at the reserve value, where it takes the value $F(R_{it})$ rather than zero. From here, as any draw below R_{it} contributes to $\tilde{G}(b)$ at every b, we then have that $\Pr\left(b_{it}^{(1)} \leq b\right) = \tilde{G}(b)^{m_{it}}$ holds without conditioning on the number of active bidders.

Across all processes, whether or not they produce a sale, the maximum bid satisfies $\Pr\left(b_{it}^{(1)} \leq b\right) = \tilde{G}(b)^{m_{it}}$. A sale occurs when at least one party clears the reserve, which has probability $1 - F(R_{it})^{m_{it}}$ by (7) applied to the parties past the screen, which are the m_{it} parties of this appendix. The distribution we observe is therefore

$$G^{\text{win}}(b) \;=\; \Pr\left(b_{it}^{(1)} \leq b \mid \text{sale}\right) \;=\; \frac{\tilde{G}(b)^{m_{it}} - F(R_{it})^{m_{it}}}{1 - F(R_{it})^{m_{it}}}, \tag{32}$$

and inverting (32) returns the object that (31) needs:

$$\tilde{G}(b) \;=\; \Big[\, F(R_{it})^{m_{it}} \;+\; \big(1 - F(R_{it})^{m_{it}}\big)\, G^{\text{win}}(b) \,\Big]^{1/m_{it}}. \tag{33}$$

Equation (33) nests the uncorrected case. Setting $F(R_{it})^{m_{it}} = 0$ returns $\tilde{G}(b) = \big[G^{\text{win}}(b)\big]^{1/m_{it}}$, the maximum order statistic inversion applied directly to winning bids, which is the estimator that ignores the reserve value. The correction is the additive term inside the bracket, and its size is the probability that a process yields no sale.

Table 17 reports the size of the discrepancy at the median of the observed winning-bid distribution, setting $F(R_{it}) = 0.6$ for illustration. The correction is invisible when the engaged pool is large and substantial at the pool sizes the merger filings indicate.

Table 17: Corrected and uncorrected $\tilde{G}$ at $G^{\text{win}} = 0.5$

	$m_{it} = 2$	$m_{it} = 5$	$m_{it} = 20$
No-sale probability $F(R_{it})^{m_{it}}$	0.360	0.078	0.00004
Uncorrected $\big[G^{\text{win}}\big]^{1/m_{it}}$	0.707	0.871	0.966
Corrected, equation (33)	0.825	0.884	0.966

Current implementation. Applying (33) requires $F(R_{it})^{m_{it}}$, the probability that a process yields no sale. We are unable to estimate F and R_{it} separately in order to compute it, since F is the object being recovered. The current implementation instead treats the no-sale rate as a model primitive. The merger filings that supply the engagement probabilities in Section 5.4 also record sale processes that ran and produced no transaction, which could microfound it, but doing so asks a great deal of the filing data and requires assumptions we are not prepared to defend.

Equation (33) corrects for processes that ran and failed to clear the reserve. It does not address which compounds were marketed at all, which is the initiation margin of Section 4.3, nor which completed transactions enter our data. Those are separate selections and we treat them separately.

G Revenue at Risk: Timing, Placebo and Estimator Checks

This appendix holds three checks on the revenue-at-risk results of Section 5.2. The first two ask whether the acquisition response has the timing that anticipation implies. The third asks whether the count model is doing the work.

Timing. Timing is consistent with anticipation without pinning it down. In a revenue-weighted event study that places each drug's pre-cliff sales in a window relative to the current quarter, the acquisition response is positive through the last eight quarters before exclusivity ends, 0.248 with a standard error of 0.236 at five to eight quarters out and 0.287 with a standard error of 0.290 at one to four, and 0.265 with a standard error of 0.207 when the two windows are pooled. No individual window is significant and we cannot reject that the response before the cliff equals the response after, $p = 0.47$. We read the profile as showing a response concentrated in the two years before the cliff, and we do not date it more finely.

Table 18: Acquisitions and entry by window relative to the cliff

	(1) Acquisitions	(2) Small-firm entry	(3) Incumbent entry
Pre-cliff sales of drugs losing exclusivity, by window (USD bn)			
25 to 32 quarters before	-0.811***	0.027	0.201
	(0.200)	(0.128)	(0.261)
17 to 24 quarters before	0.172	-0.159	0.204
	(0.410)	(0.270)	(0.175)
9 to 16 quarters before	-0.765**	0.151	-0.162
	(0.362)	(0.232)	(0.178)
1 to 8 quarters before	0.265	0.281	-0.223
	(0.207)	(0.191)	(0.170)
0 to 7 quarters after	0.133	0.007	-0.148
	(0.278)	(0.258)	(0.185)
8 to 15 quarters after	-0.262	0.170	-0.799***
	(0.382)	(0.220)	(0.247)
Mean of dependent variable	0.170	0.541	0.296
Observations	1,400	1,400	1,400
Year FE	Yes	Yes	Yes
Market FE	Yes	Yes	Yes

[1] As in the twelve-window table, with windows pooled to eight quarters. The 1 to 8 quarters window is the exposure measure of the main table.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Revenue that has already lost exclusivity. One placebo supports the anticipation reading. Revenue that has already lost exclusivity does not move acquisitions, 0.240 with a standard error of 0.361, while revenue still at risk in the same specification carries 0.456 with a standard error of 0.174. Incumbents buy ahead of

the cliff and not after it, which is what franchise defense predicts and a firm-wide cash motive does not.

Table 19: Revenue still at risk against revenue that has already lost exclusivity

	(1) Acquisitions	(2) Small-firm entry	(3) Incumbent entry
Revenue at risk, 8 quarters (USD bn)	0.456***	0.221***	-0.132
	(0.174)	(0.073)	(0.169)
Revenue that lost exclusivity in the past 8 quarters (USD bn)	0.240	-0.050	0.076
	(0.361)	(0.212)	(0.201)
Controls			
Market sales (USD bn)	-0.074	-0.041	0.175
	(0.188)	(0.081)	(0.132)
Likelihood of approval	0.005	-0.115	0.354**
	(0.176)	(0.101)	(0.145)
Mean of dependent variable	0.170	0.541	0.296
Observations	1,400	1,400	1,400
Year FE	Yes	Yes	Yes
Market FE	Yes	Yes	Yes
p-value, before = after	0.471	0.129	0.121

[1] Parent indication by quarter, 2006q1 to 2019q4, 25 markets. Poisson regressions; coefficients are the proportional change in the rate per USD billion at risk. Standard errors clustered by market in parentheses. Controls are market sales (USD bn) and the likelihood of approval.
[2] The second regressor is the sales (measured over the four quarters before the end of exclusivity) of drugs in the market whose exclusivity ended within the past 8 quarters. Column (1): incumbent acquisitions of development-stage drugs. Column (2): small-firm first Phase II trials naming fewer than four markets. Column (3): incumbent first Phase II trials.
$^{*}\ p < 0.10,\ ^{**}\ p < 0.05,\ ^{***}\ p < 0.01$.

Horizon. Table 20 re-estimates the main specification with revenue at risk redefined over twelve and sixteen quarters ahead in place of the headline eight, for both outcomes. The acquisition response concentrates at the short horizon, 0.394 at eight quarters against 0.079 at twelve, which is what a dated cliff implies: an incumbent buys when the expiry is near, and sales expiring three or four years out enter the longer-horizon measures while carrying little of the response. Entry responds over a longer window, still 0.160 at twelve quarters before fading by sixteen, consistent with entrants positioning earlier in the cliff's approach than acquirers transact.

Estimator. Tables 21 and 22 compare least squares, Poisson and a zero-inflated negative binomial on the main specification. The three agree. The zeros in these data are what a count model with this mean produces rather than excess mass: the fitted Poisson implies 1,201 zero market-quarters for acquisitions against 1,211 observed, and 786 against 775 for entry. The inflation equation is accordingly close to degenerate, and for entry it returns the Poisson coefficient to three decimals under every inflation specification we tried. We report the model because it is familiar, not because the data call for it.

Table 20: Revenue at risk at eight, twelve and sixteen quarters

	Incumbent acquisitions			Small-firm entry		
	(1)	(2)	(3)	(4)	(5)	(6)
Revenue at risk, 8 quarters (USD bn)	0.394***			0.179**		
	(0.136)			(0.074)		
Revenue at risk, 12 quarters (USD bn)		0.079			0.160***	
		(0.254)			(0.060)	
Revenue at risk, 16 quarters (USD bn)			-0.153			0.078
			(0.165)			(0.086)
Controls						
Market sales (USD bn)	-0.073	-0.072	-0.012	-0.045	-0.065	-0.054
	(0.182)	(0.196)	(0.189)	(0.116)	(0.124)	(0.116)
Likelihood of approval	0.011	0.030	0.043	-0.116	-0.116	-0.111
	(0.176)	(0.175)	(0.182)	(0.104)	(0.102)	(0.102)
Mean of dependent variable	0.170	0.170	0.170	0.688	0.688	0.688
Observations	1,400	1,400	1,400	1,400	1,400	1,400

[1] Parent indication by quarter, 2006q1 to 2019q4, 25 markets. Poisson regressions; coefficients are the proportional change in the rate per USD billion at risk. Standard errors clustered by market in parentheses. Controls are market sales and the likelihood of approval; all columns carry year and market fixed effects, the main specification of the acquisitions and entry tables.

[2] Each column redefines revenue at risk as the sales of approved drugs in the market whose exclusivity ends within the stated number of quarters, in USD billions per quarter. The headline tables use eight quarters.

$^{*}\ p < 0.10,\ ^{**}\ p < 0.05,\ ^{***}\ p < 0.01.$

Table 21: Estimator comparison: incumbent acquisitions

	Incumbent acquisitions of development-stage drugs		
	(1)	(2)	(3)
Revenue at risk, 8 quarters (USD bn)	0.091**	0.394***	0.362***
	(0.039)	(0.136)	(0.138)
Controls			
Market sales (USD bn)	-0.016	-0.073	-0.066
	(0.048)	(0.182)	(0.166)
Likelihood of approval	-0.026	0.011	-0.029
	(0.021)	(0.176)	(0.177)
Estimator	OLS	Poisson	ZINB
Inflation equation	–	–	Market sales
Overdispersion α			0.000
Mean of dependent variable	0.170	0.170	0.170
Observations	1,400	1,400	1,400

[1] Parent indication by quarter, 2006q1 to 2019q4, 25 markets. Every column has year and market fixed effects and controls for market sales (USD bn) and the likelihood of approval. Standard errors clustered by market in parentheses. The dependent variable counts acquisitions by a Big Pharma or Large Biotech incumbent of drugs in Phase II, Phase III or NDA review in the market that quarter.

[2] Column (1) is ordinary least squares in levels, so its coefficient is events per USD billion at risk. Columns (2) and (3) are multiplicative, so their coefficients are proportional changes in the count. The zero-inflated negative binomial in column (3) reports the count equation; its inflation equation is a logit on market sales, and the coefficient is conditional on the market-quarter not being in the always-zero state rather than being the unconditional mean the other columns estimate.

[3] The counts are not zero inflated in the sense the third model is built for: the Poisson fit in column (2) implies 1,192 market-quarters with no acquisition against 1,198 observed, and 785 with no entry against 775 observed. The zero-inflated model also fails to converge under several inflation specifications, including revenue at risk in the inflation equation; the variants attempted are reported in the log.

$^{*}\ p < 0.10,\ ^{**}\ p < 0.05,\ ^{***}\ p < 0.01.$

Table 22: Estimator comparison: small-firm first Phase II trials

	Small-firm first Phase II trials		
	(1)	(2)	(3)
Revenue at risk, 8 quarters (USD bn)	0.380***	0.179**	0.179**
	(0.109)	(0.074)	(0.074)
Controls			
Market sales (USD bn)	0.070	-0.045	-0.045
	(0.160)	(0.116)	(0.116)
Likelihood of approval	-0.041	-0.116	-0.116
	(0.040)	(0.104)	(0.104)
Estimator	OLS	Poisson	ZINB
Inflation equation	–	–	Market sales
Overdispersion α			0.000
Mean of dependent variable	0.688	0.688	0.688
Observations	1,400	1,400	1,400

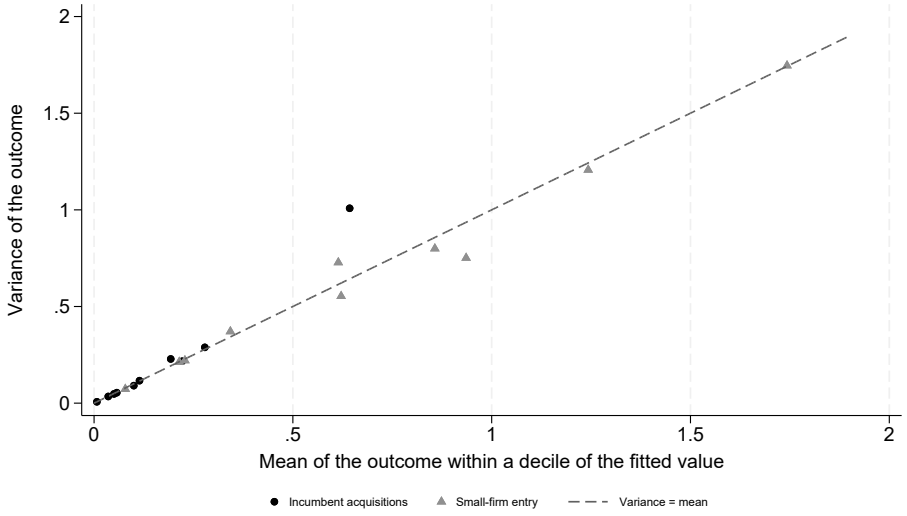
[1] Parent indication by quarter, 2006q1 to 2019q4, 25 markets. Every column has year and market fixed effects and controls for market sales (USD bn) and the likelihood of approval. Standard errors clustered by market in parentheses. The dependent variable counts first Phase II trials by firms that are neither Big Pharma nor Large Biotech, counting each market the trial names.

[2] Column (1) is ordinary least squares in levels, so its coefficient is events per USD billion at risk. Columns (2) and (3) are multiplicative, so their coefficients are proportional changes in the count. The zero-inflated negative binomial in column (3) reports the count equation; its inflation equation is a logit on market sales, and the coefficient is conditional on the market-quarter not being in the always-zero state rather than being the unconditional mean the other columns estimate.

[3] The counts are not zero inflated in the sense the third model is built for: the Poisson fit in column (2) implies 1,192 market-quarters with no acquisition against 1,198 observed, and 785 with no entry against 775 observed. The zero-inflated model also fails to converge under several inflation specifications, including revenue at risk in the inflation equation; the variants attempted are reported in the log.

$^{*}\ p < 0.10,\ ^{**}\ p < 0.05,\ ^{***}\ p < 0.01.$

Figure 12: The variance of each count tracks its mean



Notes: Market-quarters are sorted into deciles of the fitted value from the Poisson regression in column (2) of Table 3 (and its entry counterpart), and the figure plots the variance of the outcome within each decile against its mean. The dashed line is variance equal to mean. Ordinary least squares in levels assumes instead that the variance is constant, and it places 193 of 1,400 fitted acquisition counts and 49 of 1,400 fitted entry counts below zero, which a counterfactual that lowers revenue at risk makes more frequent. Least-squares and Poisson fitted values are close in sample; the case for the count model is the estimand and the zero bound, not the in-sample fit.

The firm level. Table 23 reports the firm-level regressions referenced in Section 5.2. An incumbent's own revenue at risk does not predict its own acquisitions at any of the four horizons, nor when the regressor enters in levels rather than as a share of the firm's oncology sales, nor for control-transferring alliances. Estimating the level specification at the longer horizons, which the table does not display, gives the least precise estimates of the set, 0.040 with a robust standard error of 0.026 at twenty quarters and 0.027 with 0.019 at thirty-two. The firm-level null is therefore a failure to detect a response rather than evidence that none exists. We read it alongside the market-level estimates rather than as a separate result, since the panel holds about 20 incumbents and a firm-level design has little power at that sample size.

Table 23: Incumbent acquisitions and the firm's own revenue at risk

	M&A					Alliances
	(1) $h = 8$	(2) $h = 12$	(3) $h = 20$	(4) $h = 32$	(5) $h = 8$	(6) $h = 8$
Share of own oncology sales at risk	0.014					0.000
	(0.056)					(0.052)
Share of own oncology sales at risk		-0.025				
		(0.042)				
Share of own oncology sales at risk			0.017			
			(0.036)			
Share of own oncology sales at risk				0.005		
				(0.028)		
Own revenue at risk (USD bn)					0.016	
					(0.028)	
Mean of dependent variable	0.040	0.040	0.040	0.040	0.040	0.039
Observations	1,419	1,419	1,419	1,419	1,419	1,419

[1] Incumbent by quarter, firm and year fixed effects. The dependent variable counts the incumbent's acquisitions of development-stage drugs (columns 1 to 5, M&A; column 6, control-transferring alliances). Heteroskedasticity-robust standard errors in parentheses. We report robust rather than cluster-by-firm standard errors because the panel holds about 20 incumbents, and at that number of clusters the cluster-robust estimator is downward biased: on the level regressor in column (5) it returns 0.008 against a robust 0.028 on the same coefficient. The inference in the share specifications does not turn on the choice, as each is far from significance under either estimator.

[2] The regressor is the share of the incumbent's own oncology sales from approved drugs whose exclusivity ends within h quarters; column (5) uses the level in USD billions.

$^{*}\ p < 0.10,\ ^{**}\ p < 0.05,\ ^{***}\ p < 0.01.$

H The Competitor-Approval Event Study in Levels

Table 24 reports the stacked event study of Section 5.5 with the outcomes in levels: dollars of attributed sales, patient-months, and the share of the cancer's treated patients in points. The stacks, window, fixed effects and clustering are those of Table 7. The levels estimates agree with the log version in sign and in implied magnitude once each coefficient is set against the treated pre-period mean, and they are less precise for sales and the point share, because a levels regression weights the largest incumbents. The share of patients is the object the self-commercialization payoff uses, and it is proportional by construction, which is why the text reports the log and share forms.

Table 24: The competitor-approval event study in levels

	Sales in c (USD m)		Patient-months in c		Share of c	
	(1)	(2)	(3)	(4)	(5)	(6)
Post x treated	-21.984	-8.339	-304.670***	-77.305*	-0.004	-0.003
	(15.123)	(5.236)	(110.624)	(46.052)	(0.005)	(0.002)
Control cancers	clean	all	clean	all	clean	all
Treated pre-period mean	101.64	101.64	2,206.59	2,206.59	0.06	0.06
Observations	31,974	155,156	31,970	155,143	31,970	155,143
Clusters	81	84	80	83	80	83

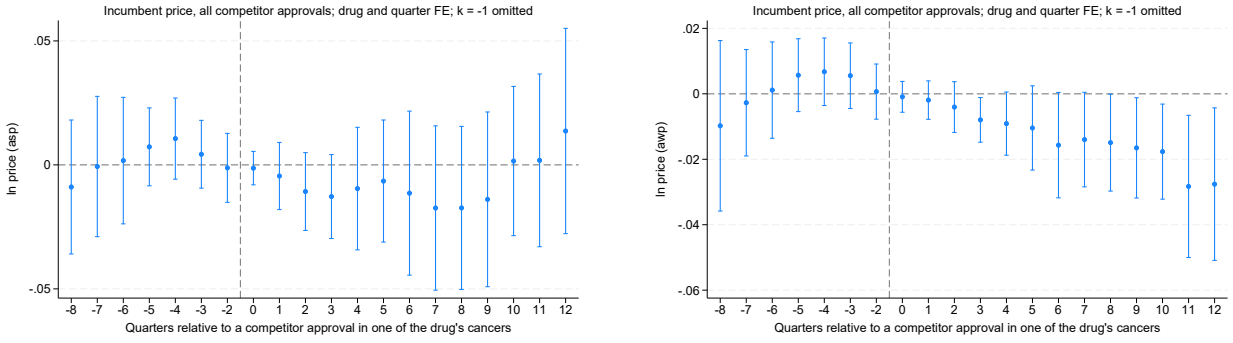
The competitor-approval event study of Section 5.5 with the outcomes in levels rather than logs: US sales attributed to the cancer in millions of 2015q1 dollars, patient-months among the cancer's patients, and the drug's share of the cancer's treated patients. Same stacks, window, fixed effects and clustering as the log version. The treated pre-period mean converts each coefficient to an implied percent. * p<0.10, ** p<0.05, *** p<0.01.

I Incumbent Prices and the Competitor Count

This appendix reports the price evidence behind the choice in Section 5.5 to summarize competition among approved drugs with a competitor count. Two price series enter. For infused and injected drugs, which carry most oncology revenue, we use the CMS Part B average sales price (ASP), the manufacturer's average price to all US purchasers net of most concessions, quarterly with a two-quarter lag, one HCPCS code per molecule, net of the statutory add-on, covering 56 drugs from 2009q1 to 2019q4. For branded oral drugs we use RED BOOK average wholesale price (AWP), a list price chained across NDCs, covering 55 drugs. Under Medicare Part B, providers are reimbursed ASP plus six percent, and prior work shows that privately negotiated physician services are set as a percent markup from ASP (Clemens and Gottlieb, 2017).

Figure 13 plots the distributed-lag coefficients of log price on the number of competitor approvals k quarters earlier in any cancer the drug is approved in. The unit is the drug rather than the cancer, as there is no indication-specific pricing in the US. Averaged over the twelve quarters after an approval, the ASP response is -0.007 and statistically indistinguishable from zero. Oral list prices keep rising throughout, about one point a year more slowly than they otherwise would.

Figure 13: Incumbent prices around a competitor's approval



(a) Infused drugs: average sales price (ASP)

(b) Oral drugs: list price (AWP)

Notes: Distributed-lag coefficients on the number of competitor approvals k quarters earlier in any cancer the drug is approved in, $k = -8$ to 12 with bins beyond, $k = -1$ omitted; drug and quarter fixed effects; standard errors clustered by drug. Left: CMS Part B average sales price per dosage unit, net of the statutory add-on, 56 infused and injected drugs, 2009q1 to 2019q4. Right: RED BOOK average wholesale price per unit for branded oral drugs, 55 drugs, 2009q1 to 2019q4. Bars are 95 percent intervals.

Table 25 reports the cross-sectional counterpart, incumbent revenue and volume on the number of approved competitors in the cancer. The competitor count is imprecise once indication effects absorb the cross-section, which is one reason the entrant's share comes from the event study rather than from these regressions (Appendix J.4).

Table 25: Incumbent revenue and volume on the number of approved competitors

	ln sales in c		ln patient-months in c			share of cancer c		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Competitors approved in c	-0.010	-0.010	-0.009	-0.009	-0.008	-0.001	-0.001	-0.000
	(0.019)	(0.020)	(0.014)	(0.013)	(0.020)	(0.001)	(0.001)	(0.001)
Pair and quarter FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Age-in-cancer FE	No	Yes	No	Yes	No	No	Yes	No
Drug $imes$ quarter FE	No	No	No	No	Yes	No	No	Yes
Observations	3,664	3,664	3,701	3,701	2,233	3,748	3,748	2,259
Drugs	103	103	101	101	37	101	101	38

Competitors approved in c = running count of other drugs with an FDA approval in the same cancer, excluding drugs sold by a company that also sells the focal drug. Sample: drug x cancer pairs where the focal drug is approved in c and selling, 2009q1-2019q4. Sales in c = US sales x Marketscan share of the drug's patient-months in cancer c; patient-months and share of cancer c are Marketscan counts. Age = quarters since the focal drug was approved in c and selling. Drug x quarter fixed effects use drugs approved in two or more cancers. SEs clustered by drug. * p<0.10, ** p<0.05, *** p<0.01.

J Measurement Detail for the Estimated Model

This appendix collects the construction of the objects Section 6 takes to the data: match quality and the sample rules behind the choice set, the bid index, the hedonic price, and the alternatives carried for the entrant's share. The robustness and diagnostic exercises the results reference are in Appendix K.

J.1 Match Quality

Match quality q_{ij} is built from three nested classifications of a compound's mechanism, from coarse to fine: the EphMRA ATC level-4 class, 72 classes defined by modality and target family, such as "protein kinase inhibitor, ALK" or "monoclonal antibody, CD20"; the target family from Evaluate's mechanism-of-action fields, 404 values; and the specific action, 584 values over 2,410 matched compounds. At each tier the measure is the leave-one-out share of same-class compounds in the full development panel, all firms, with a Phase II or later record in j, shrunk toward the tier above it. For a class c at any tier,

$$q_{ij}^{c} = \frac{n_{cj}^{-i} + \alpha\, q_{ij}^{\text{parent}}}{n_{c}^{-i} + \alpha}, \tag{34}$$

with prior weight $\alpha = 5$, where the parent of the specific action is the target family, the parent of the target family is the ATC class, and the parent of the ATC class is $\bar{\pi}_j$, the share of all compounds developed in j. The headline q_{ij} is the specific-action tier. Most specific actions are held by a handful of compounds, so the shrinkage matters, and a compound whose exact mechanism is rare borrows first from mechanisms aimed at the same target and then from its broad class rather than standing on its own few observations. The ATC-class tier alone, the coarsest measure with any mechanism content, is the alternative reported beside it in Section 7.3.

Both pieces of the construction are standard. The leave-one-out share is a conditional co-occurrence probability on a class-by-indication incidence matrix, the quantity the relatedness literature calls proximity and uses to predict what a unit does next (Hidalgo et al., 2007). The back-off in (34) is hierarchical shrinkage of a noisy fine cell toward the coarser cell that contains it, as in the county and commuting-zone estimates of Chetty and Hendren (2018). What is new here is the application to the class-by-indication network rather than either component.

The recursion is also what keeps a first-in-class compound in the estimation. A compound whose mech-

anism is genuinely new has no class-mates to average over, and a measure defined as a share of same-class compounds would be undefined or zero for it in every market, which would read as scientific infeasibility everywhere rather than as absence of evidence. Equation (34) replaces each tier by its parent when the class is empty or holds one compound, so such a compound inherits the profile of its target family, then of its ATC class, and reaches $\bar{\pi}_j$ only when all three are uninformative. The measure then returns the indication's own development rate, which is the correct prior for a compound about which the class structure says nothing. Of the 2,356 compounds in the reference panel, 1,834 resolve at the specific-action tier, 242 at the target family, 164 at the ATC class, and 116 at the base rate. In the estimation sample the corresponding counts are 866, 130 and 98 of 1,195 cases, with 37 at the base rate and a further 64 compounds that fall outside the reference panel and take $\bar{\pi}_j$ as well. For those 101 cases q_{ij} varies across indications only through $\bar{\pi}_j$, which the alternative constants absorb, so they inform the coefficient through the decision of whether to enter at all rather than through the choice among markets. Wu and Ono (2025), estimating a conditional logit over cancer indications, instead drop compounds whose mechanism has only one observed indication. The recursion keeps them, at the cost of a measure that carries no information about where they go.

J.2 The Choice Set and the Treatment of Basket Trials

A first Phase II trial that names k indications is k entries, one per named indication, each in full. There is no solid-tumor approval to enter, so a compound trialing in three cancers has entered three markets, and in the estimation a basket trial is k cases, each facing the full choice set. The registry placeholders are not markets and leave the estimation. The parent category "hematologic cancer", which a trial names when it names no specific blood cancer, has nothing approved for it and no revenue, so the payoff terms built from approved drugs do not exist there; the same holds for "other solid tumors". For the latter the ClinicalTrials.gov record of the dating trial often names the cancers the basket enrolled, and the entry is assigned to them, which places 27 of 57 placeholder entries. The 30 the registry cannot place are dropped, as are the eight compounds whose trials name more than four markets, tumor-agnostic baskets whose market choice the record cannot pin down. The choice set holds the 27 remaining indications and the outside option.

J.3 The Bid Index and the Hedonic Price

The bid index regresses the surplus bid on the drug's phase, the deal's form (a whole-company indicator), rights and royalty terms, orphan status, an indicator that the compound's analyst likelihood of approval ever sat above its indication-year average, the market's sales and pipeline, and year and indication effects. The acquirer's own position in the market is excluded, because it is the margin the counterfactual acts on.

The hedonic price A_{jt} regresses total potential deal value, the upfront payment plus development and sales milestones at their nominal amounts for a license and the company deal value for a whole-company acquisition, on every priced deal in the Section 3.3 sample, against the pool mix, market sales, the deal's form, phase, rights and royalty terms, orphan status and drug count, and a year factor in three-year bins. It is predicted at each indication-year's pool mix and market sales with the deal characteristics at their sample means. The bins replace free year effects, which with about fifteen priced deals a year put sampling noise into the price. The milestones are not risk-adjusted at the deal level, because the data do not break them down by trigger, and Section 7.2 reports the risk-adjusted price that splits the difference.

How the Two Components Load on the Pool Mix

Tables 26 and 27 show how the two components load on the pool mix. The hazard rises with the high-type share: 81 percent higher per unit of H/K with year effects, with a standard error of 0.28 on the log rate, and it roughly doubles under indication effects. The incumbents' revenue at risk, which raises the number of acquisitions in an indication in Section 5.2, does not raise the rate per compound, -0.15 per \$1 billion with a standard error of 0.22. It is correlated with the stock of small-firm compounds in the indication rather than with the hazard, so the count response of Section 5.2 runs through the number of compounds at risk rather than through the chance that any one of them is bought. The price does not load on the pool mix at all once the whole-company deals are in the sample: \$104 million per unit of H/K on total deal value with year bins, with a standard error of \$269 million, and \$609 million with indication effects, with a standard error of \$721 million, with the upfront columns equally uninformative. Indication and year constants still explain 88 percent of the fitted hazard, so most of the level is cross-sectional. Both facts are what the design anticipates. These regressions measure the payoff's levels, and it is the auction, not the hedonic, that supplies the price response to the pool in the counterfactual.

Table 26: The acquisition hazard and the high-type share of the pool

	(1) Year FE	(2) + indication FE
High-type share of the pool H/K	0.813^{***}	1.353^{***}
	(0.280)	(0.517)
Market revenue (USD bn per quarter)	0.054	0.030
	(0.052)	(0.085)
Incumbents' revenue at risk (USD bn)	-0.149	-0.056
	(0.220)	(0.178)
Indication-years	592	592
Log likelihood	-546.9	-526.6

Poisson regression of incumbent acquisitions of small-firm Phase I to NDA compounds on the high-type share of the pool, market revenue and the incumbents' revenue at risk (sales on products within eight quarters of patent expiry), exposure the mean stock of such compounds in the indication-year. Coefficients are proportional changes in the hazard. Standard errors clustered by indication.
$^{*}\ p < 0.10,\ ^{**}\ p < 0.05,\ ^{***}\ p < 0.01$.

Table 27: The hedonic price and the high-type share of the pool: total deal value and upfront payment

	Total deal value		Upfront payment	
	(1) Year bins	(2) + indication FE	(3) Year bins	(4) + indication FE
High-type share of the pool H/K	103.62	609.04	-11.02	-52.52
	(269.11)	(720.72)	(52.59)	(138.37)
Market sales (USD bn per quarter)	32.22	38.29	9.44	11.37
	(25.36)	(32.39)	(6.18)	(7.10)
Priced deals	337	337	246	246
R-squared	0.114	0.190	0.261	0.375

Total potential deal value (upfront plus milestones, nominal; the company deal value for whole-company deals) and the upfront payment, USD m 2015q1, on every priced deal 2006 to 2019: exclusive licenses plus the whole-company deals whose price can be tied to one compound. Controls: a whole-company indicator, phase at the deal, worldwide rights, royalty in the deal, orphan status, drugs in the deal, and a year factor in three-year bins. The acquirer's own position in the market is excluded by design. Standard errors clustered by indication.
$^{*}\ p < 0.10,\ ^{**}\ p < 0.05,\ ^{***}\ p < 0.01$.

J.4 Alternatives for the Entrant's Share

The entrant's share s in equation (23) comes from the stacked event study of Section 5.5. Two alternatives are carried through the results. The clean-control estimate, 10.6 percent, uses only the events whose control cancers had no approval of their own within four quarters. The equal-share benchmark sets $s = 1/(n_{jt} + 1)$ for a market with n_{jt} approved products. A cross-sectional alternative, regressing an incumbent's quarterly US sales in a cancer on the number of approved competitors there, is not used for s: the competitor count is imprecise once indication effects absorb the cross-section, and it answers a different question than the one the entry decision asks, which is what one more approved product takes from the market (Section 5.5).

K Entry and Auction Robustness

This appendix holds the robustness and diagnostic exercises behind the estimation results of Section 7: the entry model estimated on entrants alone and on aggregate shares, the variation the entry coefficients require, the simulation of the counterfactual ratios, the alternative bid variables with the paired prices from the filings, the slack in the measured reserve, and what the measured hazard does and does not separate.

K.1 Entry Estimated on Entrants Alone

Table 28 drops the outside option from the choice set, so the coefficient governs where a compound enters conditional on entering rather than both margins at once. Column 1 is the headline match-quality measure, which gives 0.037 with a standard error of 0.010 on this sample against 0.032 in the body, so conditioning on entry neither creates nor removes the response. Column 2 replaces the specific-action tier with the coarse EphMRA ATC level-4 class and finds 0.013 with a standard error of 0.007, the same sign at a third the size, which is the attenuation a blunt feasibility measure produces when one class lumps together mechanisms whose feasible indications differ. Column 3 prices the acquisition at the upfront payment and finds 0.22 with a standard error of 0.06, larger because the upfront-based payoff is about a sixth of the total-value payoff in level.

K.2 Aggregate-Share Estimators

Table 29 reports the aggregate-share estimators on 392 indication-years, 2006 to 2019. The coefficient on the acquisition payoff is small and imprecise in every column: -0.0002 per \$1 million of expected payoff by Poisson with year effects and -0.004 with indication effects, with standard errors of 0.002 and 0.005, and 0.001 and 0.000 in the two log-odds columns, with standard errors of 0.002 and 0.005. Crowding carries the cross-indication pattern, 0.78 log points per log unit without indication effects and 0.48 with, and promise loads only without indication effects. At the indication-year level the payoff varies mostly with the constants, and the aggregate estimators have little variation left to work with.

K.3 What the Entry Coefficients Require

Two assumptions make the variation of Section 6.5 informative about λ_1. The first is that a compound's mechanism-specific feasibility affects where it enters only through the payoff terms it scales and through

Table 28: Entry at the compound level, entrants only

	(1) Entrants, q	(2) Entrants, q^{ATC}	(3) Entrants, upfront
$q\times$ acquisition payoff	0.0366***		
	(0.0100)		
$q\times$ own payoff	0.0049		0.0030
	(0.0048)		(0.0057)
$\ln q$	0.3311***		0.3436***
	(0.0611)		(0.0559)
Crowding (log)	-0.0771	0.0134	-0.0507
	(0.1454)	(0.1378)	(0.1506)
Promise (log)	-0.0033	-0.0003	-0.0039
	(0.1103)	(0.1063)	(0.1120)
$q^{ATC}\times$ acquisition payoff		0.0126*	
		(0.0071)	
$q^{ATC}\times$ own payoff		0.0094	
		(0.0076)	
$\ln q^{ATC}$		0.7384***	
		(0.1233)	
$q\times$ acq. payoff, upfront			0.2177***
			(0.0584)
Cases	633	633	633
Choice rows	17,091	17,091	17,091
Clusters (years)	24	24	24
Log likelihood	-1835.0	-1874.5	-1837.4

Market choice among entrants, the headline entry model with the outside option dropped from the choice set, so the coefficient governs where a compound enters conditional on entering. Column 1 is the headline match-quality measure, column 2 the coarse EphMRA ATC level-4 class alone, and column 3 prices the acquisition at the upfront payment instead of total deal value. Standard errors clustered by decision year. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 29: Entry on indication-year shares

	Entrants: Poisson count		Log odds vs no entry, to 2015	
	(1) Year FE	(2) + indication FE	(3) Year FE	(4) + indication FE
Acquisition payoff Π^A (USD m)	-0.0002	-0.0035	0.0017	0.0004
	(0.0023)	(0.0049)	(0.0022)	(0.0046)
Own payoff $\pi^{app}V^{own}$ (USD m)	0.0009	0.0001	0.0006	0.0009
	(0.0007)	(0.0020)	(0.0010)	(0.0018)
Crowding (log)	0.7860***	0.3834	0.5166***	0.1457
	(0.0659)	(0.2382)	(0.0475)	(0.2977)
Promise (log)	0.1473*	0.0428	0.0827	-0.0168
	(0.0845)	(0.0831)	(0.0707)	(0.0848)
Indication-years	378	378	270	270
Log likelihood	-503.4	-489.9	-257.6	-240.9
R-squared			0.593	0.640

Columns 1 and 2: Poisson regression of first Phase II trials by small firms on the indication-year payoff terms, which is the multinomial logit over indications on aggregate counts; year effects absorb the denominator. Columns 3 and 4: log odds of entering the indication relative to no entry, Laplace-smoothed, decision years 2006 to 2015 so that the outside option is not right-censored. The acquisition payoff is the discounted probability of acquisition over the program's horizon times the hedonic total deal value (USD m); the own payoff is the Phase II approval probability times V^{own} (USD m). Crowding is log(1 + small-firm Phase II to NDA programs in the indication the prior year); promise is log(1 + non-industry trials started in the prior year). Standard errors clustered by indication (29).

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

$\ln q_{ij}$, so that feasibility carries no direct taste for an indication once those are in the model. The second is that the composition of the acquirer pool affects entry only through the acquisition payoff, conditional on the indication's scientific promise and the alternative constants.

The second is the more demanding of the two, because a market where many incumbents hold overlapping positions may be a market that independently looks promising to an entrant. Three features of the specification speak to it. The promise control measures non-industry trial activity in the indication directly. The aggregate-share estimators, reported with and without indication effects, bracket λ_1 from the pool-mix side. And the compound-level model identifies the coefficient from match quality, which the pool mix does not drive. What would settle the question is an instrument for the payoff itself, and the conclusion states the plan.

One restriction of the functional form deserves the same visibility. The outside option sits in the same logit as the 27 markets, so the counterfactual's aggregate-entry effect is the entry coefficient passed through the logit's substitution structure, not a separately estimated response. A nested logit would loosen that link. We estimated one with hematologic and solid-tumor nests, and the dissimilarity parameters had nothing identifying them once the alternative constants were in. The plain logit stays, and the aggregate-entry effect of Section 7.4 carries its restriction that a compound easily priced out of its best market is also a compound easily priced out of development.

K.4 Simulating the Counterfactual Ratios

The two ratios of equation (22) are simulated indication-year by indication-year. Each draw takes an engaged count from equation (17), from the full pool K_t in the baseline and from the permitted pool L_{jt} under the ban. Each engaged party is then a high type with probability H_{jt}/K_t in the baseline, and every engaged party is a low type under the ban, with valuations drawn from the recovered F_H and F_L.

Within a process, each engaged party passes the diligence screen independently, and competition is the equilibrium bid among the screen-passers facing the bid-stage share F. A party alone past the screen is not told so and shades against a phantom rival rather than dropping to the reserve (Section 4.2). The simulated price is the winning bid returned to dollars, the reserve plus the fitted surplus plus the equilibrium bid on the residual scale. A sale requires a party to pass both walk-away margins, which happens with probability $1-0.65^m$ in a process with m engaged parties, and that probability is the denominator of η^{π}_{jt} in the baseline and its numerator under the ban.

Table 30 reports the composition effect summarized in Section 7.2, holding the engaged count fixed and drawing every party from the low-type distribution instead of the sample's mix.

Table 30: The composition effect at a fixed engaged count, bid-stage share $F = 0.35$

	Expected winning bid (USD m)			Expected maximum valuation (USD m)		
Engaged parties m	Baseline	Restricted	Change (%)	Baseline	Restricted	Change (%)
2	631.4	543.6	-13.9	2463.9	2078.8	-15.6
3	891.2	727.3	-18.4	3013.7	2546.1	-15.5
5	1373.6	1086.0	-20.9	4041.0	3419.0	-15.4

Notes: Baseline draws m engaged parties from the inversion sample's type mix (81 percent high type); restricted draws all m from the low-type distribution. Prices are expected winning bids from the symmetric first-price equilibrium at the level reserve + fitted surplus + residual-scale bid, in 2015q1 $m. The expected maximum valuation is shown for comparison and is not a price.

Table 31 reports the ban's auction-stage effect across the walk-away grid of Section 7.2. The price ratio runs from 0.57 with all of the walk-away at the diligence screen to 0.77 with all of it at the bid stage, so the price effect of the restriction is between 23 and 43 percent and the working split sits inside that range at 32. The probability-of-sale ratio does not move at all. It is 0.82 in every cell, because a sale needs one engaged party through both margins and the filings fix the product of the two pass rates at every split, so F cannot touch it. What moves that ratio is the permitted pool shrinking from K_t to L_{jt}, which thins the distribution of the engaged count, and that is a separate channel from anything the split governs. The combined ratio therefore runs from 0.49 to 0.65, with 0.58 at the working value, and the entry counterfactual of Section 7.4 uses the working cell.

Table 31: The ban's auction-stage effect across the walk-away split

F (bid stage)	Price base (USD m)	Price ratio	Pr(sale) ratio	Combined
0.00	284.5	0.573	0.821	0.492
0.25	416.8	0.626	0.820	0.535
0.35*	551.7	0.681	0.820	0.579
0.50	881.2	0.746	0.820	0.630
0.65	1391.2	0.766	0.820	0.645

Notes: Each row inverts the surplus bids at the stated bid-stage share F and simulates the restriction at the same split, so the inversion and the counterfactual use one value of F throughout. Columns are means over 588 indication-years at the calibrated engagement rate $p = 0.16$: the expected winning bid in the baseline in 2015q1 $m, its ratio under the ban, the ratio of the probability that at least one engaged party bids, and the product of the two ratios. The price base rises with F because a larger bid-stage share places more of the value distribution below the reserve and the inversion recovers higher valuations; Section 7.2 reports the bid-fit check that bears on which split the bids support. * Working value.

K.5 The Bid Variable and the Paired Prices

This appendix reports two checks on the recovered valuation level that the auction-stage results summarize. The first uses the other prices. The upfront payment alone is what the seller receives with certainty, and it is priced for 246 deals. On the same 246 deals we build a risk-adjusted price, the upfront plus half the milestone schedule. Most milestones never pay, so the nominal total overstates the price and the upfront alone discards the contingent payments entirely. The middle price is closer than either endpoint to what a deal actually transfers. Table 32 runs the same reserve rule, grid and engagement rate on all three bids. On the upfront the bid index explains 53 percent of the variance, the value-to-bid ratio is 4.7 at the mean and 4.0 at the median, against 3.5 and 2.5 at the risk-adjusted price and 3.9 and 2.7 on total value, and the reserve discount calibrates to 0.40 rather than one. No price brings the ratio near the takeover benchmark, and the counterfactual ratios are close across the three bids: a combined ratio of 0.58 on total value, 0.61 at the risk-adjusted price and 0.62 on the upfront. The second check uses the paired prices coded from the merger filings, 22 processes with the winner's price and a rival's best offer in the same unit (Appendix C). Where the winner was on top, its price exceeded the runner-up's by a median of 10 percent, with an interquartile range of 2 to 20 percent, and across the 37 processes in which the winner revised its offer the rise from first offer to signing was 12 percent at the median. In seven processes the runner-up's price exceeded the winner's, which the filings attribute to certainty of closing, timing, or a proposal that was stale or never firm. Neither spread is a valuation. Each side's price bounds its value from below, so the gap between them bounds the winner's rent only in the sense of Haile and Tamer (2003). Both are the order of the slack in Gorbenko and Malenko (2014).

The comparison with the license prices is not like for like, and the difference cuts in the model's favor. The benchmark deals are whole-company takeovers of listed targets whose market price floors every bid. A license has no such floor, its reserve is the seller's continuation value, which we measure at a small fraction of the winning bid, and the acquirer's valuation includes the commercialization the seller cannot fund. A valuation several times the upfront payment is therefore not itself evidence against the model. What the ratio measures is the division of the surplus. A value-to-bid ratio of 4.7 says the winning incumbent keeps about four fifths of its own valuation as rent and the seller receives one fifth. That is a statement about competition in the licensing market, and it may be true, since a licensing canvass often ends with one party pricing. But a symmetric inversion cannot separate thin competition from unobserved asset heterogeneity in the residual

on winning bids alone, which is why the level stays out of the paper.

Table 32: The bid variable: total deal value, the risk-adjusted price and the upfront payment

	Total deal value	Risk-adjusted ($\delta = 0.5$)	Upfront payment
Priced deals (licenses and whole-company)	337	246	246
Inversion sample, after trimming	126	100	96
Mean bid in the inversion sample (USD m)	760.7	462.1	142.0
Mean recovered value over mean bid	3.88	3.47	4.65
Median value-to-bid ratio	2.72	2.46	4.03
Fit gap (USD m)	125.2	53.6	23.7
Rank correlation, bid and value	1.000	1.000	1.000
Price ratio under the ban	0.681	0.719	0.734
Pr(sale) ratio under the ban	0.820	0.820	0.820
Combined ratio	0.579	0.608	0.619

Notes: The auction stage estimated three times, with total potential deal value (upfront plus development and sales milestones, nominal; the company deal value for whole-company deals), with the risk-adjusted price (upfront plus half the milestone schedule, on the 246 deals reporting both components), and with the upfront payment alone as the bid, all in 2015q1 $m, on the priced-deal sample of Section 3.3. Same reserve rule (ϕ calibrated so 95 percent of pool deals clear the reserve; 1.00 on total value and the risk-adjusted price, 0.40 on the upfront), same grid over the walk-away split and engagement rate $p = 0.16$; rows report the working cell, a bid-stage share $F = 0.35$. The value-to-bid ratios are the mean recovered valuation over the mean winning bid and the median of the deal-level ratio, in the trimmed inversion sample. The fit gap and rank correlation are defined as in Table 10.

K.6 How Far the Reserve Sits Below the Price

Section 6.2 states that the reserve is slack by design. This appendix gives the magnitudes, and the comparison depends on which price the reserve is held against, because the three measures are not equally certain money. Across deals by pool members the reserve averages $34 million. The upfront payment, the only component paid regardless of what happens next, averages $163 million against it, a factor of 4.8 at the means and 6.1 at the median deal. The risk-adjusted price averages $551 million, a factor of 16.2 and a median of 19.1. Total potential deal value averages $900 million, a factor of 25.9 and a median of 32.2, but much of that is milestone money that never pays, so the upfront comparison is the conservative one and it already makes the point: the seller's accept threshold is not what any plausible offer struggles to clear.

Three features of the construction keep the remaining gap from being surprising. The reserve values the compound at the entrant's share of its market, 3.1 percent, while the acquirer brings a franchise, a sales force and a higher chance of approval, and that wedge is why the biotech sells rather than developing alone. The reserve is evaluated in the compound's lead market only, so a multi-indication compound's other markets are missing from the seller's outside option while the price covers the whole asset. And the priced sample is selected toward large disclosed deals by pool members, which raises the price side of every ratio above what a randomly chosen transfer would show.

The commercialization discount ϕ of equation (18) is calibrated against the same comparison and is constrained from one side only. The requirement that the winning bid clears the reserve in 95 percent of pool deals rules out values too high. When bids sit far above continuation values, as they do on total deal value and the risk-adjusted price, every value up to one satisfies it, so ϕ lands at one as an upper bound rather than an estimate. On the upfront payment alone the requirement bites and pins ϕ at 0.40. Little rides on where in that range ϕ sits, since forcing $\phi = 0.40$ everywhere leaves every estimated counterfactual ratio unchanged to the second decimal.

K.7 What the Measured Hazard Does and Does Not Separate

The acquisition hazard π^A_{jt} is measured as a single rate, and equation (11) factors the model's counterpart into the probability that a sale process is initiated and the probability that the best offer clears the reserve. The factorization is not estimated separately. The measured hazard is the product at whatever process cost the world has, so estimation imposes nothing on κ; what the data do not identify is the split between the two factors. In the counterfactual the ban's ratios move only the conditional piece, which holds the initiation factor at its baseline. Proposition 4.4 signs that omission. The option value governing initiation is an integral over the upper tail of the winning-bid distribution, which a ban on the strongest bidders compresses, so the initiation factor weakly falls and the simulated fall in the hazard understates the full response. The entry effects we report are therefore conservative on this channel, and separating the factorization is left to future work.